Evaluation of 22nm Memristor-Based Main Memory (MBMM) in Commodity DIMM Architectures

Abstract

The memory supply-demand gap of late 2025, driven by the reallocation of DRAM wafers to High-Bandwidth Memory (HBM) for AI **[1], [2]**, necessitates alternative technologies for commodity main memory. This research evaluates 22nm Memristor-based Non-Volatile Memory (NVM) as a DRAM replacement in standard DIMMs. Utilizing a cross-layer pipeline bridging NVSim **[3]** and NVMain 2.0 **[4]**, I characterize 1T1R (transistor-gated) and 1S1R (selector-gated) architectures across six workloads - drawn from five benchmarks, with AlexNet split into read-dominant and write-dominant phases - spanning compute-bound, memory-streaming, and AI-inference behavior. Under bandwidth-saturating workloads with high memory-level parallelism (MLP - many independent memory requests in flight at once), multi-rank configurations achieve substantial latency reductions through rank-level interleaving. However, single-threaded latency-bound workloads exhibit a "Flatline Paradox": insufficient MLP leaves ranks idle, preventing latency or power scaling regardless of chip count.

The key quantitative results are: (1) In wall-clock latency, 1T1R SLC ReRAM operates within 1.50x of DDR5-4800 under compute-bound workloads - where it runs 49x faster than the legacy PCM baseline (13-16x under sustained streaming) - while trailing DDR5 by 4.6x under high-parallelism AI inference (a memory-latency ratio under the cache-less trace capture of Section 2.1, not a projected application slowdown); the selector-gated 1S1R SLC follows 1T1R at a ~1.5x average-latency cost (1.3-1.7x across the suite) while offering 1.9x DDR5's die-level density (3.8x as MLC). (2) Under the repaired power model, selector-gated 1S1R ReRAM emerges as a credible low-power path past DRAM: a full module draws 1.12 W - 1.8x DDR5's 0.623 W conservative calibration floor (itself now realized under a restored idle-gating mechanism, Section 3.1.6 item 5), and within 11% of outright parity at the vendor spec-limit ceiling - with zero refresh cost, and because that draw is 97% static, essentially all of it is exposed to idle-power gating - a mechanism now mechanically restored and simulated for every technology, though ReRAM's own power-down energy remains an explicit, disclosed placeholder equal to its standby draw, pending real device characterization no literature search located (Appendix A), so no simulated power benefit is yet claimed for it. Power parity is therefore still a projection contingent on that future characterization - bounding arithmetic on measured static power, not a simulated result - while DDR5 spends 33-45% of its module power on refresh it can never shed - a concrete opening for gating-capable memory controllers and selector-first DIMM designs. The contrast that proves the point: the transistor-gated 1T1R module leaks 50.9 W (65-78x DDR5) - infeasible ungated at DIMM scale - so the selector's 47x leakage discipline, not raw cell speed, decides architectural viability, and 1S1R SLC is the flagship configuration. These power results rest on a fidelity audit of the simulation flow (Section 3.1.6) that found fourteen silent failure modes in the standard NVSim-to-NVMain toolchain - a mixed-clock-domain efficiency metric, two silently ignored parameter families (leakage and access energy), a technology-blind static-power default, single-rank power reported as system power, a disabled power-down model, weak trace provenance, DDR5 refresh timing and supply parameters inherited unrescaled from a DDR3-era template, a PCM baseline running at twice its cited clock basis, heterogeneous host-CPU frequencies that replayed each technology against a different offered load and admitted trace population, an unsourced DDR5 CAS/RCD/RP timing placeholder, and unsourced ReRAM MLC read/write latency and energy penalty multipliers attributed to a citation that does not support them (later found to include a further, independent data error in the MLC read-latency figure, also corrected) - of which this work repaired twelve, validating every repair against device-level anchors (0.0% error) or exact predicted arithmetic, and re-running the affected simulations (the DRAM and PCM baselines model standard idle behavior; ReRAM figures are worst-case ungated). (3) DIMM write endurance scales linearly with module capacity: at the modeled 8 GB SLC module, worst-case sustained streaming (LBM) yields 1.1 years - below the 5-10 year server replacement target - but at the 64-128 GB capacities where ReRAM's density advantage is actually realized, worst-case SLC lifetime reaches 9-17 years (the slower-writing selector variant, 12-25), while compute-bound and read-dominant workloads clear the target severalfold to a hundredfold even at 8 GB. Endurance is thus a capacity- and workload-dependent design constraint rather than a categorical barrier. (4) Read latency is invariant to a ±20% ReadVoltage perturbation, with Power-Delay Product changing less than 4%, confirming robustness to supply voltage variation and operating-point selection.

ReRAM write-latency penalties are workload-dependent and are confined to write-active workloads, positioning selector-gated 22nm SLC ReRAM as a latency-competitive, density-superior DDR5 alternative for compute-bound workloads - and within a disclosed 1.9-2.1x of DDR5 under sustained streaming - with a clear research path - restoring idle-power gating to both the simulation stack and the architecture - standing between it and outright power parity. High-parallelism AI inference remains DDR5 territory.

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1. Introduction & Background

1.1. Project Context

- **The External Memory Wall**: Traditional DRAM is struggling to meet the bandwidth requirements of modern processors due to physical scaling limits, specifically capacitor leakage and transistor short-channel effects **[5]**.
- **Economic Drivers**: By late 2025, the massive reallocation of DRAM wafers to AI-focused HBM production created a severe supply-demand gap for commodity main memory **[1], [2]**.
- **The Objective**: This project investigates whether memristor-based non-volatile memory (NVM) can effectively substitute traditional DRAM in standard DIMM modules to alleviate this constraint.
- **The Endurance Constraint**: However, deploying NVM as main memory introduces a critical challenge: endurance. Because physical wear-out limits are the primary bottleneck of memristive technologies, a viable DRAM alternative necessitates architectural interventions to prevent premature device failure.

1.2. ReRAM Fundamentals

Before the engineering choices, the device itself: a memristor (ReRAM cell) is a two-terminal metal-insulator-metal stack - typically an HfOx or TaOx oxide - whose electrical resistance *is* its stored state. A programming voltage grows or ruptures a conductive filament of oxygen vacancies through the insulating oxide, switching the cell between a low-resistance state (LRS) and a high-resistance state (HRS); a read applies a small sense voltage that leaves the filament undisturbed [14]. Because the filament persists without power, storage is non-volatile - no refresh, no standby charge maintenance - and because state is resistive rather than capacitive, the cell has no capacitor to scale, which is precisely the wall DRAM's planar cell faces [5]. Writes physically stress the filament, so endurance is finite (Section 3.1.4), and filament formation is stochastic, which is why the resistance targets and sensing margins below are engineering choices rather than free parameters.

- **Process Selection:** I standardized on 22nm FinFET LOP (Low Operating Power). This node allows for logic-compatible integration **[8]** while the LOP profile aligns with the thermal and power constraints of high-capacity main memory replacement. The choice also frames the entire comparison conservatively: commodity DDR5 is fabricated on 10 nm-class DRAM nodes (1γ [17]) that sit at the end of the capacitor's scaling roadmap [5], whereas 22nm is a mature, logic-compatible CMOS node with several shrink generations of headroom still ahead of it - the density figures in this evaluation are therefore a floor on ReRAM's potential, not a ceiling.
- **Resistance Design:** Cells are calibrated to high resistance targets ($10^5\Omega$ LRS and $10^9\Omega$ HRS) **[7]** to mitigate IR-drop on 1024-cell bitlines, ensuring reliable sensing margins.

- **Device Structures & Sneak-Path Mitigation:** I evaluate two distinct hardware topologies to handle array sneak-paths:
  - **1T1R (Transistor-based):** Utilizes CMOS access transistors to completely eliminate sneak-paths. While this results in a logic-compatible but larger bit-cell area - a commonly-used, transistor-drive-limited engineering estimate of approximately $20F^2$ (derivation and demonstrated-endpoint caveats in Appendix A) - modern commercial foundries currently mass-produce 22nm embedded ReRAM (eReRAM) macros at megabyte scale **[24]**, inherently validating its manufacturability far beyond the 1Mb subarrays evaluated here.
  - **1S1R (Selector-based):** Suppresses local sneak-paths via highly non-linear thresholding. While academic 1S1R test-chips frequently restrict subarray sizes to manage leakage, industry milestones - such as Crossbar Inc.'s fabrication of a 4Mb cross-point array utilizing a super-linear threshold selector **[9]** - demonstrate the physical viability of massive crossbar scaling. Building on this foundation, combining the NVSim non-linear selector model **[3]** with the aforementioned ultra-high resistance calibration **[7]** provides a theoretical architectural projection capable of scaling to 1024-cell bitlines. This reduces the cell to an idealized $4F^2$, enabling a significant footprint reduction for high-density DIMMs.
- **Area Realism over Financial Metrics:** To evaluate module cost and density objectively across these topologies, I utilize area-per-bit (F^2) rather than financial cost-per-bit, isolating architectural evaluation from fabrication economics and market volatility.

1.3. Related Work

The architectural case for replacing DRAM with a non-volatile main memory was first made at scale by three concurrent ISCA 2009 studies, all targeting phase-change memory. Lee et al. architected PCM as a scalable DRAM alternative, narrowing a raw 1.6x delay / 2.2x energy deficit to within 1.2x / 1.0x through row-buffer reorganization and partial writes [36]; Qureshi et al. hybridized a large PCM store behind a small DRAM buffer (~3% of PCM capacity) with lazy writes and fine-grained wear leveling [37]; and Zhou et al. attacked durability directly, extending projected PCM lifetime to 13-22 years via redundant-bit-write removal and row-shifting wear leveling [38]. That generation established the field's enduring template - hide the write penalty, level the wear, buy density where latency cannot be hidden - but it evaluated DDR2/DDR3-era baselines, and its commercial embodiment, Intel's Optane/3D XPoint, was ultimately discontinued on cost economics rather than physics [23].

For the resistive technology evaluated here, the closest prior work is Xu et al.'s HPCA 2015 study of crossbar ReRAM as main memory, which characterized the IR-drop- and sneak-current-induced, data-dependent RESET latency of large crossbars and recovered to within ~10% of an ideal DRAM-only system through split-phase RESET and compression-based encoding [39]. Kültürsay et al. ran the parallel exercise for STT-RAM, finding an unmodified STT-RAM main memory uncompetitive but a lightly optimized one performance-comparable to DRAM at ~60% lower memory energy [40]. On the empirical side, Izraelevitz et al. published the first comprehensive third-party characterization of the only NVM main-memory module ever shipped

- the Optane DC PMM - and found its behavior substantially more nuanced than the "slow, persistent DRAM" abstraction earlier emulation-based studies assumed [32]: a standing caution for every simulation-only evaluation, this one included (Section 2.2, Validation Scope). Concretely, on real silicon [32] measured idle random-read latency of 305 ns against 81 ns for local DRAM on the same platform (~3x, their Section 3.1.1), and a single-DIMM maximum bandwidth of 6.6 GB/s read / 2.3 GB/s write (their Abstract) - real, measured numbers this book's own simulated ReRAM/PCM figures are never directly benchmarked against, since Optane's selector-gated PCM architecture and this study's technologies differ in both material and device topology. They are reported here as the field's one genuine real-hardware anchor point, explicitly distinct from every simulated result in Section 3.

Against that body of work, this book's contribution is fivefold: (a) the baseline is commodity DDR5 with vendor-calibrated power, not the DDR2/DDR3-era DRAM of the founding studies; (b) the ReRAM inputs are a device-anchored 22nm NVSim characterization propagated into cycle-accurate NVMain simulation, rather than abstract latency/energy multipliers; (c) transistor-gated 1T1R and selector-gated 1S1R are compared head-to-head at a matched process node, making the selector's leakage discipline - not a single technology's viability - the object of study; (d) the simulation toolchain itself is audited and repaired (Section 3.1.6), where prior studies take the NVSim-to-NVMain wiring on trust; and (e) endurance is projected from measured per-subarray write counts under real traces rather than analytic write rates. The complementary boundary is equally explicit: this work models the crossbar at NVSim's calibrated array-level abstraction (Section 1.2) and does not reproduce [39]'s circuit-level, data-dependent write timing - the two levels of analysis answer different questions, and integrating them is part of the agenda in Section 4.1.

Every technology evaluated in this book is judged on the same three axes: **latency** (Section 3.1.1), **density** (Section 3.3), and **endurance** (Section 3.1.4) - with power as a fourth, closely related dimension (Sections 3.1.2-3.1.3). This framing is stated explicitly here because it also gives a home for admitting when an axis is *not* evaluated for a given technology - for instance, this study's PCM baseline was never carried through an endurance projection (Section 3.1.4), a disclosed scope choice rather than a missing result.

2. Infrastructure & Methodology

2.1. Decentralized Structure & Simulation Stack

- **The Bridge:** I developed a cross-layer pipeline that automatically translates device-level physical metrics from NVSim (Area, absolute Latency in nanoseconds, Energy) into cycle-accurate architectural parameters for NVMain 2.0 (JEDEC timing constraints such as Column Address Strobe latency (tCAS), Row-to-Column Delay (tRCD), Row Precharge time (tRP), and dynamic energy-per-access coefficients).
  Core memristive cell parameters and resistance targets were rigorously calibrated against established silicon fabrication literature. A full table of these electrical parameters and their corresponding references is provided in **Appendix A**.

- **Simulation Pipeline**: The pipeline follows a structured sequence: hardware characterization in NVSim → diagnostic extraction to JSON → system configuration generation → cycle-accurate trace execution in NVMain. NVSim characterizes exactly **one 1 Gb** chip (Table 1); NVMain never re-touches cell-level physics; it replicates that same chip **1, 8, 16, or 64** times per configuration and wraps the result in its own controller, decoder, and interconnect - components NVSim has no concept of. The diagram below illustrates this end to end.

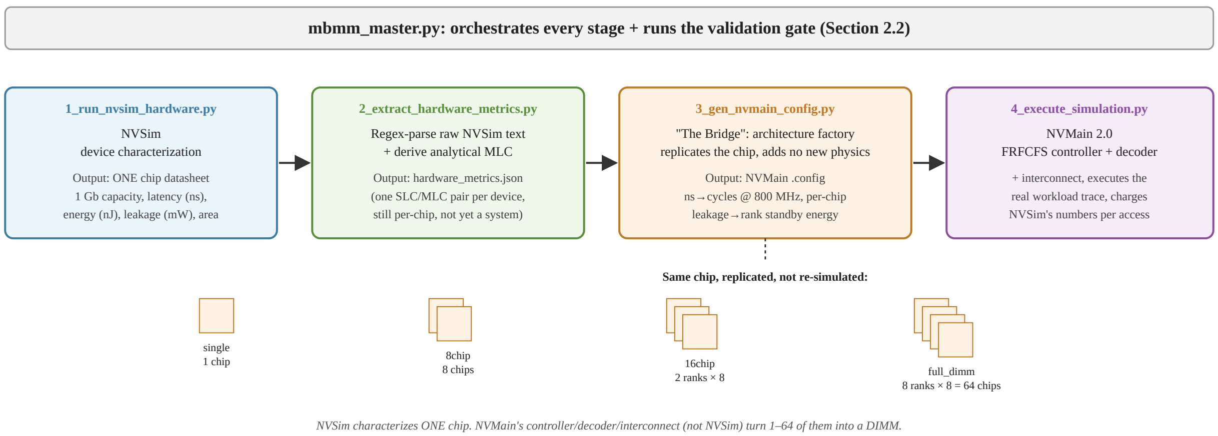


Diagram: NVSim's device characterization (one chip) becomes an NVMain system via replication and NVMain's own controller logic - not part of the numbered Figure/List of Figures sequence, since it illustrates pipeline architecture rather than a simulation result.

- **Decoupled Benchmarking**: To ensure simulation stability, I bypassed brittle integrated wrappers in favor of a robust trace-based pipeline utilizing gem5 v25.1 memory access logs. I transitioned to 602.gcc and 619.lbm using gem5's X86O3CPU (Out-of-Order). The audit of Section 3.1.6 later established that these traces were captured without L1/L2 caches or warmup (raw CPU-to-memory stream, first 10M instructions), so they represent maximum memory pressure rather than cache-filtered traffic - a conservative choice for endurance bounds, and a documented limitation for absolute latency magnitudes.

2.2. Validation & Protocol

- **The Gate-Keeper**: All architectural conclusions are gated by the **mbmm_master.py** script. This script mandates a validation flow of cleaning build artifacts, recompiling the C++ engines, and passing a sanity-check simulation before data is accepted.

- **Critical C++ Repairs**: To modernize the legacy simulation engines, I applied legacy compatibility patches:

  - > **NVSim**: Enforced -std=c++11 to resolve namespace collisions between MemoryType::data and std::data in modern GCC.

> ▸ **NVMain**: Repaired a fatal memory deallocation bug in FlipNWrite.cpp (mismatched new/delete) that triggered SIGSEGV failures.

> ▸ **Build Migration**: Migrated the legacy build system from Python 2 to **Python 3** using 2to3.

> ▸ **MLC Analytical Modeling**: Prior to simulating Multi-Level Cell (MLC) scaling, I established the physical and architectural differences between SLC and MLC - a technique with established multi-bit-per-cell precedent in the literature, demonstrated up to 3 bits per cell [13]. Recognizing that NVSim lacks native support for the complex Analog-to-Digital (ADC) sensing and Iterative Step-and-Verify (ISPV) programming required for MLC, I implemented an Analytical Penalty Method (1.5x read latency, 3.263x write latency, 1.1x read energy, 3.0x write energy, measured on the same EMBER macro's two publications - Upton et al. [6], ESSCIRC 2023, Table I, for the read-energy figure, and Levy et al. [31], IEEE JSSC 2024, Section V, for the read-latency and write-side figures - see Appendix A for the full derivation) rather than relying on NVSim's native MLC implementation, which triggers a floating-point exception in Mat.cpp's ISPV sensing logic.

- **Validation Scope**: The validation flow above establishes *internal* consistency, not hardware correlation. Input parameters are anchored to real silicon and vendor documents - the EMBER macro's published measurements [6], [31] and the vendor DDR5 IDD datasheets of Section 2.3 - and every repair in Section 3.1.6 is verified against NVSim's device-level output or exact predicted arithmetic (0.0% error). No end-to-end result, however, is validated against a measured ReRAM DIMM or an instrumented DDR5 system - no ReRAM main-memory module exists to measure. The “0.0% error” claims throughout this book are therefore pipeline-fidelity guarantees (the system level faithfully reflects its device-level inputs), not accuracy guarantees against physical hardware.

2.3. Hardware Baselines & Memory Models

- **The DDR5-4800 Baseline Reference:** The DDR5 configuration is modeled strictly after the JEDEC Solid State Technology Association, Standard JESD79-5 (DDR5 SDRAM) [10]. I engineered the timing parameters for a 4800 MT/s module (tCAS-tRCD-tRP of 40-39-39, per the SK hynix DDR5-4800 speed-bin decoder - Section 3.1.6, item 12 - at 1.1V) and explicitly modeled DDR5's defining architectural upgrade: two independent 32-bit sub-channels per DIMM and a Burst Length of 16. This model inherently imposes the rigorous, real-world JEDEC bus serialization and complex bank-group switching penalties that define modern commodity DIMM performance. Refresh is modeled explicitly to the same standard: an all-bank refresh command every tREFI = 3.906 μs (the JEDEC 32 ms retention window spread over 8,192 commands), each occupying tRFC = 295 ns, at the DDR5 supply of 1.1 V, with refresh energy accumulated in dedicated rank-level counters - the source of the refresh decomposition in Section 3.1.2. These refresh and voltage parameters were corrected during the fidelity audit (Section 3.1.6, items 7-8) after being found inherited, unrescaled, from a DDR3-1333 template configuration.

- **The Microsoft PCM 2009 Baseline:** A legacy non-volatile memory baseline, named for its architectural framing in Lee et al. **[11]**, used to benchmark modern 22nm ReRAM against older NVM scaling constraints. Unlike ReRAM, PCM was never run through NVSim's device-characterization pipeline: its concrete timing and energy parameters are inherited unmodified from NVMain's own bundled example configuration (**pcm_microsoft_2009.config**), whose header attributes those specific numbers to a different device paper - Choi et al.'s 20nm, 1.8 V, 8 Gb PRAM (ISSCC 2012) **[41]** - not to Lee et al. This project did not independently re-derive or verify those inherited numbers. It runs at its cited derivation basis of CLK = 400 MHz - restored during the fidelity audit (Section 3.1.6, item 9) after a stray clock override was found running it twice as fast as its timing parameters intend - and runs a 33.3M-cycle budget at that basis - the matched-host window of Section 3.1.6, item 11: 83.33 ms wall-clock and an identical admitted trace population for every configuration. Neither its inherited config nor its cited source papers specify a particular access-device architecture (transistor- or selector-gated) - this baseline is a black-box timing/energy model, not a characterization of any specific PCM cell topology. This is a disclosed scope gap, not a hidden assumption: the real, shipped commercial embodiment of PCM - Intel/Micron's Optane/3D XPoint, architecturally confirmed by Intel/Numonyx's own IEDM 2009 device paper **[42]** and independently corroborated by a professional physical teardown of a retail unit - is a selector-gated 1S1R cross-point design (an Ovonic Threshold Switch in series with the phase-change element, no access transistor), not modeled here at all. Real, measured performance numbers from that shipped product are reported alongside this study's simulated results in Section 1.3.
- **The ReRAM DIMM Interface (800 MHz):** All ReRAM configurations run behind an 800 MHz (1600 MT/s) DIMM interface - a deliberate modeling choice, not a device limit. The interface clock sets only command granularity and the channel's bandwidth ceiling; the NVSim-characterized cell latencies are specified in nanoseconds and are clock-independent by construction (a 32.13 ns 1T1R read occupies 26 cycles at 800 MHz and would occupy 77 at 2400 MHz - the same wall-clock time). The rate is positioned between two precedents: twice the 400 MHz basis of NVMain's PCM reference lineage [11], and below the DDR4-class DDR-T interface of Intel's shipped Optane persistent-memory DIMMs [32] - the industry precedent that NVM DIMM interfaces trail the contemporary DRAM PHY rather than match it. Where the choice matters is the bandwidth-bound AI regime of Section 3.1.1: part of the 4.6x queueing deficit reflects this interface asymmetry against dual-channel DDR5-4800, so the reported gap is conservative against ReRAM, and pairing the same media with a faster PHY is an available mitigation, not a hidden cost. *This 800 MHz figure itself has no independent ReRAM-interface citation* - it is positioned only by relationship to the two precedents above, not derived from a ReRAM PHY specification. A targeted literature search (EMBER [6]/[31]'s “100 MHz” is confirmed, by both papers' own text, to be the macro's internal sense/write-circuit clock, not an external interface rate - no interface clock is stated for either device) found no independent ReRAM interface-speed citation to support it; the only

real, datasheet-verified ReRAM chip I/O clocks located (Fujitsu's MB85AS8MT and MB85AS4MT SPI-interface parts, 10 MHz and 5 MHz respectively) run 80-160x *slower* than 800 MHz, actively contradicting rather than merely failing to support this figure. 800 MHz should therefore be read as a deliberate, disclosed modeling assumption positioned between two DRAM/PCM-adjacent precedents, not a literature-derived ReRAM interface speed.

- The 2D and 3D DRAM Control Configurations: NVMain 2.0 ships with native 2d_dram (legacy planar, DDR3-era, 666 MHz) and 3d_dram (Wide I/O-style Through-Silicon Via, 1333 MHz) example models. These are idealized textbook abstractions rather than models of purchasable products: unlike the DDR5-4800 model, which faithfully pays JEDEC bus serialization, Burst Length 16 transfer, and bank-group switching penalties, the example configurations bypass real-world interface overheads. They were retained in the simulation matrix as pipeline control variables - useful for verifying clock-domain normalization and for bounding theoretical throughput - but they are excluded from every reported cross-technology comparison, which is restricted to technologies with commodity referents: DDR5-4800, PCM, and the four ReRAM configurations. The 2D/3D runs are preserved in the released dataset (Appendix C) for completeness.

Table 1 collects the NVSim device-level characterization that the system simulation inherits - the quantitative form of the trade-off profile stated above: the selector costs 1.6x on reads and 2.3x on writes, and repays it with a 47x lower leakage floor and an 8.7x smaller die.

Table 1: NVSim device-level characterization per 1 Gb chip (22nm FinFET LOP, 1.4 V ReadVoltage nominal).

Config.	Cell size	Read lat. (ns)	Write lat. (ns)	Read energy (nJ)	Write energy (nJ)	Leakage (mW/ chip)	Die area (mm^2)
1T1R SLC	$20F^2$	32.13	32.28	1.19	1.74	794.7	19.80
1T1R MLC*	$20F^2$	48.20	105.33	1.31	5.21	794.7	19.80
1S1R SLC	$4F^2$	52.13	72.89	6.77	2.47	16.9	2.28
1S1R MLC*	$4F^2$	78.20	237.85	7.45	7.41	16.9	2.28

**MLC rows apply the Analytical Penalty Method (1.5x read latency, 3.263x write latency, 1.1x read energy, 3.0x write energy - Section 3.1.6, item 13); leakage and die area are unchanged. Sources: results/hardware_metrics.json; Sections 2.1-2.2.*

2.4. Workload Traces & Benchmarks

To rigorously evaluate the memory architectures across different access patterns, I selected a diverse suite of workloads spanning traditional single-threaded execution, pure bandwidth stress tests, and modern parallel AI:

- **SPEC CPU2017 (602.gcc & 619.lbm):** Generated via gem5 [26], from the SPEC CPU2017 suite [28]. gcc was selected as a compute-bound, control-flow heavy workload with irregular memory accesses. In contrast, lbm (fluid dynamics) was selected as a memory-bound workload to test sustained, sequential memory throughput.
- **STREAM Benchmark:** An industry-standard synthetic diagnostic [27] used specifically to measure the practical upper bound of sustained memory bandwidth and expose baseline latency bottlenecks.
- **SCALE-Sim ML Traces (AlexNet & GPT-2):** Generated via the SCALE-Sim systolic array simulator. These workloads were selected to represent modern, highly parallel AI operations. AlexNet is split into IFMAP (read-dominant) and OFMAP (write-heavy "Write-Torture") to isolate the ReRAM write-latency penalty, while GPT-2 evaluates massive transformer-based memory-level parallelism (MLP).

2.4.1. Diagnostic Trace Dictionary

The following specific traces were executed across all simulated architectures to isolate distinct memory behaviors:

- **gcc_spec2017 (Compute-Bound):** A standard CPU compilation workload whose control-flow-heavy instruction mix generates sparse, irregular memory traffic (measured mix: 236,562 reads / 170,800 writes per window - 58/42). **Purpose:** It leaves the main memory mostly idle, perfectly exposing the pure *Static Leakage Power* of the DIMMs. Note that the gem5 traces are uncached raw CPU-to-memory streams (Section 3.1.6, item 6), so this idleness stems from the workload's own compute-bound arithmetic density, not from cache filtering - its memory pressure is, if anything, overstated.
- **lbm_spec2017 (Memory-Streaming):** A fluid dynamics simulation that constantly streams large data vectors from memory (measured mix: 9,481,308 reads / 6,965,793 writes - 58/42, at roughly 40x GCC's request density). **Purpose:** It highlights the *Dynamic Energy* penalty of switching topologies, as memory is constantly active.
- **stream (Vector-Streaming)**: A standard synthetic memory benchmark designed to measure sustainable memory bandwidth and simple vector kernel execution, providing a baseline for continuous, predictable memory traffic (measured mix: 99,999 reads / 100,000 writes - an exact 50/50 copy kernel).
- **alexnet_layer1_ifmap (CNN Read-Storm):** A neural network loading its Input Feature Map. Cycle-accurate traces for this and all subsequent neural network workloads were generated using the SCALE-Sim systolic array simulator **[12]**. **Purpose:** Exposes memory controller queueing delays under massive, parallel read requests (pure-read: 184,319 reads, zero writes).

- **alexnet_layer1_ofmap (CNN Write-Torture):** A neural network writing its computed activations back to memory. **Purpose:** Explicitly exposes the physical 4× write-latency penalty of Iterative Step-and-Verify (ISPV) MLC programming (pure-write: 13,542 writes, zero reads).
- **gpt2_ifmap (Transformer-style Inference):** A General Matrix Multiply (GEMM) read trace representing transformer-style weight streaming. **Purpose:** The maximum-parallelism read test, comparing ReRAM's read-bandwidth behavior directly against modern dual-channel DDR5 (pure-read: 6,553 reads, zero writes). Provenance caveat (Section 3.1.6): unlike the AlexNet traces (confirmed SCALE-Sim TPU-v1 configuration), this trace could not be tied to a preserved SCALE-Sim run; it is treated as a representative parallel-read stress pattern rather than a validated GPT-2 model trace.

Together, the six traces span the three canonical intensity classes: compute-bound with memory mostly idle (GCC), mixed sustained streaming (LBM and STREAM, 58/42 and 50/50 read/write), and pure single-operation stress at both extremes - read-intensive (GPT-2, AlexNet IFMAP) and write-intensive (AlexNet OFMAP). Under the matched-host replay of Section 3.1.6, item 11, every technology admits the identical request mix for every trace, so these classifications hold uniformly across the comparison.

3. Results: Multi-Rank Architecture & Scaling

3.1. Granular Workload Diagnostics (Latency & Power Bar Charts)

To fully understand the system-level tradeoffs of memristor-based main memory, I isolated the physical variables of latency and power across distinct workload profiles. The evaluation is organized around the canonical ReRAM trade-off profile - slower access than DRAM, zero refresh, higher density, and individually expensive write operations - and quantifies each axis in turn: latency in Section 3.1.1, the refresh-free but leakage-bound power identity (including where the write-energy cost actually surfaces) in Section 3.1.2, their balance in Section 3.1.3, write endurance in Section 3.1.4, and density in Section 3.3.

One framing note before the numbers, so the trust boundary is explicit from the outset: every figure in this chapter passed through a simulation-fidelity audit of the NVSim-to-NVMain toolchain that found **fourteen** silent failure modes, of which **twelve** were repaired - each repair validated against device-level anchors or exact predicted arithmetic, and the affected simulations re-run - while **two** remain as disclosed permanent limitations rather than bugs: idle-power gating is disabled in the simulator source (so all ReRAM power figures are worst-case ungated), and the traces carry weak provenance (cache-less capture, and one AI trace unattributable to a preserved generator configuration). The full audit, with the repair arithmetic, is Section 3.1.6; results below cite its item numbers where a specific repair or caveat applies.

3.1.1 Latency Analysis: The MLC Write Penalty

The most critical operational constraint of ReRAM is its write latency, which is exacerbated when utilizing Multi-Level Cells (MLC) due to the required Iterative Step-and-Verify (ISPV) programming. To quantify this impact at the architectural level, I evaluated the memory configurations across compute-bound, streaming, and neural network traces. A methodological note governs how performance is read throughout this section: in fixed-window trace replay the trace dictates the offered load, and after the matched-host correction of Section 3.1.6, item 11, every configuration admits the identical request population per workload (a uniform 250M-trace-cycle cutoff at a 3 GHz host; 83.33 ms wall-clock for every technology). Five of the six workloads - GCC, STREAM, GPT-2, and both AlexNet phases - are completed in full by every technology, so there the cost of a slower memory appears purely as latency and queueing over identical populations. LBM's sustained streaming is the one service-limited case, and it is a clean, like-for-like throughput result: of an identical 16,447,102-request admission, DDR5 completes 100%, 1T1R SLC 40.0%, 1S1R SLC 28.0%, 1T1R MLC 16.3%, 1S1R MLC 9.3%, and PCM 3.9% - a delivered-throughput ordering, graded exactly by device write speed, that hardens this section's MLC verdict and quantifies the sustained-streaming cost of every resistive technology at once; its single accounting consequence is that LBM latency averages cover each configuration's completed prefix. This completion gap is not purely a device-physics ceiling: a follow-up sensitivity sweep (Appendix A) shows the memory controller's write-queue depth is a live, previously undocumented lever on it - raising it recovers part of the gap, at a real latency cost.

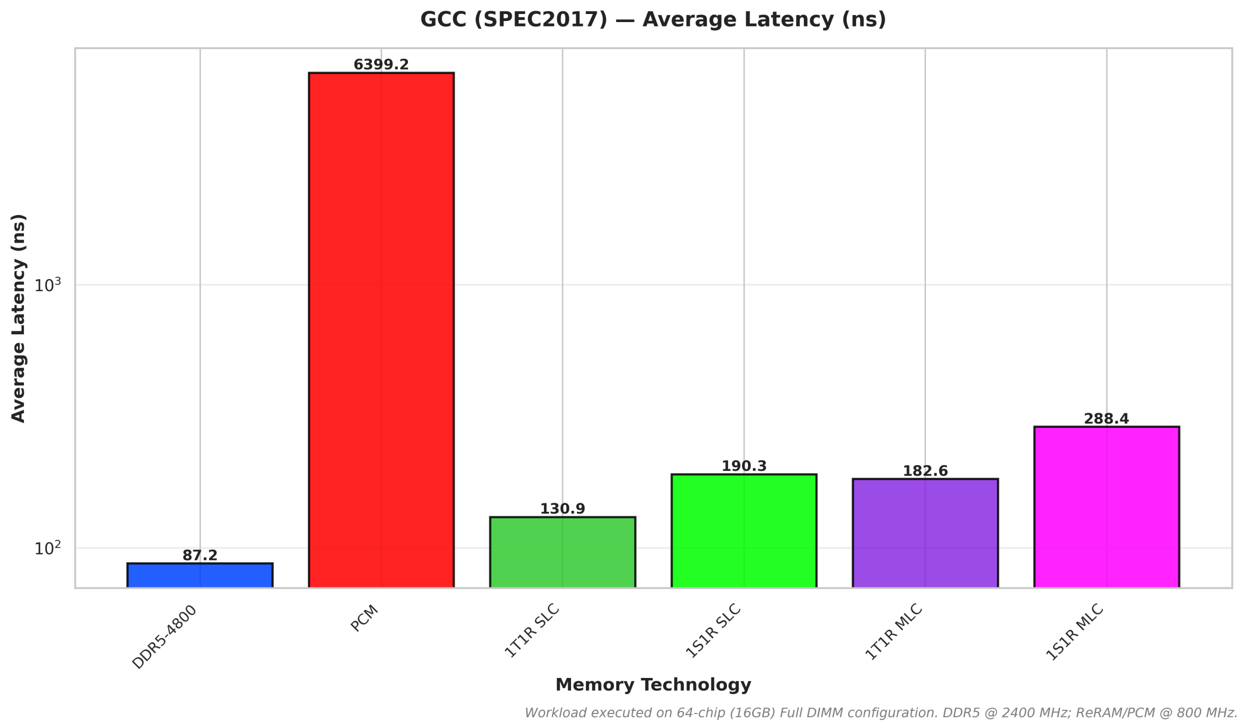


Figure 1: Average Total Latency under the compute-bound GCC (SPEC2017) workload.

Under standard compute-bound workloads with small, infrequent memory bursts like gcc (Figure 1), the 22nm 1T1R SLC ReRAM configuration demonstrates exceptional responsiveness. Averaging 104.73 cycles at 800 MHz (130.91 ns), 1T1R SLC is slower than DDR5-4800 in absolute wall-clock time (81.16 ns, or 194.78 cycles at 2400 MHz - an average that now includes DDR5's faithfully modeled refresh blocking; Section 3.1.6, item 7) - a 61% latency overhead that is the smallest cross-technology gap in the suite. The selector-gated 1S1R SLC follows at 190.35 ns, a 1.5x cost over 1T1R for the selector's slower access path: NVSim's device characterization prices a 1S1R read at 52.1 ns versus 32.1 ns for 1T1R (1.6x) and a write at 72.9 versus 32.3 ns (2.3x), because the crossbar must sense through the non-linear selector at the deliberately high 10^5 Ω resistance calibration and bias writes past the selector threshold, whereas the access transistor - the very thing that costs 1T1R its $20F^2$ cell - provides isolated, full-drive access. This gap is not a modeling artifact but the latency half of the selector trade-off, and Section 3.1.3 shows it is decisively repaid in the power term. Whether this latency proximity translates into system efficiency is a power question, and it is answered in Sections 3.1.2 and 3.1.3 under the repaired power model of Section 3.1.6.

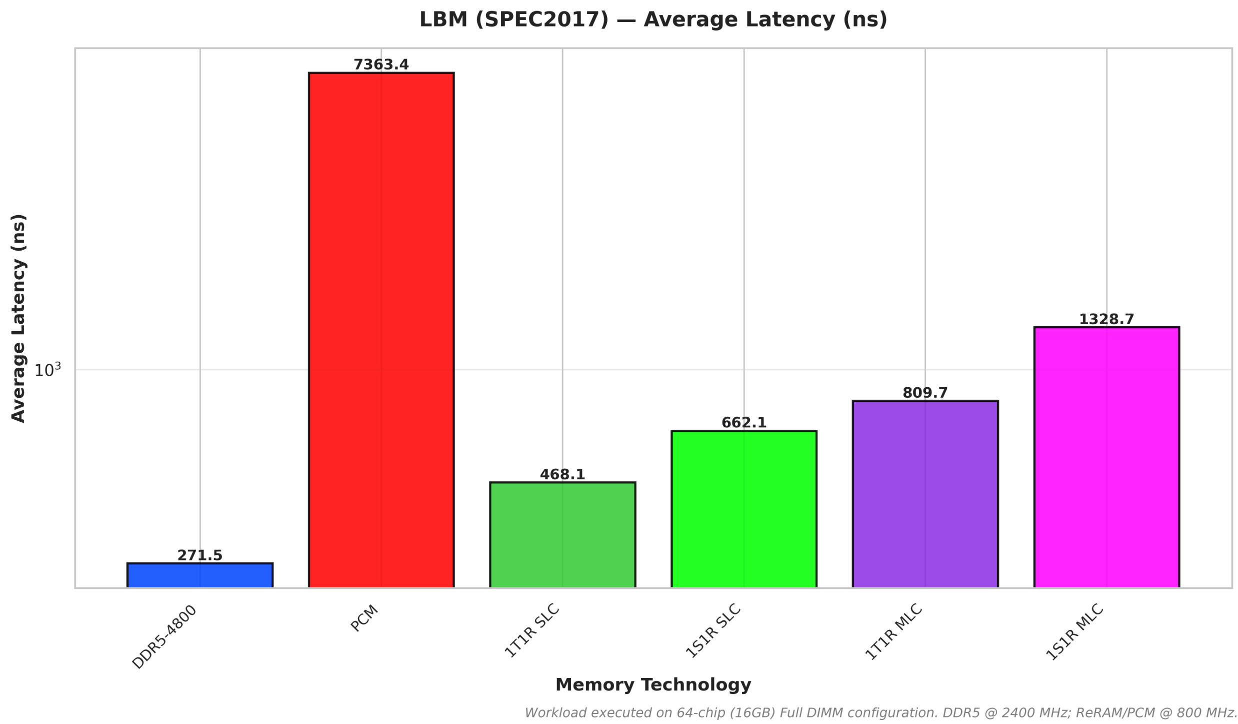


Figure 2: Average Total Latency under the continuous memory-streaming LBM (SPEC2017) workload.

To contrast the compute-bound nature of GCC, I evaluated the memory subsystem under the lbm trace (Figure 2), a fluid dynamics workload characterized by heavy, continuous memory streaming. Here, the architectural dynamics change significantly. The 1T1R SLC configuration averages 374.50 cycles at 800 MHz (468.12 ns), compared to DDR5-4800 at 600.31 cycles at 2400 MHz (250.13 ns) - a 1.9x wall-clock edge to DDR5 from its higher operating frequency. The efficiency implications under streaming are governed by each technology's leakage tier and are quantified in Section 3.1.3. This workload also exposes the limits of Multi-Level Cells: the 1S1R MLC latency spikes to 1062.99 cycles (1328.74 ns), as continuous write-pressure backs up the FRFCFS controller queue with ISPV programming operations, increasing average total latency by 2.8x relative to 1T1R SLC (2.0x relative to its own 1S1R SLC sibling).

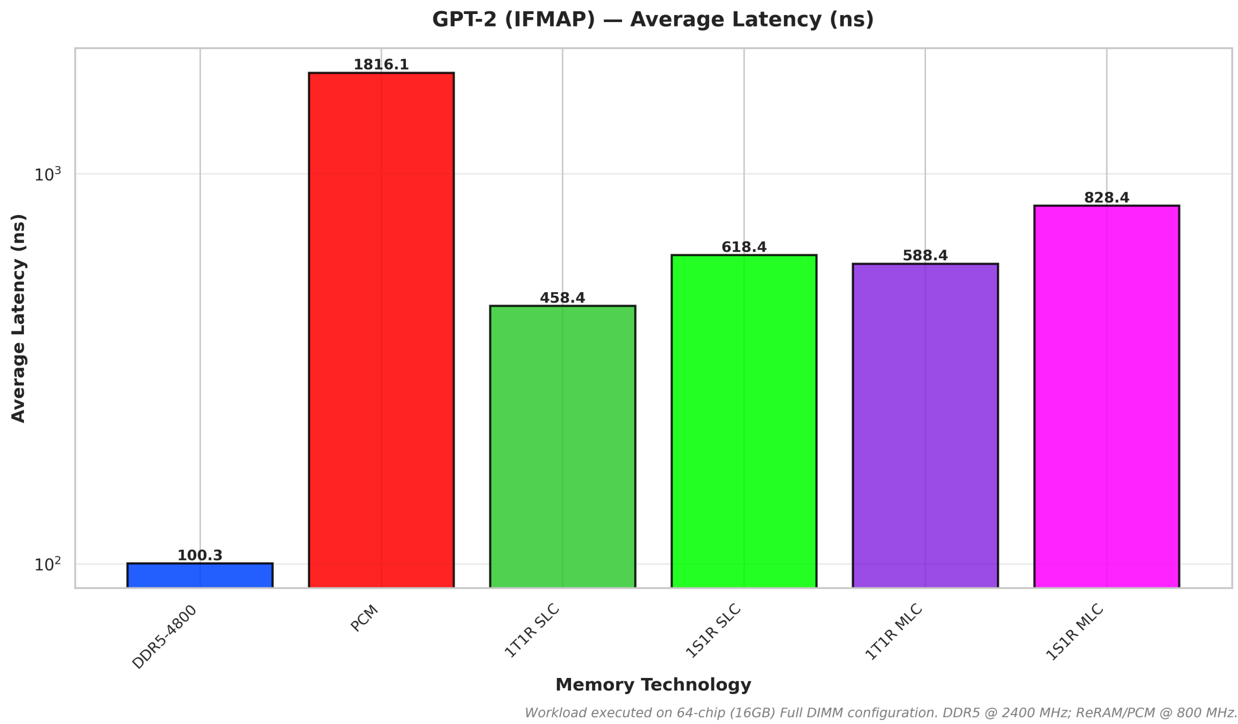
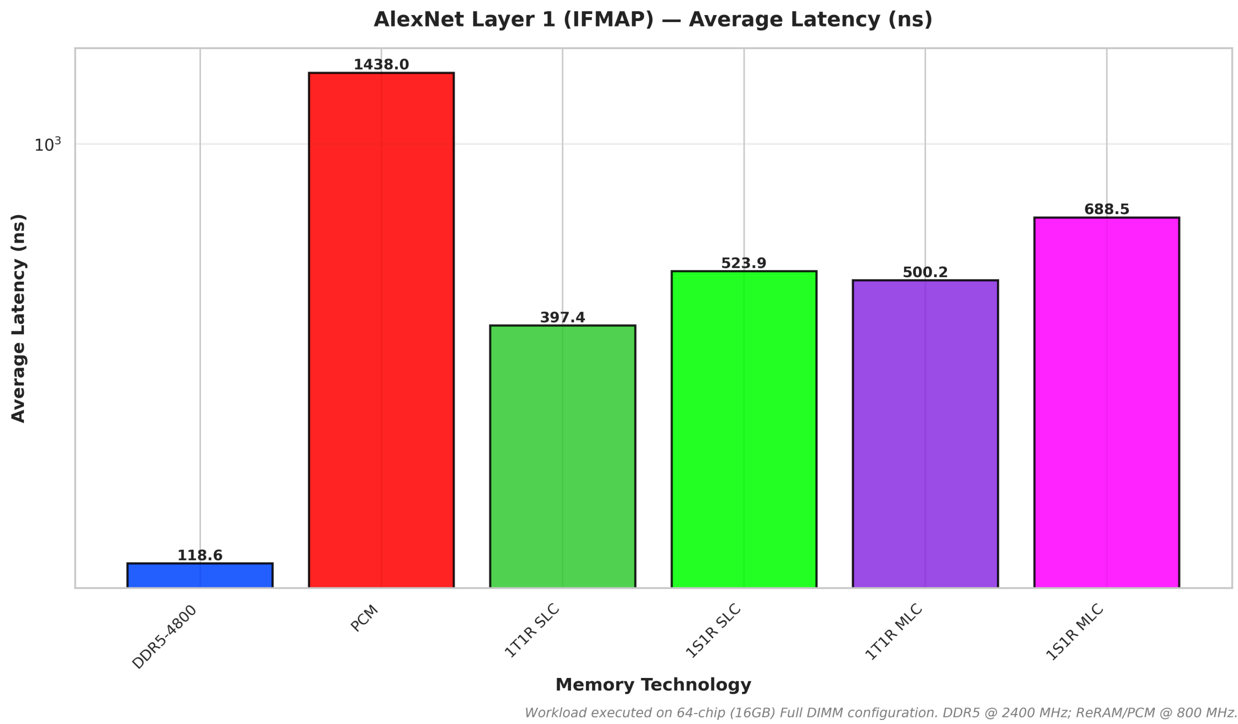


Figure 3: Average Total Latency under the parallel read-storm of GPT-2 Inference (IFMAP).

However, when the system is subjected to the massive parallel read-storms typical of Large Language Model (LLM) inference, the dynamics shift. In the GPT-2 trace (Figure 3), the 1T1R SLC configuration averages 366.73 cycles at 800 MHz (458.41 ns), trailing the DDR5-4800 baseline (240.65 cycles at 2400 MHz, 100.27 ns) by approximately 4.6x in wall-clock terms. Despite ReRAM's fast raw read speed, this data exposes the queueing delays within the memory controller. Under extreme read pressure, the ReRAM DIMM cannot clear its request buffers as efficiently as the highly-pipelined, dual-channel DDR5 interface. Under AI inference, DDR5 wins decisively on latency - with the caveat that 4.6x is a memory-latency ratio, not a projected end-to-end application slowdown (Section 3.1.6) - and the PDP analysis of Section 3.1.3 shows that the efficiency verdict compounds rather than offsets this deficit. Any ReRAM role in AI inference deployment is therefore confined to duty-cycled or standby-dominated serving scenarios, where zero-refresh retention between bursts - not sustained throughput - carries the value; under sustained inference load, DDR5 wins on latency, efficiency, and absolute power (Section 3.1.2).



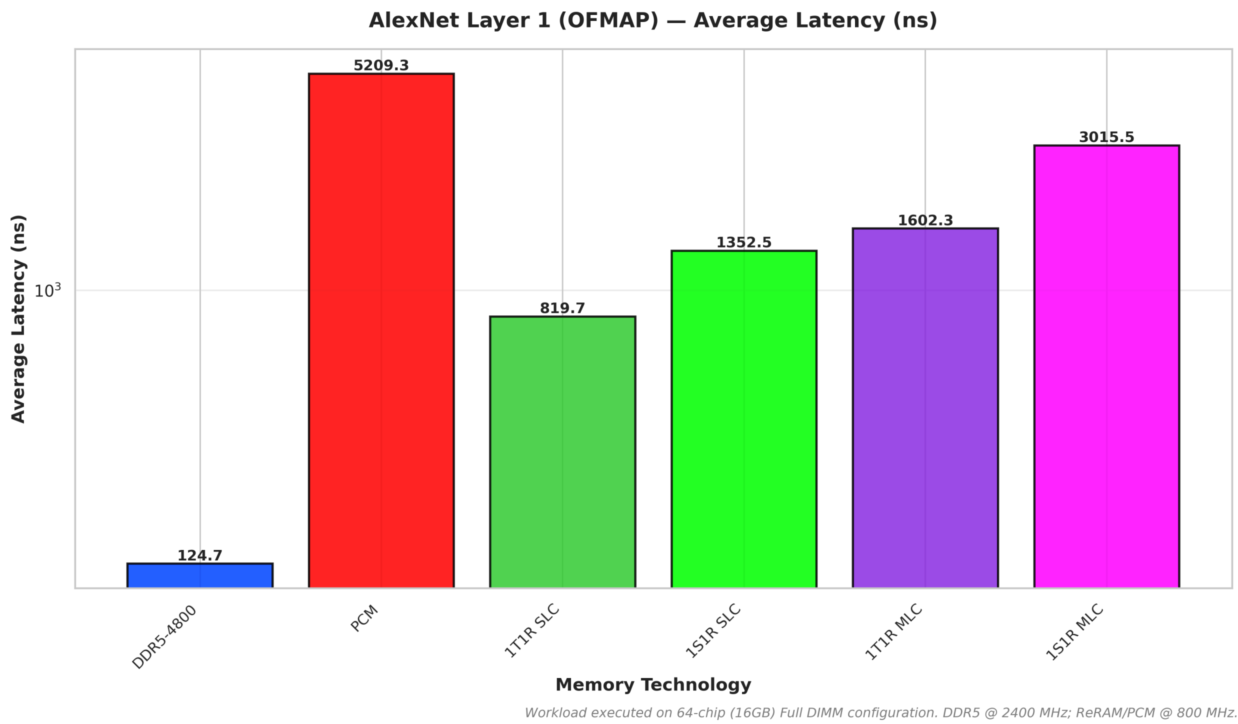


Figure 4 & 5: Average Total Latency under AlexNet Layer 1 IFMAP (pure-read) and OFMAP (pure-write) workloads, highlighting the MLC write-latency penalty.

To explicitly isolate and prove the MLC write penalty, I engineered a "Write-Torture" test by comparing the Input Feature Map (IFMAP) and Output Feature Map (OFMAP) traces of

AlexNet Layer 1 **(Figures 4 and 5)**. The data reveals a severe architectural bottleneck. Under the read-heavy IFMAP workload, 1S1R MLC averages 550.82 cycles. However, under the write-heavy OFMAP workload, the 1S1R MLC latency spikes to 2412.40 cycles - just over 2× degradation compared to its SLC counterpart (1081.98 cycles). This empirical gap confirms the analytical penalty method implemented in my simulation stack: MLC ReRAM incurs a severe latency tax during write operations, dictating that MLC topologies must be strictly relegated to read-only or read-mostly applications, such as housing frozen LLM weights.

The IFMAP/OFMAP pair also yields the suite's cleanest read-versus-write comparison at the system level, because the two traces isolate pure operation streams - IFMAP issues only reads (184,319 requests), OFMAP only writes (13,542) - and every technology replays the identical pair. The write-side degradation factor (OFMAP over IFMAP average latency) therefore ranks write tolerance directly: DDR5 is nearly read/write-symmetric at 1.08x, 1T1R SLC degrades 2.1x, 1S1R SLC 2.6x, 1T1R MLC 3.2x, 1S1R MLC 4.4x, and PCM 3.6x. Two device-level facts from Table 1 sit beneath that ranking. The selector prices writes intrinsically: a 1S1R write costs 1.4x its read at the device (72.9 vs 52.1 ns) before ISPV programming multiplies it toward 3.0x for MLC (237.9 vs 78.2 ns). The access transistor, by contrast, makes the 1T1R device almost perfectly symmetric (32.3 vs 32.1 ns) - so 1T1R's 2.1x system-level degradation measures burst structure and bank pressure under the write stream, not cell physics, and marks the asymmetry floor that controller-level write buffering (Section 4.1) could attack. DRAM's near-symmetry is the benchmark every resistive contender chases; this ordering quantifies the fourth axis of the trade-off profile stated at the top of Section 3.1 - individually expensive write operations - as measured system latency.

The STREAM benchmark (Figure 6) applies pure vector-kernel bandwidth pressure, isolating sustained memory bandwidth from compute dependencies. The 1T1R SLC configuration averages 473.17 cycles (591.46 ns), trailing the DDR5-4800 baseline (276.04 ns) by 2.1x in wall-clock latency, with 1S1R SLC at 890.29 ns; together with the LBM result, this brackets sustained streaming as a 1.9-3.2x latency-deficit regime across the SLC ReRAM family. The efficiency consequences are quantified in Section 3.1.3.

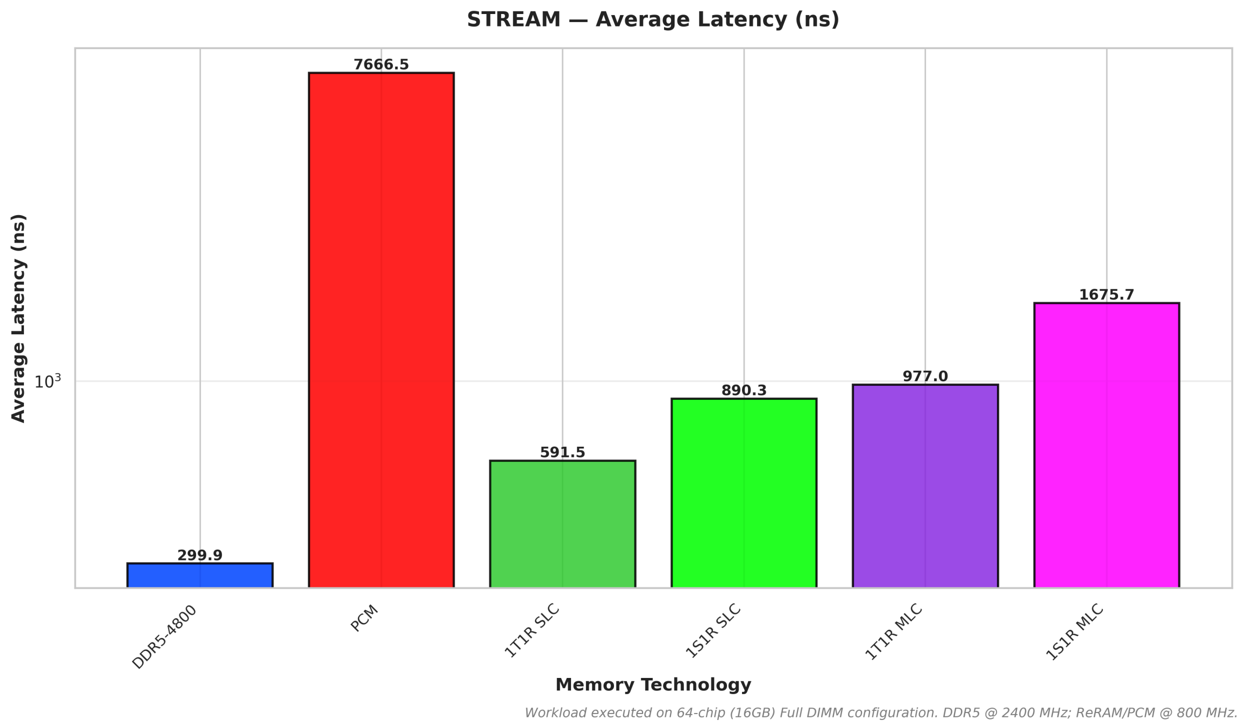


Figure 6: Average Total Latency under the STREAM memory bandwidth benchmark.

Table 2 collects the average total latencies across the full workload suite, complementing the per-workload bar charts.

Table 2: Average total latency (ns), full-DIMM configurations, 83.33 ms matched-host window (Section 3.1.6, item 11).

Workload	DDR5-4800	PCM	1T1R SLC	1S1R SLC	1T1R MLC	1S1R MLC
GCC	87.2	6,399.2	130.9	190.3	182.6	288.5
LBM	271.5	7,363.4	468.1	662.1	809.7	1,328.7
STREAM	299.9	7,666.5	591.5	890.3	977.0	1,675.7
GPT-2	100.3	1,816.1	458.4	618.4	588.4	828.4
AlexNet IFMAP	118.6	1,438.0	397.4	523.9	500.2	688.5
AlexNet OFMAP	124.7	5,209.3	819.7	1,352.5	1,602.3	3,015.5

Wall-clock nanoseconds (clock domains: ReRAM 800 MHz, PCM 400 MHz, DDR5 2400 MHz). Source: results/system_v6/processed_bar_chart_metrics.csv.

3.1.2 Power Analysis: Static Leakage Dominance

While ReRAM exhibits latency penalties under write-heavy and massive parallel read workloads, its candidate advantage has always been the energy profile: traditional DRAM requires continuous, power-intensive capacitive refresh cycles to maintain data integrity - a permanent power tax paid regardless of workload activity - whereas non-volatile retention is free. Whether ReRAM actually realizes that advantage, however, is decided by peripheral leakage, and quantifying leakage correctly required repairing the simulation toolchain first. All power figures in this section therefore come from the repaired model documented in Section 3.1.6: NVSim's per-technology leakage is wired into the standby-energy parameters NVMain actually reads (validated against device-level anchors to 0.0% error), power is accounted as the whole-module sum for every technology, and no idle-gating is applied, because NVMain's power-down machinery is disabled in source. The ReRAM figures are consequently worst-case ungated, while the DRAM and PCM baselines model their standard idle behavior - an asymmetry restated wherever cross-technology totals are compared.

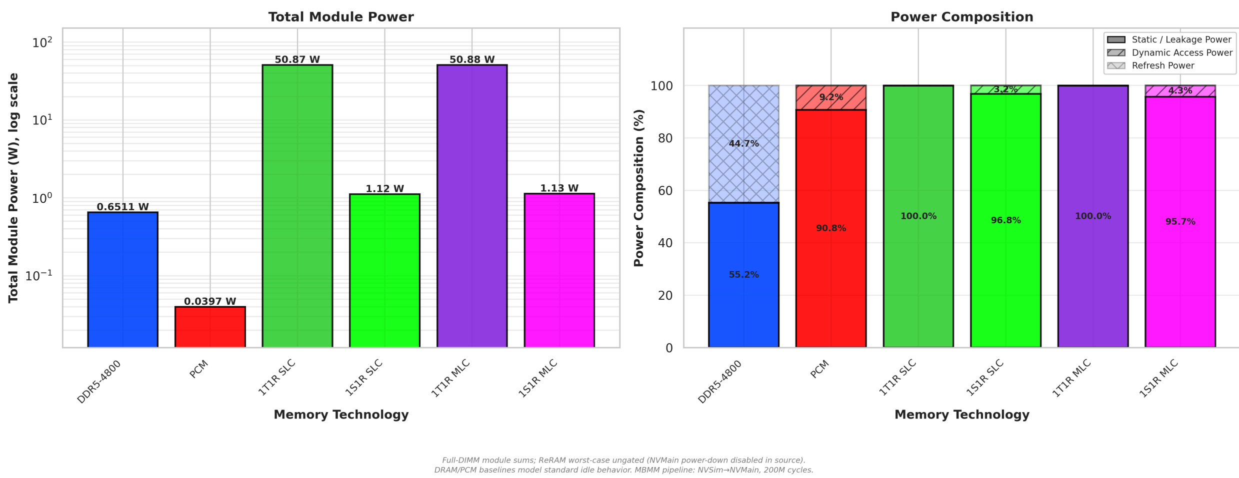


Figure 7: System Power Breakdown (Static / Dynamic / Refresh) under the compute-bound GCC (SPEC2017) workload.

In instruction-heavy, compute-bound workloads like gcc (Figure 7), the main memory subsystem remains idle for the vast majority of the execution time, making GCC the cleanest probe of each technology's standing power floor. The non-volatile opportunity shows immediately: ReRAM's refresh component is identically zero, because retention is free, while the decomposition in Figure 7 exposes DDR5's structural burden - even with its own idle-gating mechanism now restored and live (Section 3.1.6, item 5), 46.7% of its 0.623 W module-sum draw under GCC is still refresh power, energy spent merely retaining data, on top of a 53% standby/power-down floor, with actual access activity accounting for about 0.1%. Restoring idle-gating lowered DDR5's total draw (from 0.651 W) but could not touch the refresh burden itself - refresh is mandatory regardless of power-down state - so refresh's **share** of a now-smaller total actually rose. The selector-gated 1S1R module converts that opportunity into a commodity-envelope number: 1.12 W (97% static), refresh-free, with essentially every milliwatt of its draw exposed to future idle-gating. The repaired leakage wiring prices the alternative just as clearly: the transistor-gated 1T1R module leaks 50.9 W (99.99% static), roughly two orders of magnitude above DDR5, and PCM sits at 0.040 W (91% static). Real dynamic power is small for every ReRAM configuration (12-98 mW module-wide under GCC, now technology-differentiated through the repaired access energies of Section 3.1.6, item 10): ungated ReRAM power is leakage, full stop.

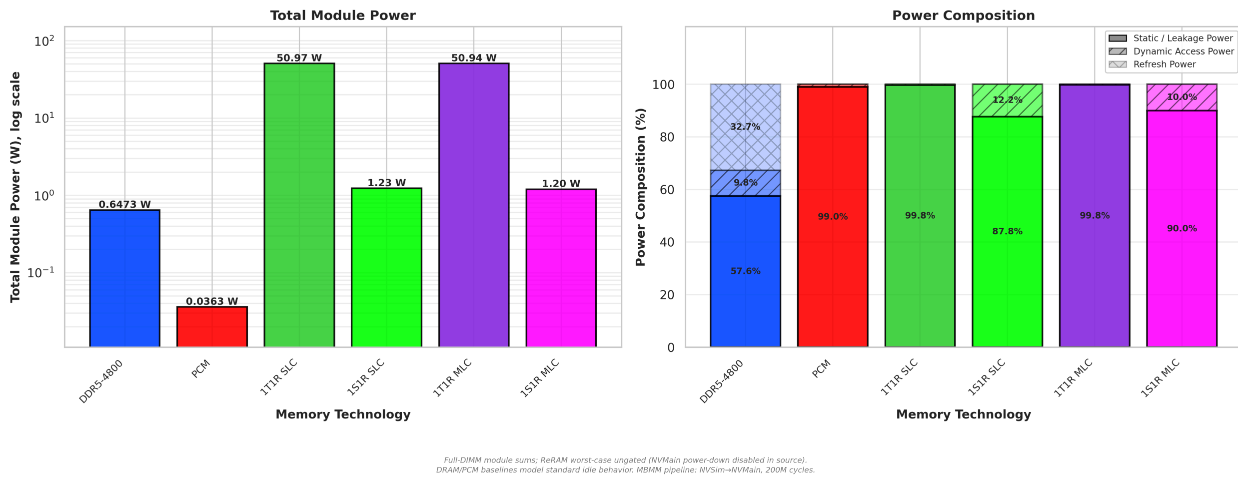


Figure 8: System Power Breakdown (Static / Dynamic / Refresh) under the highly active GPT-2 Inference (IFMAP) workload.

The repaired model also retires an artifact of this project's earlier analysis. Under the simulator's generic standby defaults, total power appeared to cluster at nearly the same level for every ReRAM configuration under any given workload - a convergence this project's earlier analysis treated as a real effect, here named the “Standby Convergence” (deliberately not “flatline”: Section 3.2's multi-rank scaling findings are an unrelated address-footprint effect, not a power phenomenon). With real per-technology leakage wired in, that convergence stands revealed as a modeling artifact; the true signature is leakage-class separation. The four ReRAM full-DIMM configurations fall into two power tiers set entirely by their standby physics - 50.9 W for the transistor-gated 1T1R family and 1.12-1.30 W for the selector-gated 1S1R family, against DDR5's (now idle-gated) 0.623 W and PCM's 0.040 W - and within each ReRAM tier, SLC and MLC variants are nearly indistinguishable (50.870 vs. 50.877 W under GCC), because dynamic energy contributes only milliwatts. What survives from the earlier observation is the intra-family invariance: write intensity never spikes power, and MLC imposes no power penalty relative to SLC under active workloads. The 47x leakage gap that NVSim characterizes at the device level (794.7 vs. 16.9 mW per chip) now propagates faithfully to the system level, where it becomes the single most consequential number in this evaluation.

This confirms Static Leakage Dominance end-to-end: peripheral and access-device leakage governs ReRAM's power identity at the device level (NVSim: 794.7 vs. 16.9 mW per chip) and - now measured with real leakage physics - at the system level as well. Under AI inference (GPT-2, Figure 8), the most parallel workload in the suite, accumulated dynamic read energy raises the 1S1R SLC module only from 1.12 W to 1.23 W - roughly a 10% excursion - while DDR5 spans 0.65-0.78 W across the entire suite, so the standby floor decides the cross-technology ranking under every workload. The architectural consequence is decisive: 1S1R lands within 1.7x of DDR5 at the conservative calibration floor (within 11% of parity at the ceiling) with zero refresh cost and its gating headroom unexercised - a credible successor profile - while an always-on 1T1R DIMM, at 65-78x DDR5's power, is infeasible without gating. One honest bound on that comparison: both ends of the DDR5 band are vendor *specification* currents (Section 2.3), not measured typicals - a typical-current module sits below the spec-limit ceiling - so the 11%-of-parity figure is the most favorable end of a disclosed range, not an expected-case estimate. The selector's leakage discipline is thereby elevated from a device curiosity to the deciding architectural requirement.

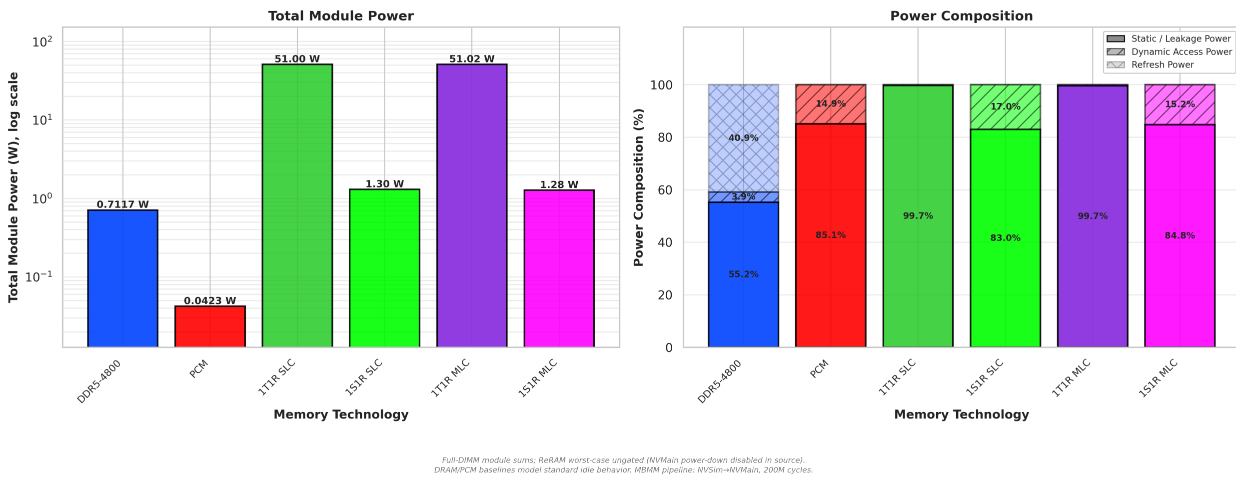


Figure 9: System Power Breakdown (Static / Dynamic / Refresh) under the continuous memory-streaming LBM (SPEC2017) workload.

This leakage dominance also exposes a genuine tug-of-war in the dynamic term under a workload like lbm, write-heavy in service-time impact despite being read-majority by request count (58/42 read/write, Section 2.4.1) - its slower write latency dominates total device-busy time even though writes are the minority of requests (Figure 9; components this small cannot be resolved visually against the 50.9 W static tier - the values come directly from the NVMain energy counters). The MLC configuration's extended Iterative Step-and-Verify (ISPV) writes still bottleneck the memory controller relative to SLC, so it processes fewer requests per second - but with the repaired per-technology access energies (Section 3.1.6, items 10 and 13), each MLC write also costs 3.0x its SLC counterpart in energy, and under lbm the energy penalty outweighs the throughput throttling: 1T1R MLC dynamic power reaches 162.1 mW against SLC's 140.6 mW. At module scale, both variants remain pinned to the same 50.9 W static tier, reconfirming that ReRAM DIMMs are fundamentally bounded by their static peripheral leakage rather than dynamic cell activity.

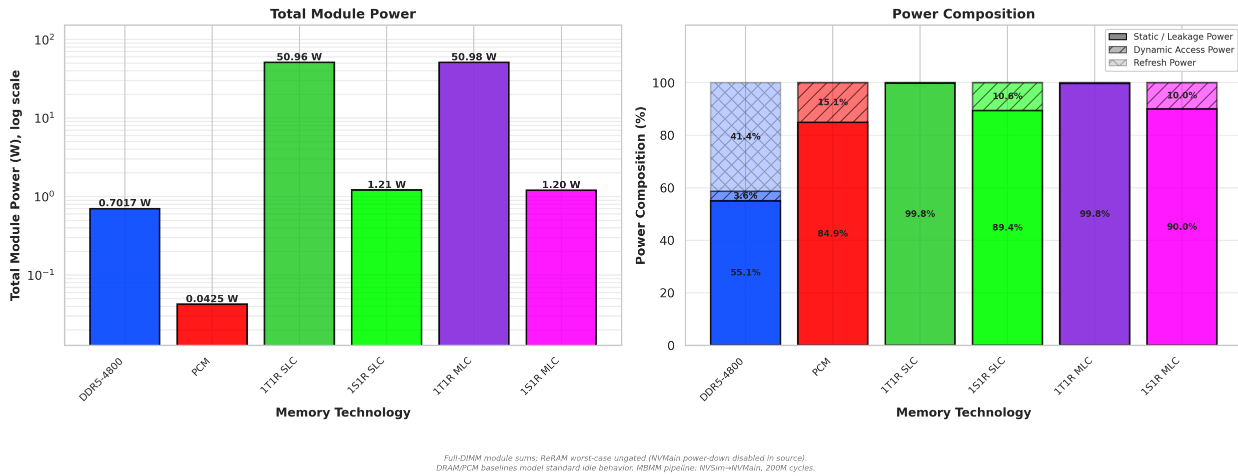
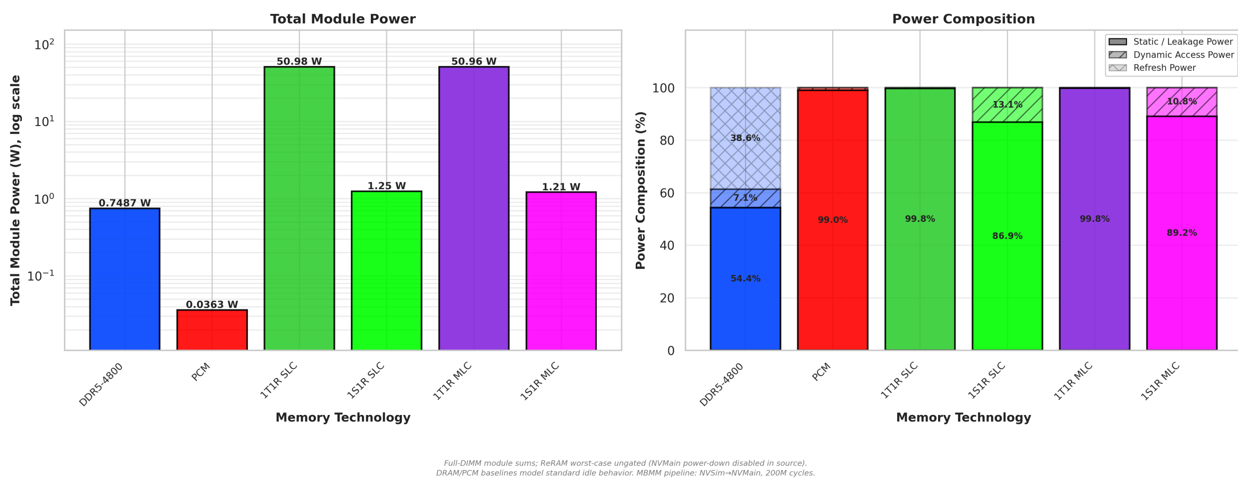


Figure 10: System Power Breakdown (Static / Dynamic / Refresh) under the STREAM benchmark.

The standard stream benchmark (Figure 10) follows the same direction as lbm rather than reversing it: 1T1R SLC dynamic power reaches 99.5 mW while MLC now draws more, at 121.7 mW, because the corrected write-latency penalty (3.263x, Section 3.1.6, item 13) throttles MLC throughput less than the earlier unsourced 4x estimate implied, so more energy-costly writes complete per second than the write-side energy penalty alone would suggest. The two effects - operations per second versus energy per operation - only trade the lead in favor of SLC on read-dominated traffic (the AlexNet IFMAP and GPT-2 results below), where MLC's much smaller corrected read-energy penalty (1.1x, against the write side's 3.0x) dominates instead; in every case the module total remains anchored to its technology's leakage tier.



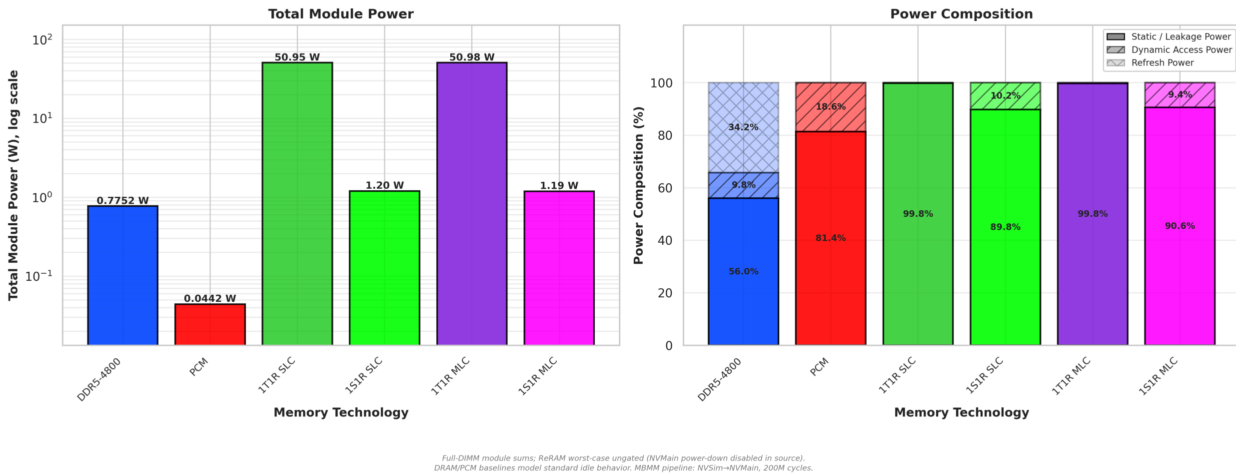


Figure 11 & 12: System Power Breakdown (Static / Dynamic / Refresh) under AlexNet Layer 1 IFMAP (Read-Heavy) and OFMAP (Write-Heavy) workloads.

Finally, to ensure the Static Leakage Dominance principle holds across diverse neural network operations, I evaluated both the read-intensive AlexNet IFMAP and the write-intensive AlexNet OFMAP workloads (Figures 11 and 12). A ReRAM write is individually an expensive operation: NVSim prices a 1T1R write at 1.74 nJ against 1.19 nJ for a read, delivered as ~100 uA-class programming currents, and MLC writes stretch into 3.263x-long ISPV sequences - so write-heavy traffic is precisely where one would expect ReRAM power to spike. It mostly does not. Despite the severe latency penalty incurred during OFMAP writes (as established in Section 3.1.1), no configuration exhibits a power spike large enough to threaten the static tier - OFMAP dynamic power is lower than IFMAP dynamic power for 1T1R SLC (87.7 vs. 122.2 mW) and for both 1S1R variants (120.6 vs. 163.3 mW SLC; 112.1 vs. 131.4 mW MLC), because slow ISPV write operations throttle request throughput enough to offset the higher energy per write. 1T1R MLC is the exception: its corrected, less-severe write-latency penalty throttles throughput less than the SLC comparison would suggest, so OFMAP's higher per-write energy edges narrowly ahead of IFMAP's (119.5 vs. 99.5 mW) - a genuine but small reversal (both values sit three orders of magnitude below the 50.9 W static tier) that does not disturb the section's conclusion: extended write durations never translate into a power spike large enough to matter; total power tracks the technology's leakage tier, perturbed only marginally by request throughput.

Table 3 collects the whole-module power totals, making the leakage-class tiers directly comparable across every workload.

Table 3: Total module power (W), full-DIMM, repaired model, module-sum semantics.

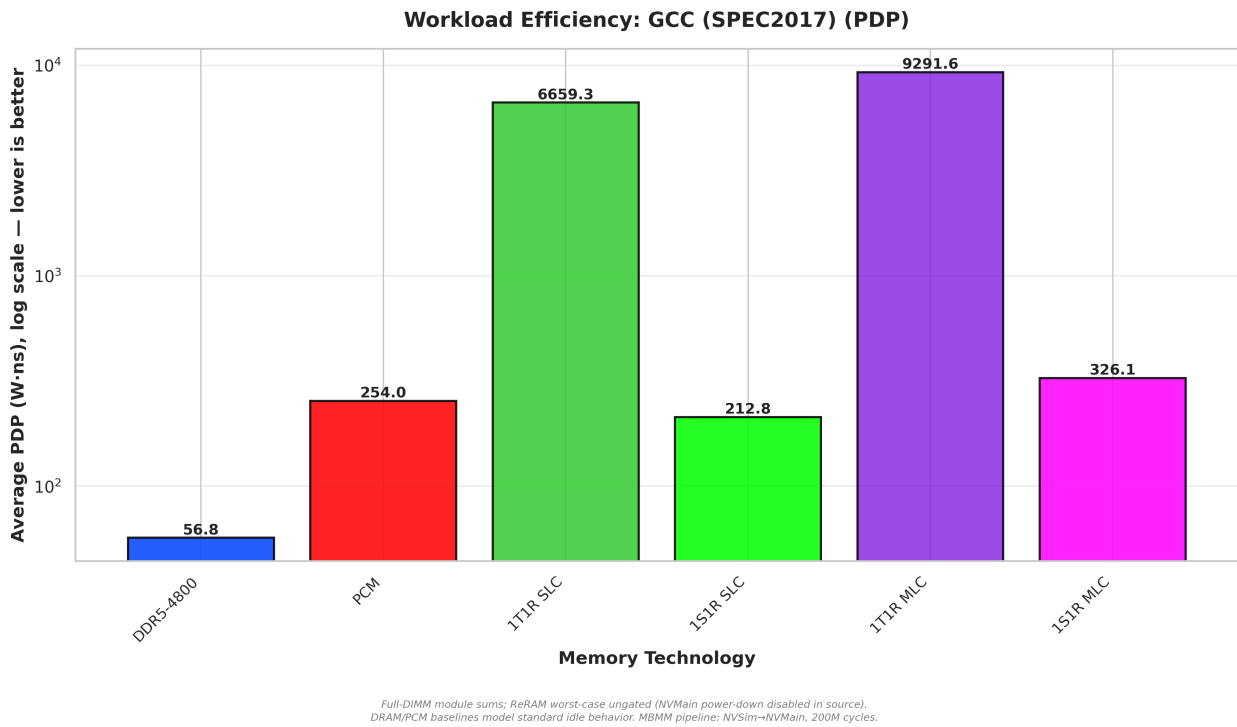
Workload	DDR5-4800	PCM	1T1R SLC	1S1R SLC	1T1R MLC	1S1R MLC
GCC	0.623	0.040	50.870	1.118	50.877	1.130
LBM	0.707	0.042	50.998	1.304	51.020	1.276

Workload	DDR5-4800	PCM	1T1R SLC	1S1R SLC	1T1R MLC	1S1R MLC
STREAM	0.698	0.042	50.957	1.210	50.980	1.202
GPT-2	0.640	0.036	50.965	1.233	50.944	1.202
AlexNet IFMAP	0.745	0.036	50.980	1.245	50.958	1.213
AlexNet OFMAP	0.757	0.044	50.948	1.205	50.977	1.194

DDR5 figures reflect the restored idle-gating mechanism (Section 3.1.6, item 5) - real, JEDEC-datasheet-backed power-down savings, 0.6-4.2% lower than the previously-reported ungated-equivalent figures depending on workload. ReRAM figures are unchanged: the mechanism is mechanically live for ReRAM too (89% of full-DIMM GCC cycle-slots are now spent in a power-down state, confirmed directly from the simulator's own counters) but its power-down energy is an explicit, disclosed placeholder equal to existing standby energy ("assume no savings, pending real characterization" - no credible ReRAM power-gating figure exists in the literature, Appendix A), so no numeric change results by design - this is not an oversight. PCM shows zero power-down activity in either run, likely because its FRFCFS-WQF write-queue-flush controller keeps its request queue non-empty far more of the time than the plain FRFCFS controller ReRAM/DDR5 use, starving the power-down entry condition - an open follow-up item, not yet root-caused to full confidence. Source: results/system/processed_bar_chart_metrics.csv (re-run 2026-09-05).

3.1.3 Power-Delay Product (PDP): The Architectural Sweet Spot

While isolated latency and power metrics define physical boundaries, true architectural viability is measured by the balance between them. I utilized the Power-Delay Product (PDP = Total System Power [W] × Average Request Latency [ns]; units W·ns = nJ) to rank configurations. All PDP values in this section are computed from the repaired, module-sum power model of Section 3.1.6 with no idle gating applied: the ReRAM values are therefore worst-case ungated, while the DRAM and PCM baselines model standard idle behavior. This asymmetry must accompany any reading of the cross-technology ranking.



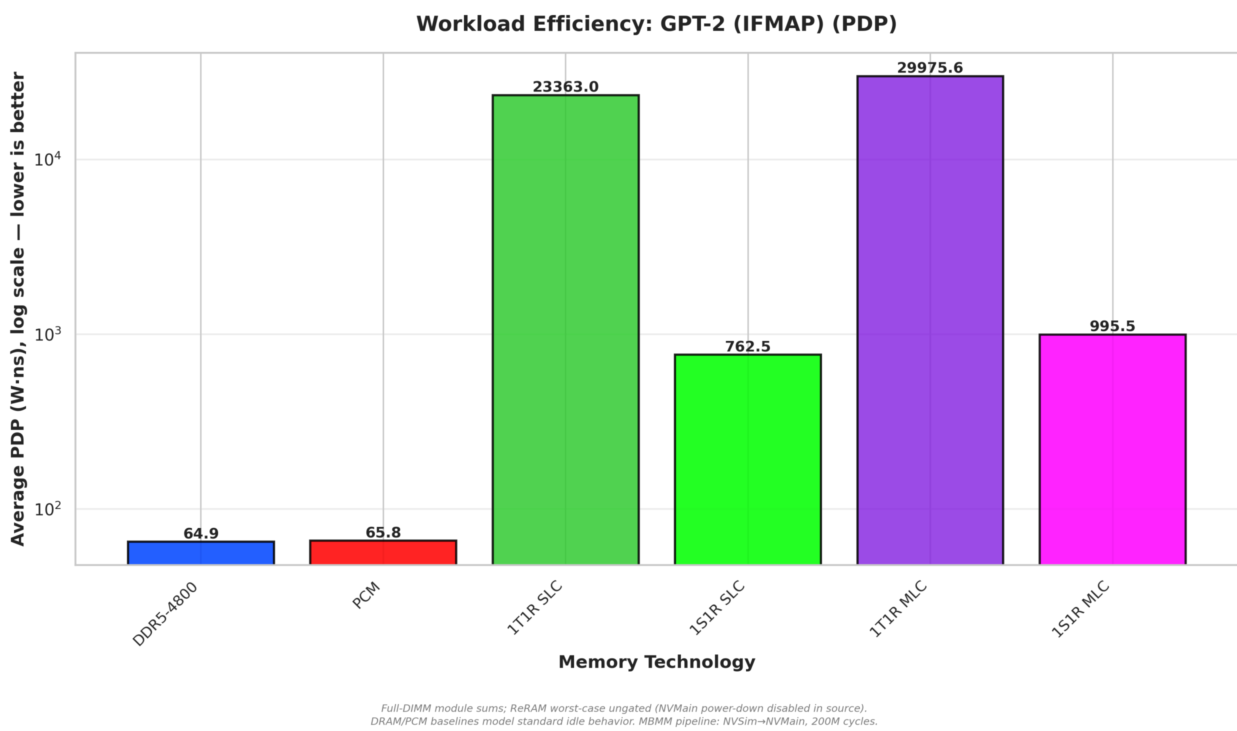
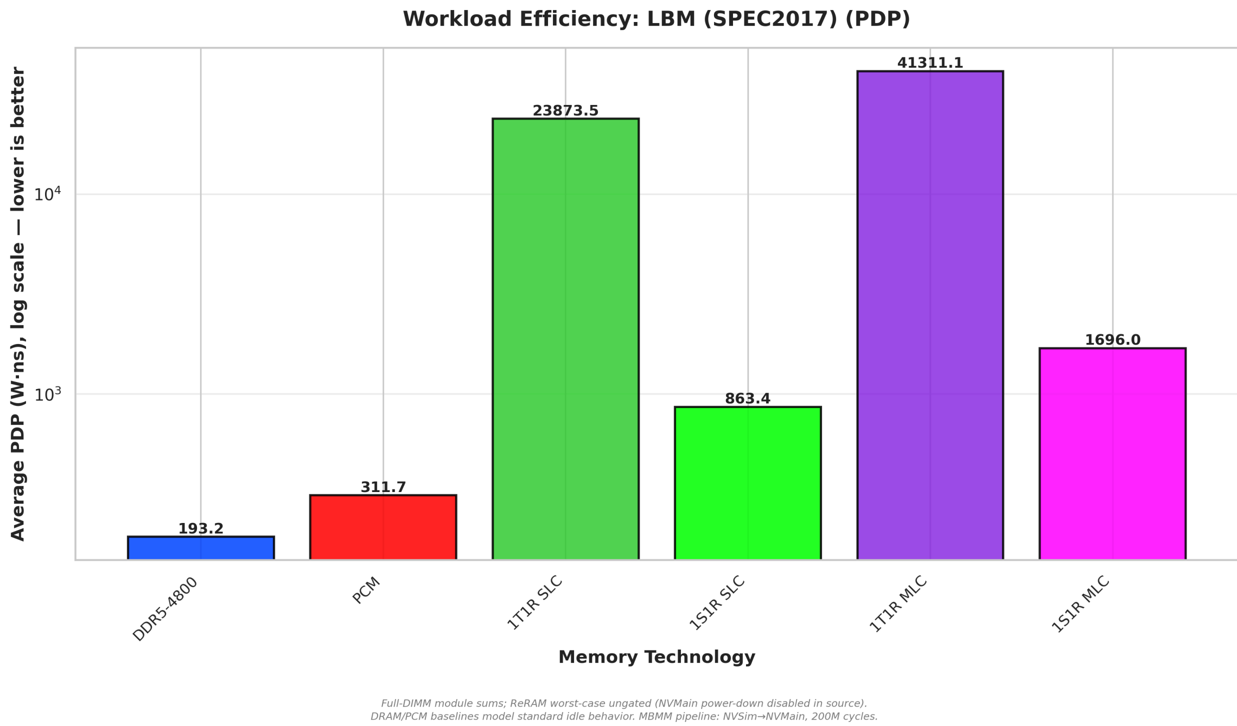


Figure 13 & 14: Power-Delay Product under single-threaded compute (GCC SPEC2017) and massively parallel AI inference (GPT-2 IFMAP) workloads.

The repaired power model inverts the intra-ReRAM hierarchy that the earlier technology-blind analysis suggested. Once each family carries its real leakage, the selector-gated configurations dominate: under GCC (Figure 13), 1S1R SLC posts 215.0 W·ns against 1T1R SLC's 6,928.8 - a 32x advantage - because the 47x leakage gap in the power term dwarfs the 1.6x latency cost of the selector's slower access path. The same inversion holds under every workload in the suite. Against the baselines, DDR5 - now benefiting from its own restored idle-gating mechanism (Section 3.1.6, item 5) - posts 56.2 W·ns under GCC and 64.3 under GPT-2 (Figure 14); 1S1R SLC, whose power-down mechanism is mechanically active but delivers no modeled savings (Section 3.1.2), trails DDR5 by roughly 3.8x under GCC, and the 1T1R family remains uncompetitive at any operating point regardless. Cell-level density still costs efficiency within each family - MLC multiplies the delay term (323.9 vs. 215.0 W·ns for 1S1R under GCC) - but the deciding term is now leakage class, not cell type.



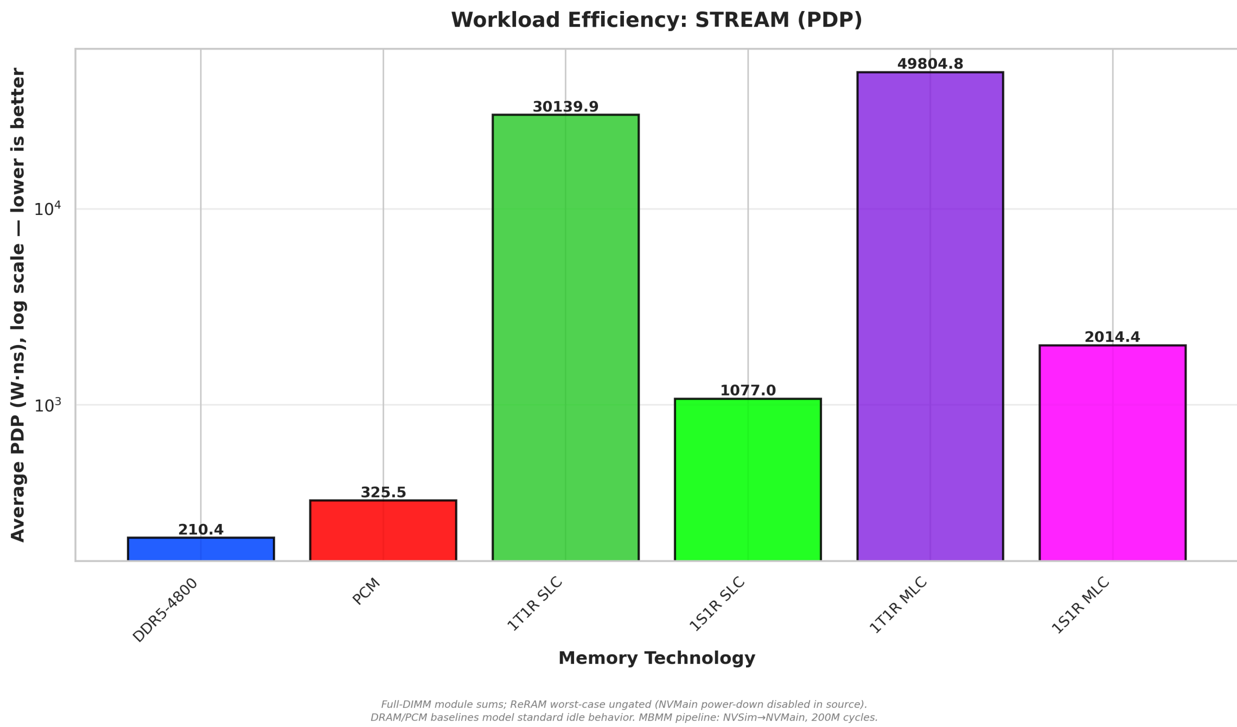
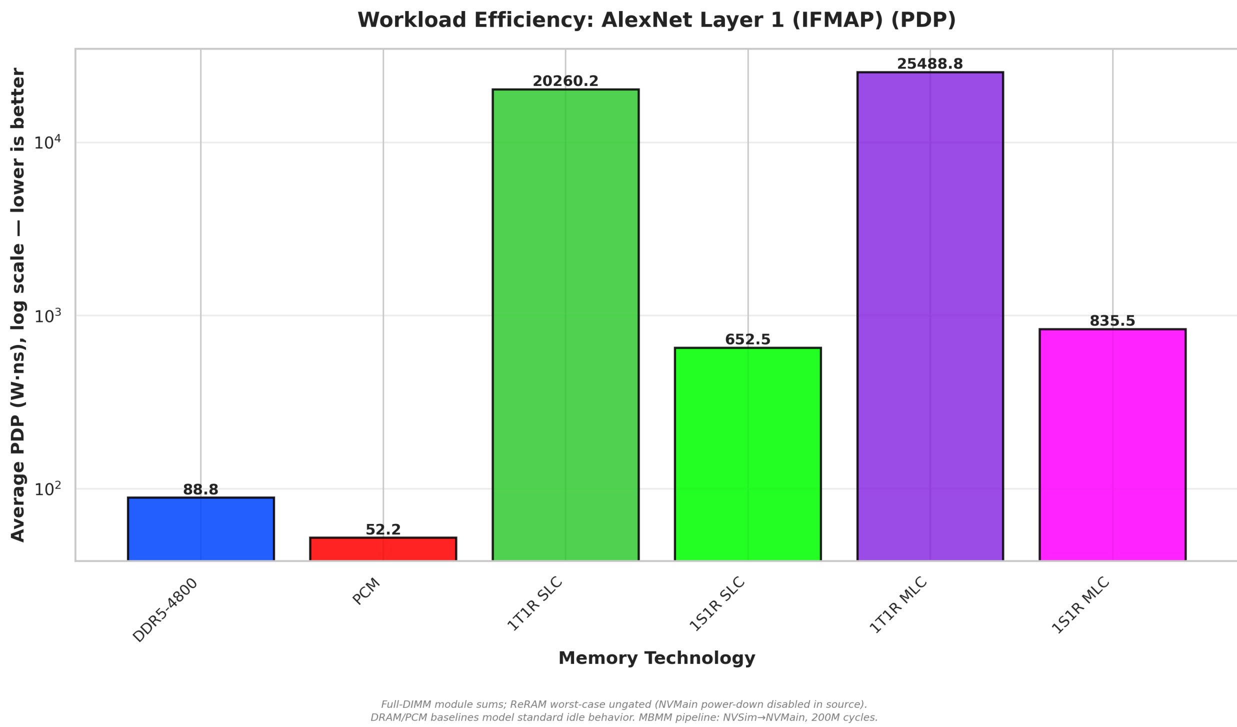


Figure 15 & 16: Power-Delay Product under heavy continuous memory-streaming workloads (LBM SPEC2017 and STREAM).

Under maximum bandwidth pressure (Figures 15 and 16), the repaired ordering persists: 1S1R SLC posts 862.3 W·ns under lbm and 1,077.1 under stream, with its MLC sibling at roughly 1.9x those values (1,695.7 and 2,014.6 W·ns) as streaming amplifies every write-side penalty, and the 1T1R family remains more than an order of magnitude further adrift (23,952 W·ns for 1T1R SLC under lbm). DDR5, now benefiting from its restored idle-gating mechanism, holds a ~4.5-5.2x edge over 1S1R SLC in the streaming regime (192.1 and 209.0 W·ns).



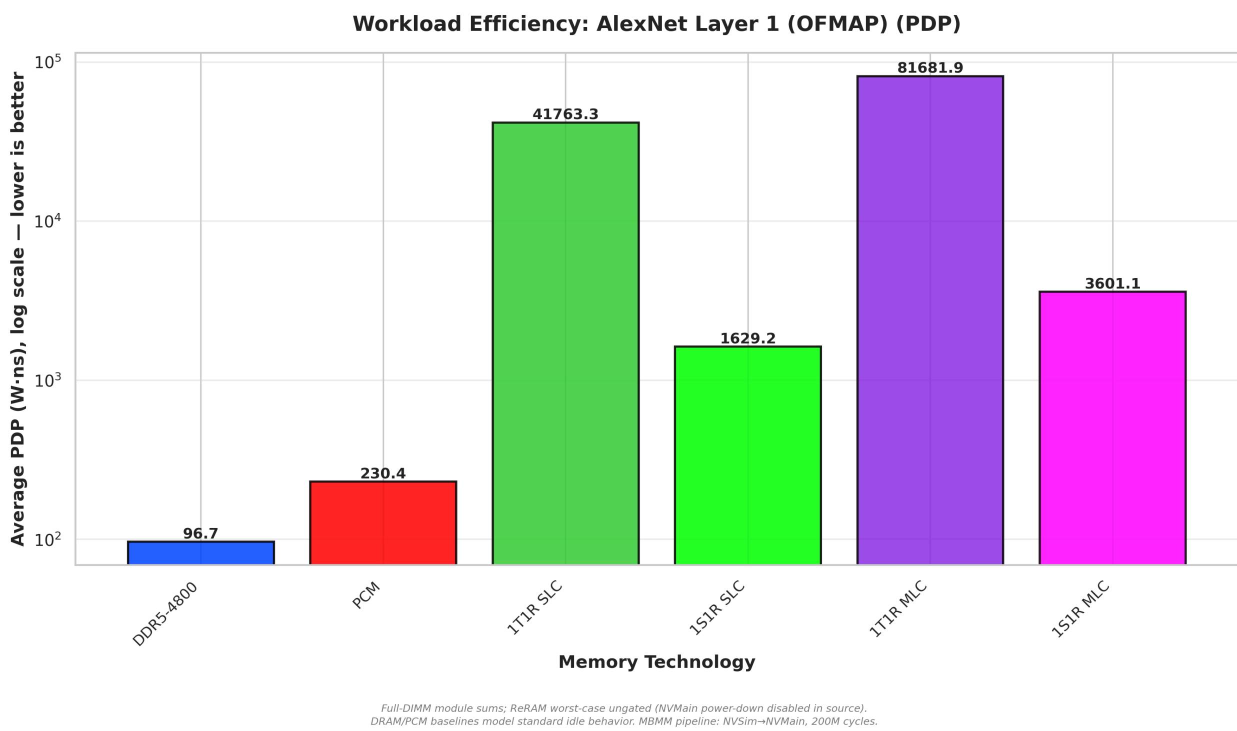


Figure 17 & 18: Power-Delay Product under AlexNet Layer 1 IFMAP (Read-Heavy) and OFMAP (Write-Heavy) workloads, highlighting the MLC write-latency penalty on system efficiency.

Finally, the PDP diagnostics explicitly map the efficiency limits of density scaling within the viable selector family. Under the read-heavy AlexNet IFMAP (Figure 17), moving from 1S1R SLC to the ultra-dense 1S1R MLC topology incurs a 1.3x PDP increase (651.6 to 836.0 W·ns). Under the write-heavy AlexNet OFMAP trace (Figure 18), the 1S1R MLC PDP spikes to 3,601.3 W·ns - 2.2x its SLC sibling's 1,629.4. This confirms a strict efficiency tax for density: MLC solves the physical capacity problem, but its iterative write penalties still degrade overall system efficiency under active write stress, even though the corrected ISPV penalty parameters (Section 3.1.6, item 13) show the tax is smaller than the earlier unsourced estimate suggested.

Table 4 collects the Power-Delay Products with the suite geometric means, the tabular form of Figures 13-18 and 27.

Table 4: Power-Delay Product (W·ns = nJ), full-DIMM, and six-workload geometric means.

Workload	DDR5-4800	PCM	1T1R SLC	1S1R SLC	1T1R MLC	1S1R MLC
GCC	56.2	254.2	6,928.8	215.0	9,416.3	323.9
LBM	192.1	311.7	23,952.1	862.3	41,289.9	1,695.7
STREAM	209.0	325.5	30,147.3	1,077.1	49,812.2	2,014.6
GPT-2	64.3	65.8	23,364.9	762.5	29,977.5	995.5
AlexNet IFMAP	88.4	52.2	20,275.1	651.6	25,461.6	836.0
AlexNet OFMAP	94.5	230.4	41,771.1	1,629.4	81,692.7	3,601.3
Geometric mean	103.3	165.3	21,508.4	738.1	32,632.8	1,221.1

PDP = total module power x average request latency. DDR5's idle power-down mechanism is now restored and live (Section 3.1.6, item 5) - its PDP improved slightly (real power savings outweigh a small power-down transition-latency overhead). ReRAM's power-down mechanism is also mechanically live (89% of full-DIMM GCC cycle-slots power-gated

- *Section 3.1.2) but its power-down energy is an explicit,*

no-real-number-available placeholder equal to existing standby energy, so its power is unchanged by design; the small PDP shifts visible above (mostly under 1%, up to +4.0% for 1T1R SLC under GCC) are pure transition-latency overhead with no offsetting power benefit. PCM shows no power-down activity in either run (Appendix A). Source: results/system processed CSVs (re-run 2026-09-05).

3.1.4 Endurance Viability Analysis

A key viability question for ReRAM as a main memory replacement is write endurance: how many write cycles before cell degradation? While SLC ReRAM cells are typically rated at 10^7 cycles and MLC at 10^6 write cycles - conservative engineering targets within the wide range of endurance values reported across published metal-oxide RRAM demonstrations [14], with conservatism warranted by documented methodological inconsistencies in reported endurance data [15] - the practical lifetime under real workloads depends on the actual write frequency, not the worst-case specification. In NVMain, each recorded write event programs one 64-byte cache line, so wear is accounted per line location: the physical 8 GB SLC full-DIMM module (64 chips of 1 Gb) comprises 134.2 million cache-line locations, and lifetime is the time for any location to accumulate the rated endurance under uniform wear leveling. (NVMain's decoded address space is 64x larger than the physical module - a configuration-generator artifact that affects neither per-request timing nor energy; endurance is computed over the physical cell population, and lifetime scales linearly with module capacity.) Using the per-subarray write counts extracted from the NVMain simulation statistics, the projected lifetimes are summarized below.

The calculation itself, worked for the 8 GB SLC module under LBM (the worst case): total write budget under uniform wear leveling is 134.2×10^6 line locations $\times\ 10^7$ rated cycles/line $=$ 1.342×10^{15} total writes available across the module. LBM admits 3,269,479 writes in the 83.33 ms matched-host window, a rate of $3,269,479/0.08333\ \mathrm{s} \approx 39.2 \times 10^6$ writes/s. Lifetime is then the write budget divided by the annualized write rate: $1.342 \times 10^{15}/(39.2 \times 10^6 \times 31,536,000\ \mathrm{s/yr}) \approx 1.08$ years - matching Table 5's reported figure exactly. The same arithmetic at 64 GB (8x the line count, same write rate) gives ≈ 8.7 years.

Table 5: Projected DIMM Write Endurance Lifetime (1T1R SLC Full DIMM, physical capacity 8 GB, 66.7M cycles at 800 MHz = 83.33 ms matched-host window (Section 3.1.6, item 11), uniform wear leveling over 134.2 million 64-byte line locations; lifetime scales linearly with module capacity).

Workload	Total Writes (83.33 ms)	Write Rate	Lifetime (SLC 10^7, 8 GB)	Lifetime (SLC 10^7, 64 GB)
LBM (worst case)	3,269,479	39.2 M/s	1.08 years	8.7 years†
GCC	170,800	2.05 M/s	20.8 years	166 years
STREAM	100,000	1.20 M/s	35.4 years	283 years
AlexNet OFMAP	13,542	0.16 M/s	262 years	2,094 years

†Lifetimes scale linearly with module capacity; the 64 GB column reflects a commodity server-class module, where ReRAM's density advantage is realized. Directly measured MLC lifetimes at the physical 16 GB module under LBM: 0.39 years for 1T1R MLC (1,824,503 writes in 83.33 ms at 21.9 M/s) and 0.65 years for 1S1R selector MLC (1,090,741 writes at 13.1 M/s); even at 128 GB these reach only 3.09 and 5.18 years respectively. All lifetimes assume uniform wear leveling.

The results reframe endurance as a capacity- and workload-dependent constraint. At the modeled 8 GB module, the worst-case sustained streaming workload (LBM) yields a projected lifetime of only 1.1 years for SLC - below the 5-10 year server replacement target (enterprise server useful life currently averages 5.4 years and is trending toward six to seven years [16]) - while every other workload clears the target - by 2-4x for compute-bound GCC (20.8 years) and by an order of magnitude or more beyond it (35.4 years for STREAM, 262 years for AlexNet OFMAP). Because uniform wear leveling spreads a fixed workload write rate over the module's entire cell population, lifetime scales linearly with capacity: at the 64-128 GB module sizes where ReRAM's density advantage is actually realized, worst-case SLC lifetime reaches 9-17 years (the slower-writing selector variant, 12-25), meeting the target at 64 GB and exceeding it at 128 GB. MLC is harsher: at its physical 16 GB module the measured LBM lifetimes are 0.39 years (1T1R) and 0.65 years (1S1R), and even at 128 GB they remain below or marginal to the target - an independent endurance argument for restricting MLC to read-dominant deployments that converges with the write-latency argument of Section 3.1.1. Endurance is therefore not a categorical barrier but a design constraint that binds only for small modules under sustained write streaming, directly motivating the wear-leveling controller and write-coalescing cache proposed in Section 4.2. Note additionally that these lifetimes assume ideal uniform wear leveling; hot-spot write patterns without such a controller would reduce projected lifetime proportionally. That proportionality also bounds the exposure rather than leaving it open-ended: a controller that lets the hottest region absorb twice its uniform share of writes simply halves every figure above - the 128 GB SLC configurations still clear the server target (8.7 years for 1T1R, 12.4 for the selector), while the halved 64 GB figures (4.3-6.2 years) sit at the target's lower edge - so imperfect leveling shifts where the capacity threshold lies, not whether one exists.

3.1.5 Design Robustness: ReadVoltage Sensitivity

A persistent concern with resistive memories is parameter sensitivity: do the efficiency conclusions hold if the actual fabricated device deviates from the target resistance? To quantify this, I ran a three-point sweep of the NVSim ReadVoltage parameter at −20%, nominal, and +20% of the 1.4V design point ({1.12V, 1.40V, 1.68V}) for the 1T1R SLC Full DIMM configuration under LBM and GCC, the two workloads that bracket the bandwidth-bound and compute-bound extremes.

The results, shown in Figure 19, reveal a key robustness property: read latency is completely invariant to a +/-20% ReadVoltage perturbation. The current-sensing amplifier operates within its saturation margin at all three voltages, producing identical read latency (32.13 ns) regardless of supply. Only read energy scales with applied voltage: read energy is 22.4% lower at 1.12V and 27.5% higher at 1.68V relative to the 1.4V baseline. At the full-DIMM system level, where static peripheral leakage dominates over per-operation read energy, the per-read energy variation is negligible: total PDP changed by less than 4% across the full +/-20% sweep. One limitation applies: this sweep was executed before the power-model repair of Section 3.1.6 and has not been re-run under the repaired dataset. Because the repair raises the static share of ReRAM system power to 99.5% or more (Section 3.1.2), per-operation read energy becomes an even smaller fraction of the total, so the insensitivity conclusion holds a fortiori; read-latency invariance is unaffected by the repair, which changes power accounting only.

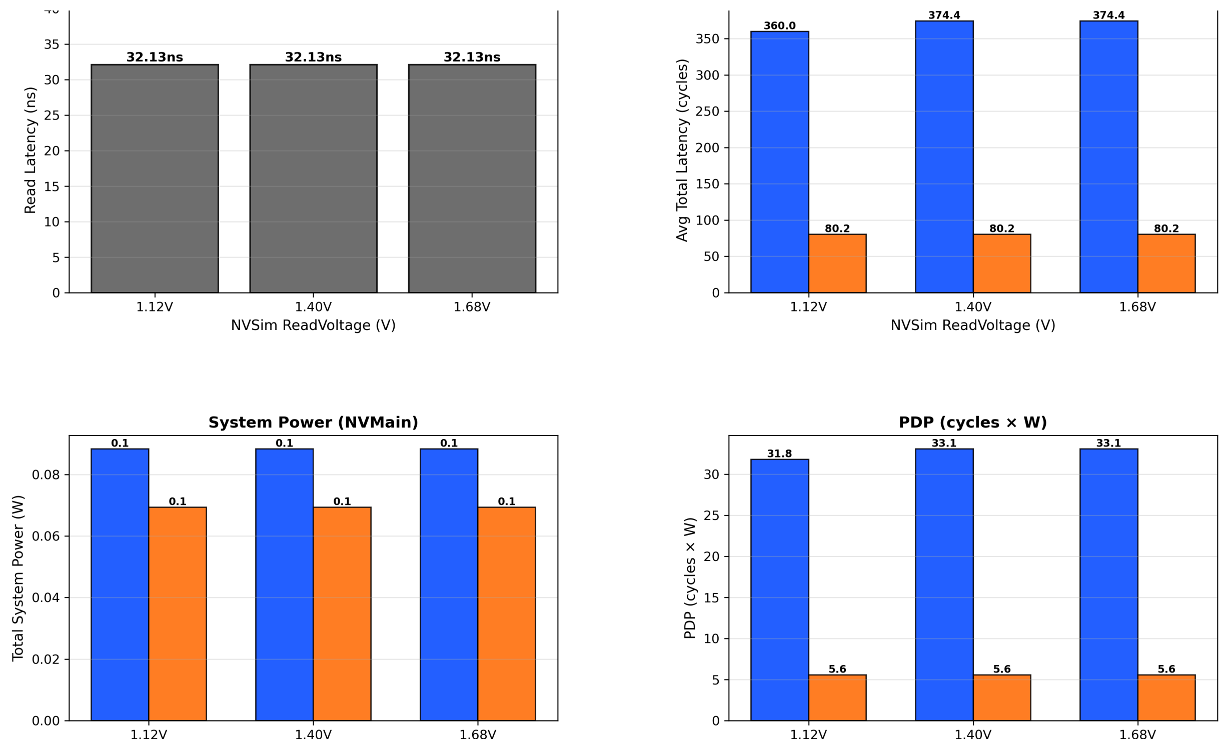


Figure 19: ReadVoltage Sensitivity Analysis - 1T1R SLC Full DIMM (LBM and GCC, +/-20% sweep). Left: NVSim read latency is invariant across all three voltages. Right: System-level PDP changes by less than 4% across the full sweep, confirming that efficiency conclusions are robust to process variation in the ReadVoltage operating point.

3.1.6 Simulation-Fidelity Audit: Where the Toolchain Could Not Be Trusted - and How It Was Repaired

A systematic audit of the NVSim-to-NVMain flow, conducted against the raw simulator sources and statistics files, found fourteen silent failure modes that bounded which conclusions this evaluation could honestly draw - across the ReRAM configurations, the metrics pipeline, and both non-ReRAM baselines. Twelve of the fourteen have since been repaired in this project's toolchain, with the affected simulations re-run; every power and PDP figure in this book derives from the resulting repaired dataset. (1) Found and fixed: the pipeline's original efficiency metric multiplied Watts by cycle counts taken from different clock domains (800 MHz ReRAM vs. 2400 MHz DDR5), biasing the comparison 3x against DDR5; all PDP values are computed in physical W·ns. (2) Found and fixed: the StandbyPower parameter written by the configuration generator was dead configuration - NVMain's source contains no reference to it, so the device-characterized leakage never reached the simulation. The repair maps NVSim's per-technology leakage onto the standby-energy parameters NVMain actually reads. (3) Found and fixed: in place of real leakage, NVMain applied generic standby-energy defaults (Eactstdby/Eprestdby), giving every non-volatile configuration the same ~68 mW-per-rank static floor - the original power results were technology-blind in their static component, even though NVSim characterizes a 47x leakage difference between 1T1R (794.7 mW/chip) and 1S1R (16.9 mW/chip). Under the repaired wiring, the simulated full-DIMM static floors (50.858 W for 1T1R, 1.082 W for 1S1R) reproduce the NVSim characterization exactly (0.0% anchor error). (4) Found and fixed: the reported "Total System Power" was the single most-loaded rank, not the module sum; because the ReRAM configurations span 8 ranks while DDR5 spans 2 ranks across 2 sub-channels, per-rank accounting distorted every cross-technology comparison. All power figures in this book are whole-module sums for every technology. (5) Found and fixed: NVMain's power-down state machine was disabled in source, so no idle-gating policy could be simulated for any technology - every ReRAM power figure in earlier drafts of this book was worst-case ungated, while the DRAM and PCM baselines modeled their standard idle behavior, an asymmetry this book previously disclosed but did not close. The dormant call site (`MemoryController::CycleCommandQueues()`'s `HandleLowPower()`) has been restored: it is a shared mechanism, so it activated for every technology at once, not ReRAM alone. DDR5 already carried real, JEDEC-datasheet-backed power-down currents, and now realizes a real, measured power reduction from them (0.651 to 0.623 W under GCC, Section 3.1.2). ReRAM has no equivalent real number: no NVSim datapoint decomposes its leakage into a gatable-periphery-vs-ungatable-crossbar split, and a dedicated literature search (Appendix A) found no citable ReRAM power-gating figure - so its power-down energy is set to an explicit, honest placeholder equal to its own existing standby energy ("assume no savings, pending real characterization"), which keeps the mechanism mechanically live (86-89% of full-DIMM cycle-slots are now power-gated, confirmed directly from the simulator's own counters) while changing no reported power number for ReRAM, by design - the alternative (a fabricated non-zero savings figure, or the naive 0.0 J/cycle NVMain default) would have been worse than the disclosed ungated baseline this replaces. PCM shows no power-down activity at all in either run, most likely because its `FRFCFS-WQF` write-queue-flush controller keeps its request queue non-empty far more of the time than the plain `FRFCFS` controller ReRAM and DDR5 use, starving the power-down entry condition - flagged as an open follow-up, not yet root-caused to full confidence. (6) Found and documented: trace provenance is weaker than the methodology narrative implied - the gem5 traces were generated without L1/L2 caches, warmup, or region selection (raw CPU-to-memory stream, first 10M instructions), which overstates memory pressure (conservatively for the endurance bounds, but distorting for absolute queueing behavior), and the GPT-2 trace could not be tied to any preserved SCALE-Sim configuration (AlexNet's TPU-v1 256x256 output-stationary provenance is confirmed). (7) Found and fixed: the DDR5 configuration's refresh machinery was inherited from a DDR3-1333 template and never rescaled when the clock was raised to 2400 MHz - refresh operations fired 3.6x too often (derived tREFI 1.085 μs vs. JEDEC 3.906 μs) at 6.6x too little cost each (tRFC 44.6 vs. 295 ns). Recalibrating to JEDEC JESD79-5 raised DDR5's refresh share from 57.9% to 71.3% of module power under GCC (a share subsequently reduced to the final 33-45% band once item 8's vendor-current calibration repriced the non-refresh components) and its average GCC latency from 69.9 to 81.2 ns, because refresh blocking is now faithfully priced. (8) Found and partially fixed: every supply parameter in the DDR5 configuration proved to be NVMain's stock built-in default rather than a datasheet value. The supply voltage - 1.5 V against DDR5's JEDEC-specified 1.1 V - was corrected, scaling all DDR5 power linearly by 0.733 (verified exact); the IDD current magnitudes were subsequently calibrated to published vendor datasheets (Micron and SK hynix 16 Gb DDR5-4800 tables) [29], [30], run as a two-vendor band - the conservative SK hynix floor is this book's headline, the Micron ceiling is reported alongside - with a documented caveat that datasheet IDD values are specification limits rather than typicals; calibrating with real currents also exposed and forced the repair of a sign artifact in NVMain's activate-energy formula (clamped at zero, worst case 2.8% of module power before the fix). (9) Found and fixed: the PCM baseline's cycle timings were derived for its cited 400 MHz basis, but a stray trailing clock override ran it at 800 MHz - twice as fast as Lee et al.'s parameters intend - and a second instance of the same bug sat in the reporting layer's hardcoded clock table; a syntax typo additionally zeroed its per-bit write energy. Restoring the 400 MHz basis roughly doubled PCM's latencies (e.g., LBM 3,732 to 7,465 ns) and corrected its power to 0.036-0.044 W. (10) Found and fixed: the configuration generator wrote ReRAM access energies under key names NVMain never reads (ReadEnergy/WriteEnergy vs. the live Erd/Ewr) - so every ReRAM configuration had silently used identical stock access-energy constants for the pipeline's entire history, despite NVSim characterizing a 5.7x read-energy difference between the families. Wiring the live keys and re-running the 96-run ReRAM matrix left every static floor, latency, and endurance counter identical while making dynamic power technology-differentiated for the first time (hand-reconciled against access counts to four significant figures). A definitive key-liveness audit of every generator-written key against the simulator's parameter readers closed this bug class permanently.

The repair was validated before its results were adopted. Pilot runs confirmed the leakage wiring end-to-end: the repaired 1T1R full-DIMM configuration measures 50.858 W of static power against a 50.858 W device-derived target (0.0% error), with real dynamic power resolving to approximately 5 mW under GCC - confirming that ungated ReRAM power is, to three significant figures, leakage. The full 120-run matrix (20 configurations × 6 workloads, including all 96 ReRAM runs) was then re-executed under the repaired model with anchor checks passing, and the endurance write counters were verified bit-identical to the pre-repair runs - the repair changes power accounting, not timing. The DDR5 baseline repairs were validated by the same discipline: the recalibrated refresh model produced exactly the predicted 21,333 refresh events per bank over the 200M-cycle window, the voltage correction reproduced the predicted linear scaling to within 4×10^{-6} relative deviation, and all non-DDR5 rows remained byte-identical across re-runs. The final calibration cycle added an independent, blind re-verification pass (a fresh session re-deriving every check from configs, sources, and raw statistics), which validated the DDR5 calibration end-to-end, refuted one suspected regression, and surfaced item (10). (11) Found and fixed: the configurations assumed heterogeneous host CPU frequencies - CPUFreq 800 MHz for every ReRAM configuration, 2 GHz for PCM, 3 GHz for DDR5 - and NVMain admits trace timestamps against a (CPUFreq/CLK)-scaled cutoff without rescaling them, so ReRAM was replayed under ~3.75x slower wall-clock request arrivals than DDR5 and each clock domain saw a different admitted trace population. Repaired by fixing CPUFreq = 3 GHz for every configuration and matching every cycle budget to the same 250M-trace-cycle admission - 83.33 ms wall-clock for all (ReRAM 66.7M memory cycles, PCM 33.3M, DDR5's existing 200M unchanged): every technology now processes the identical request population per workload, verified exactly (407,363 GCC / 16,447,102 LBM admissions across all eight configuration families). The correction moved exactly what its direction analysis predicted: admission-sensitive compute-bound latency rose (1T1R SLC GCC 98.0 to 130.9 ns, widening the DDR5 gap from 1.21x to 1.61x), trace-limited AI workloads did not move (the 4.7x GPT-2 deficit is unchanged), static power did not move (anchors exact to the last digit), and endurance write pressure rose toward each configuration's service ceiling (worst-case 1T1R SLC lifetime 24.4 to 17.3 years at 128 GB). Exact-config snapshots now ship with every results generation, closing a reproducibility gap in which on-disk configurations had drifted from the state that produced banked results. The residual limitations are items (5) and (6), plus the spec-limit character of vendor IDD tables: ReRAM totals are worst-case ungated, absolute queueing magnitudes inherit the uncached-trace caveat - and, more broadly, the pipeline is open-loop trace replay with no CPU or accelerator feedback path, so every latency reported here is a memory-subsystem quantity: the 4.6x AI-inference deficit of Section 3.1.1 is a memory-latency ratio, not a projected application slowdown, and latency-tolerant accelerators that overlap compute with memory access would experience a smaller end-to-end penalty; additionally, the completed-request and trace-player mechanism audit that exposed item (11) confirms that, after the matched-host correction, every configuration admits the identical request population per workload: GCC and the four lighter traces complete in full for every technology, so their latency averages compare identical populations, while LBM - the one genuinely service-limited workload - is completed to a technology-dependent depth (the sustained-throughput ordering of Section 3.1.1), so its latency averages cover each configuration's completed prefix, and DDR5 power is reported as a two-vendor calibration band. Within those bounds, the cross-technology power and PDP comparisons that earlier drafts of this work withheld are restored in Sections 3.1.2, 3.1.3, and 3.3.

(12) Found and fixed: the DDR5-4800 baseline's tCAS/tRCD/tRP timing parameters (34-34-34 cycles) were unsourced placeholders with no attached citation or datasheet reference. Cross-referencing SK hynix's public DDR5 SDRAM and DIMM part-number decoders (which enumerate per-speed-grade CAS-latency codes) against the standard DDR5-4800 (non-3DS) speed bin identifies the real value as 40-39-39, not 34-34-34 - a 15.7% increase in the CAS+RCD+RP component of DDR5's timing envelope. [10] (the primary JEDEC JESD79-5D standard) remains access-gated and could not be consulted directly; the SK hynix decoder is the best available public grounding. The DDR5-4800 configuration was corrected and the full 6-benchmark trace suite re-run; DDR5's total latency rose 2.3-8.6% across the suite (diluted from the 15.7% component-level change once blended with the timing parameters this fix did not touch - tRAS, tWR, tRFC, and refresh), while DDR5's power was unaffected, since this fix touches only timing, not the IDD/energy calibration of item (8).

(13) Found and fixed: the MLC read/write latency and energy penalty multipliers - applied analytically on top of NVSim's SLC characterization, since NVSim itself has no multi-level-cell model - were unsourced 3x/4x placeholders that earlier drafts of this book attributed to "EMBER Macro analytical heuristics" not actually present in either EMBER publication: the cited conference paper (Upton et al. [6], ESSCIRC 2023) reports no write-verify data split by bits-per-cell (it does give an aggregate, non-split SET/RESET pulse-energy estimate, but nothing usable as a 1-vs-2-bit/cell multiplier). A literature search for the real multipliers - including two independent research-assistant passes, both of which initially overreached with unverifiable claims that had to be retracted under direct primary-source checking - located the actual measured figures on the same EMBER macro's full journal publication (Levy et al. [31], IEEE JSSC 2024, Section V): read energy 1.0/1.1 pJ/bit at 1/2 bits per cell (from the ESSCIRC precursor's own Table I), write-verify bandwidth 12.4/3.8 Mbps and write-verify energy 0.40/1.2 nJ/bit at 1/2 bits per cell (from the JSSC follow-up, which the ESSCIRC paper's page budget omitted entirely). At this stage the read-latency multiplier was set to 1.917x, derived by pairing EMBER's own 12 ns read time with a “23 ns” figure read off the same ESSCIRC Table I - a derivation later found to be itself in error (see item (14)). The write-latency, read-energy, and write-energy multipliers - 3.263x, 1.1x, 3.0x respectively - replace the unsourced 3x/3x/3x set and remain correct; write energy happens to be numerically unchanged (3.0x both times), while the read-energy correction is the largest single change of the three (3.0x to 1.1x). All 8 MLC configurations were re-simulated across all 6 benchmark traces under the multipliers as understood at this stage (Sections 3.1.2 and 3.1.3, in their original form, detailed the resulting per-workload comparisons against SLC; those sections now reflect item (14)'s further correction).

(14) Found and fixed: item (13)'s 1.917x read-latency multiplier was itself a data error, caught by an independent citation-verification pass that read Upton et al.'s [6] Table I directly rather than trusting the prior derivation. The table's “Read Time (1 b)” row lists five macros side by side - EMBER (This Work) at 12 ns, and four competing designs, including reference [9] (T. F. Wu et al., ISSCC 2019) at 23 ns - and every entry in that row, including EMBER's own, is explicitly scoped to *1-bit/cell* operation. The “23 ns” is [9]'s own 1-bit/cell read time, not EMBER's 2-bit/cell number; EMBER's ESSCIRC paper never reports a 2-bit/cell read latency at all. Re-deriving from EMBER's own data: the JSSC 2024 follow-up (Levy et al. [31], Section V.A and Abstract) states “1 b/cell read operation with 1.0 pJ/bit energy at 2.4 Gbps, and 2 b/cell read with 1.1 pJ/bit at 1.6 Gbps” - giving a clean 1.5x read-latency multiplier (2.4/1.6 Gbps) via the same bandwidth-ratio methodology already used for the write-latency multiplier (12.4/3.8 Mbps -> 3.263x). The corrected 1.5x is milder than the erroneous 1.917x, not harsher: MLC read latency was previously over-penalized. All 8 MLC configurations were re-simulated across all 6 benchmark traces under the corrected multiplier; MLC latency fell a further 5.5-18.6% and MLC PDP fell 0.7-18.6% relative to the item (13) dataset (Sections 3.1.2 and 3.1.3 reflect the corrected figures throughout; Section 3.1.4's LBM endurance lifetimes were likewise recomputed, since faster reads let more total requests - including writes - complete within the fixed simulation window).

A dedicated NVSim sensitivity sweep, prompted by a related citation question over the HRS resistance target (Appendix A), separately confirmed that the 47x transistor-vs-selector leakage-class separation of item (3) is completely insensitive to that target across four orders of magnitude of HRS - so no further re-simulation was triggered by that finding; see Appendix A for the sweep itself.

3.2. Architectural Scaling & Memory Level Parallelism (Pareto Frontiers)

While granular diagnostics reveal the baseline physical characteristics of 22nm ReRAM, understanding its viability as a main memory replacement requires evaluating how these modules scale. To map this, I plotted multi-objective Pareto frontiers comparing Total System Power against Average Total Latency as the ReRAM architecture scales from a single chip to a 64-chip Full DIMM (8 GB SLC / 16 GB MLC).

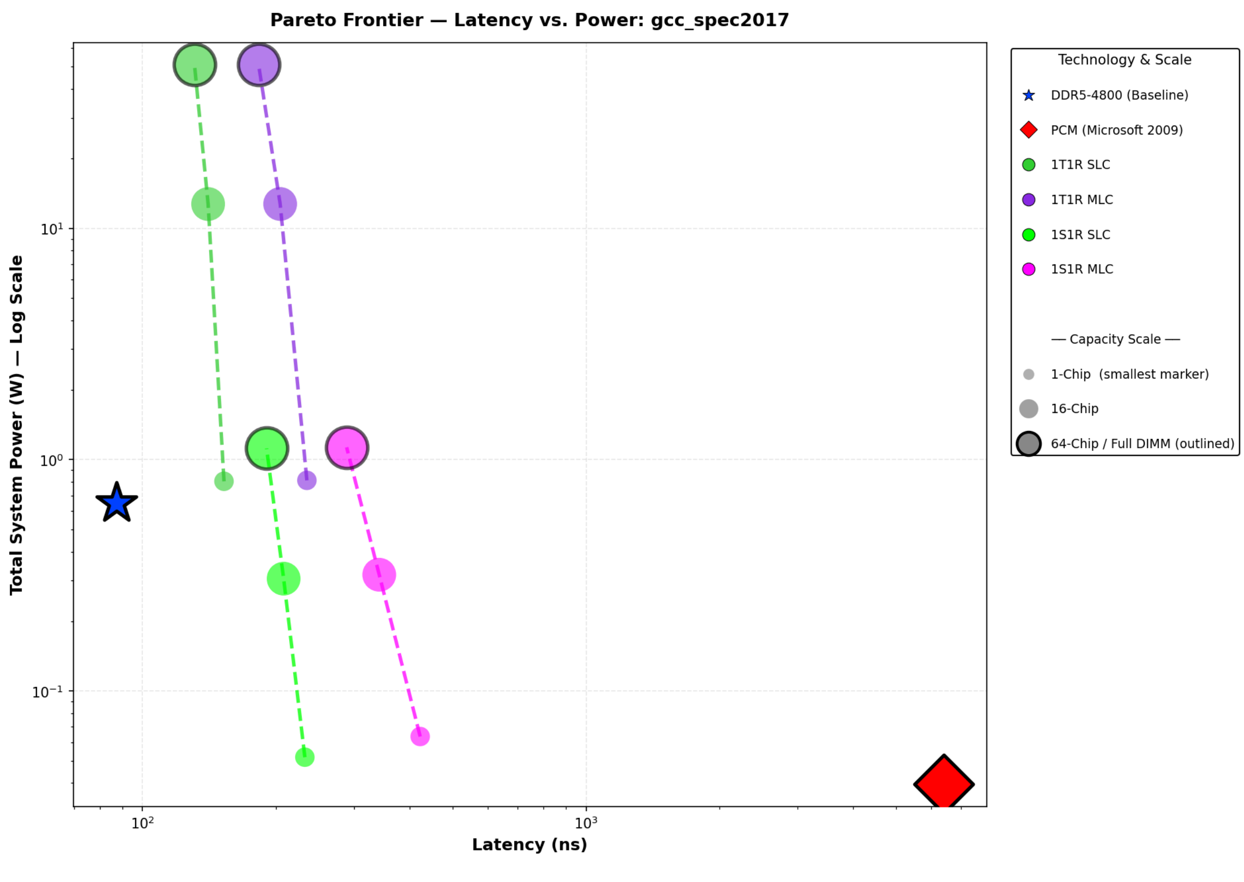
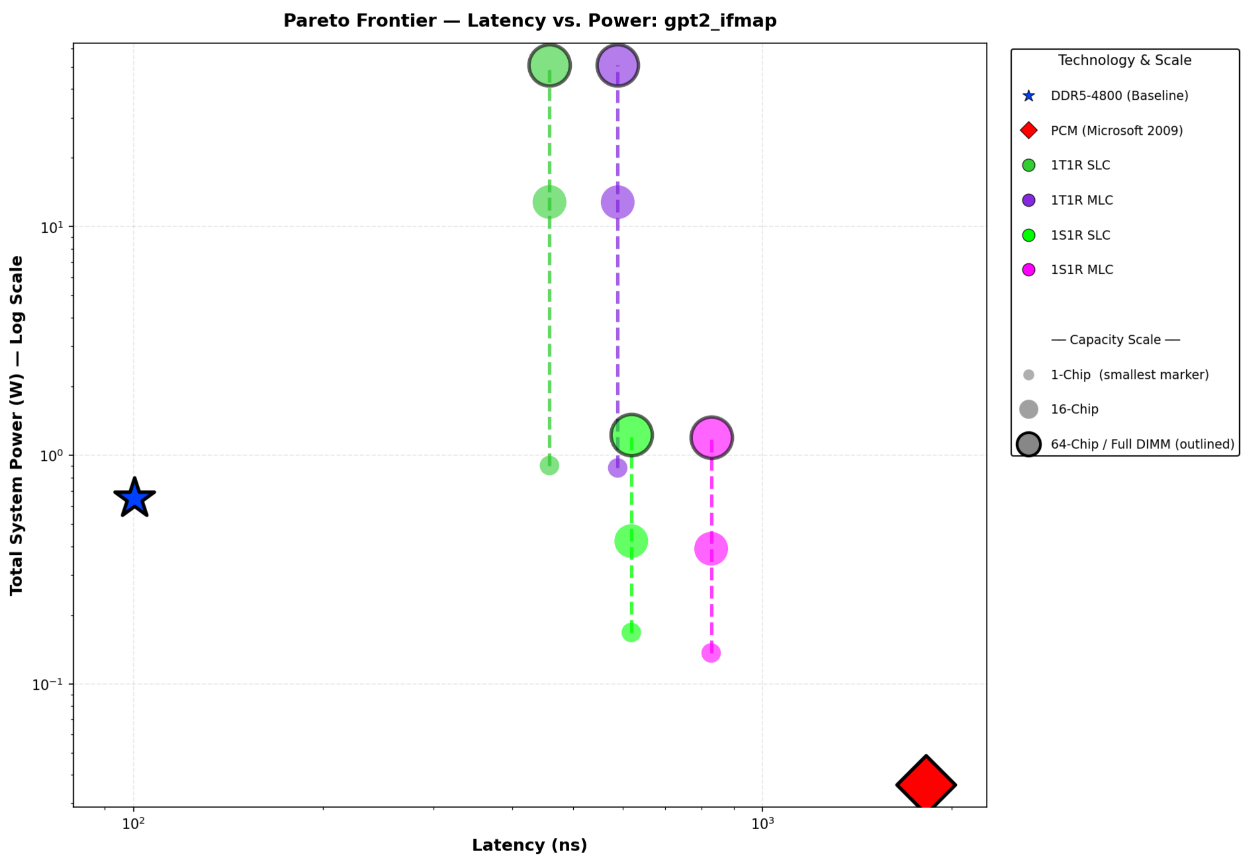


Figure 20: Pareto scaling trajectory under the single-threaded GCC workload.

Under standard CPU-bound workloads like gcc (Figure 20), scaling chip count genuinely reduces latency - 152.58 ns at a single chip to 130.91 ns at a full 64-chip DIMM for 1T1R SLC, a real ~14% improvement - but, on inspection, neither smoothly nor for the reason an earlier version of this analysis claimed. *Compute-Bound Scaling, Correctly Attributed:* single-chip and 8-chip configurations perform identically (both instantiate exactly one memory rank in this project's architecture factory - “8-chip” only widens the bus, per the address-decode analysis in Appendix A); the entire gain accrues only once rank count itself increases, at 16-chip (2 ranks) and full-DIMM (8 ranks). The mechanism is address footprint, not workload memory-level parallelism (MLP): gcc's SPEC trace touches roughly 51 MB of address space, about 800x larger than a single rank's 64 KB addressable span under this configuration's R:BK:RK:C mapping (COLS=1024, 64-byte bursts), so its traffic genuinely spreads across every additional rank the DIMM provides, and the FRFCFS controller services that spread-out traffic more concurrently as rank count grows. A dedicated queue-depth x chip-count sweep (Appendix A) confirms this gain persists proportionally at every controller QueueSize tested - it is real rank-level parallelism, not a queueing artifact. The repaired power model prices the scaling honestly regardless of mechanism: peripheral leakage grows linearly with chip count (0.80 W single-chip to 50.9 W full-DIMM for 1T1R SLC; 0.027 W to 1.12 W for 1S1R SLC), so this real latency gain still costs full leakage as it accrues - the strongest scaling argument remains the selector's leakage discipline, not workload MLP.



Ungated static power (NVMain power-down disabled); module-sum semantics; real per-technology leakage (see fidelity audit).

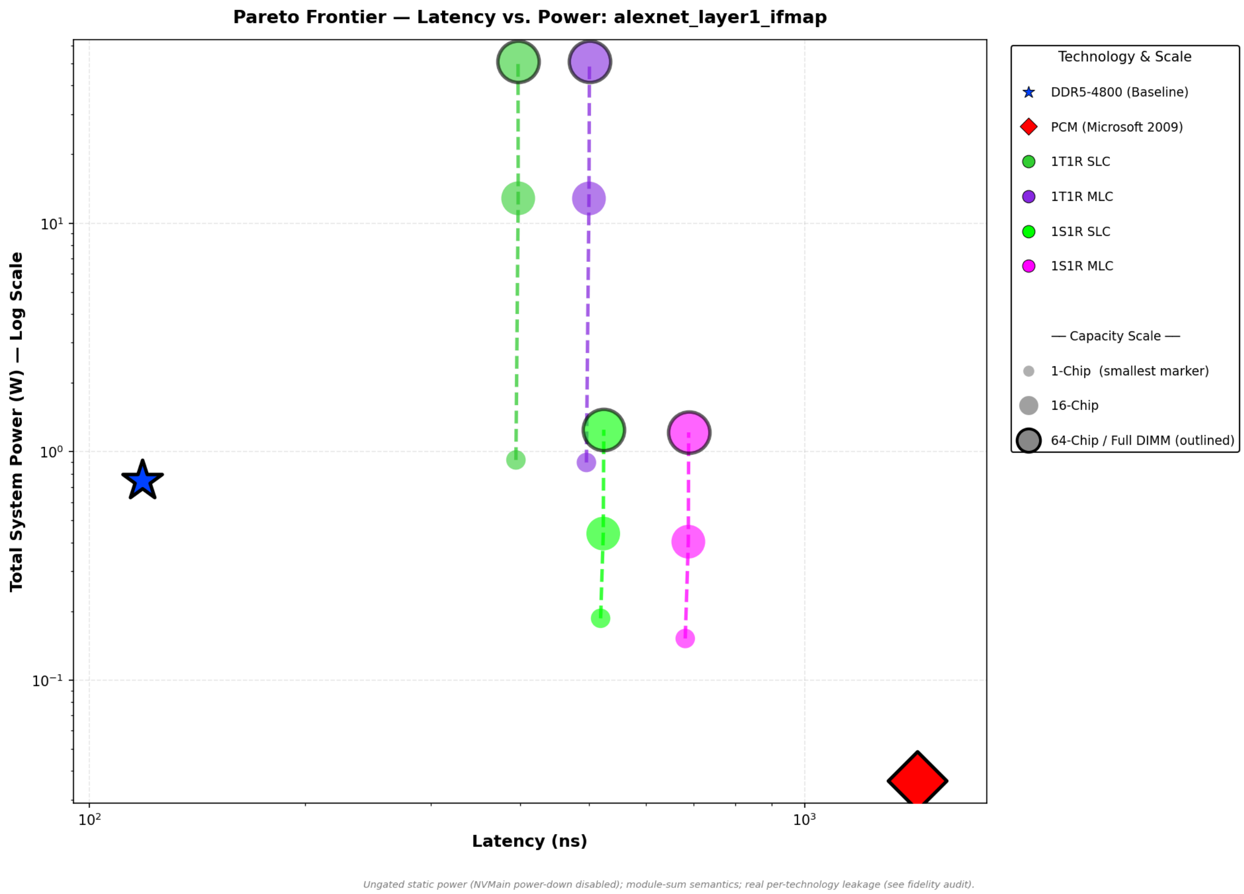


Figure 21 & 22: Pareto scaling trajectories under highly parallel AI inference workloads (GPT-2 IFMAP and AlexNet IFMAP).

The two AI-inference traces available to this study behave in the opposite direction from what an earlier version of this analysis claimed (Figures 21 and 22). *AI-Inference Is the Actual Flatline:* GPT-2 IFMAP's latency is exactly flat - 458.41 ns at every chip count from 1 to 64 - and AlexNet IFMAP is nearly as flat (394.73 to 397.41 ns, a small real degradation, not an improvement). The Pareto trajectories here move *horizontally* as chip count scales, not vertically: latency barely changes while module power still climbs with leakage, so added ranks buy these two traces essentially nothing. The cause is address footprint, not workload parallelism: both traces' captured address ranges (GPT-2 IFMAP ~64.0 KB, AlexNet IFMAP ~68.3 KB) sit at or just past the exact 64 KB per-rank addressable span this configuration's address-decode logic provides - GPT-2 IFMAP's footprint lands 6 bytes short of that boundary, so its entire request stream decodes to a single rank regardless of how many the DIMM physically provides; AlexNet IFMAP's marginally larger footprint spills a small tail of requests into a second rank, enough to produce a small, non-beneficial sensitivity rather than a genuine improvement. The same queue-depth x chip-count sweep referenced above rules out a controller-queueing explanation: latency changes with queue size, but the flatline itself is bit-for-bit identical at every queue depth tested, for both traces. Practically, this means the two AI-inference traces available to this study are too small to exercise multi-rank scaling at all - almost certainly an artifact of how they were captured (a single SCALE-Sim layer's activation map, not a full model's working set) rather than evidence that AI-inference workloads in general cannot benefit from rank interleaving; a larger, more representative AI-inference trace is future work (Section 4.1). What the comparison does still show honestly: 1T1R SLC's ungated 51.0 W module power (Section 3.1.2) sits far above the DDR5 baseline at every scale point regardless of latency, while the selector-gated family holds the same flat latency within a 1.25 W envelope - the power-tier separation is real even though the "high-MLP unlocks latency" mechanism originally claimed for it is not.

The Pareto trajectory for AlexNet OFMAP (Figure 23) reveals the most extreme manifestation of the write-penalty scaling wall. Under sustained write pressure, the 1S1R MLC configuration suffers a PDP spike to 3,601.1 W·ns against 1,629.2 W·ns for 1S1R SLC - a 2.2x write-torture tax within the selector family - while the 1T1R family, whatever its latency behavior, is priced out entirely by its 50.9 W leakage floor (41,763.3 W·ns for 1T1R SLC). This confirms that high-density cross-point topologies face fundamental efficiency limits under write-intensive AI workloads, where iterative ISPV programming operations saturate the memory controller queue.

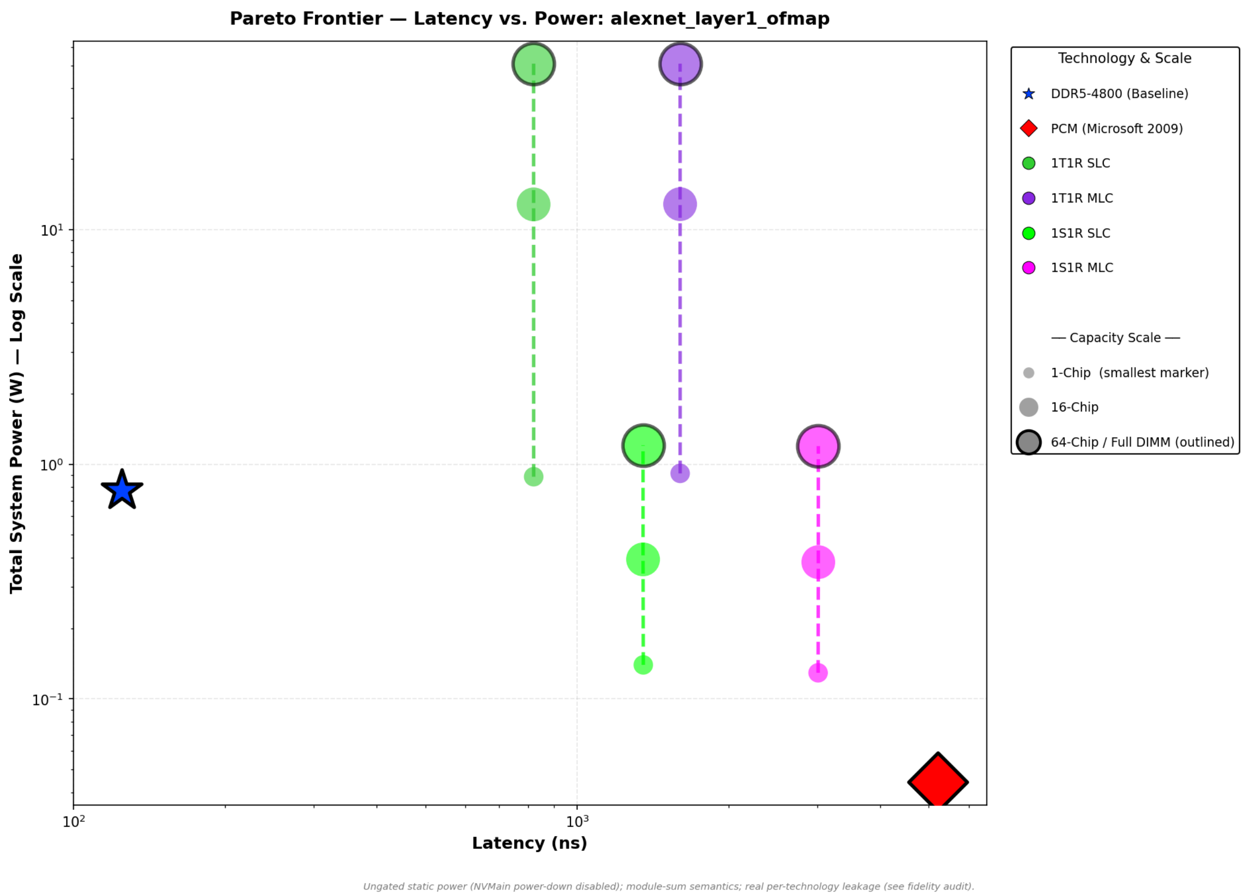
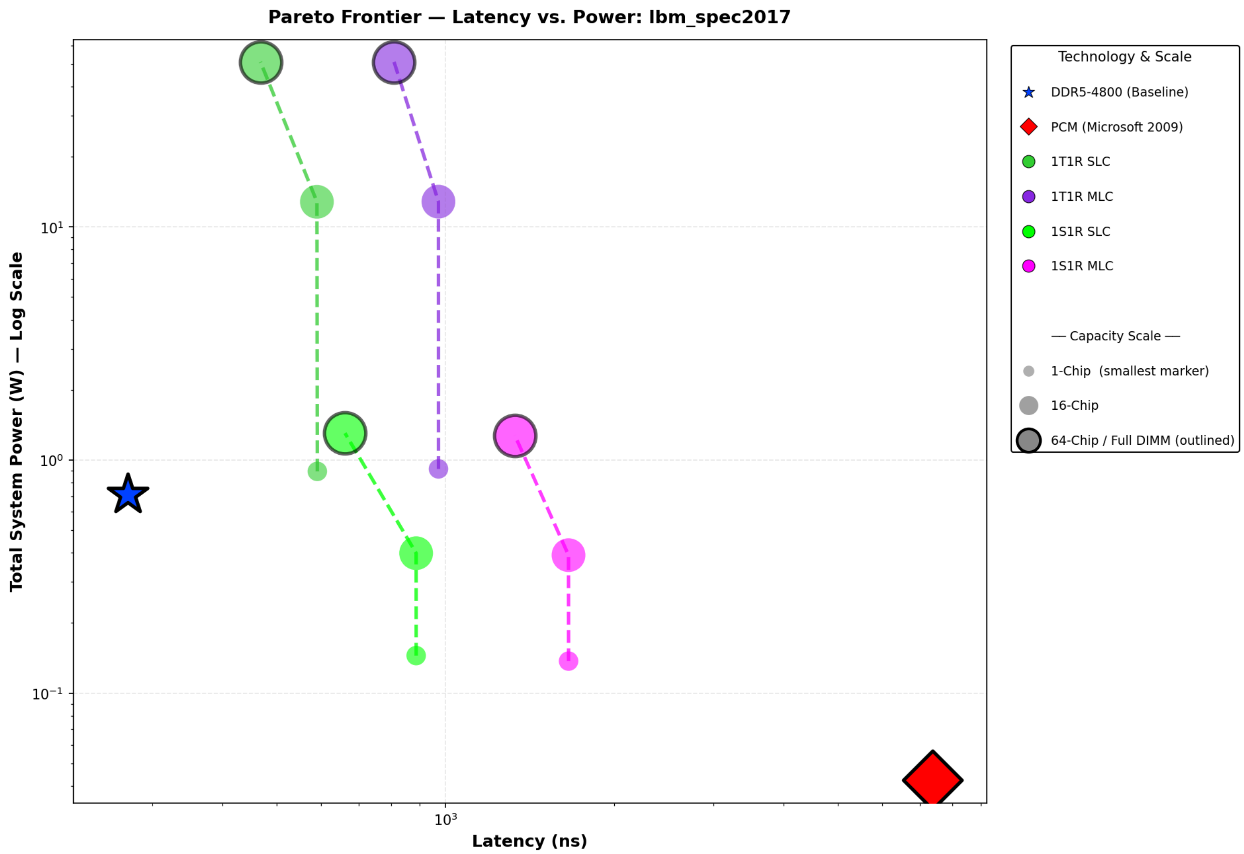


Figure 23: Pareto scaling trajectory under write-torture conditions (AlexNet OFMAP).



Ungated static power (NVMain power-down disabled); module-sum semantics; real per-technology leakage (see fidelity audit).

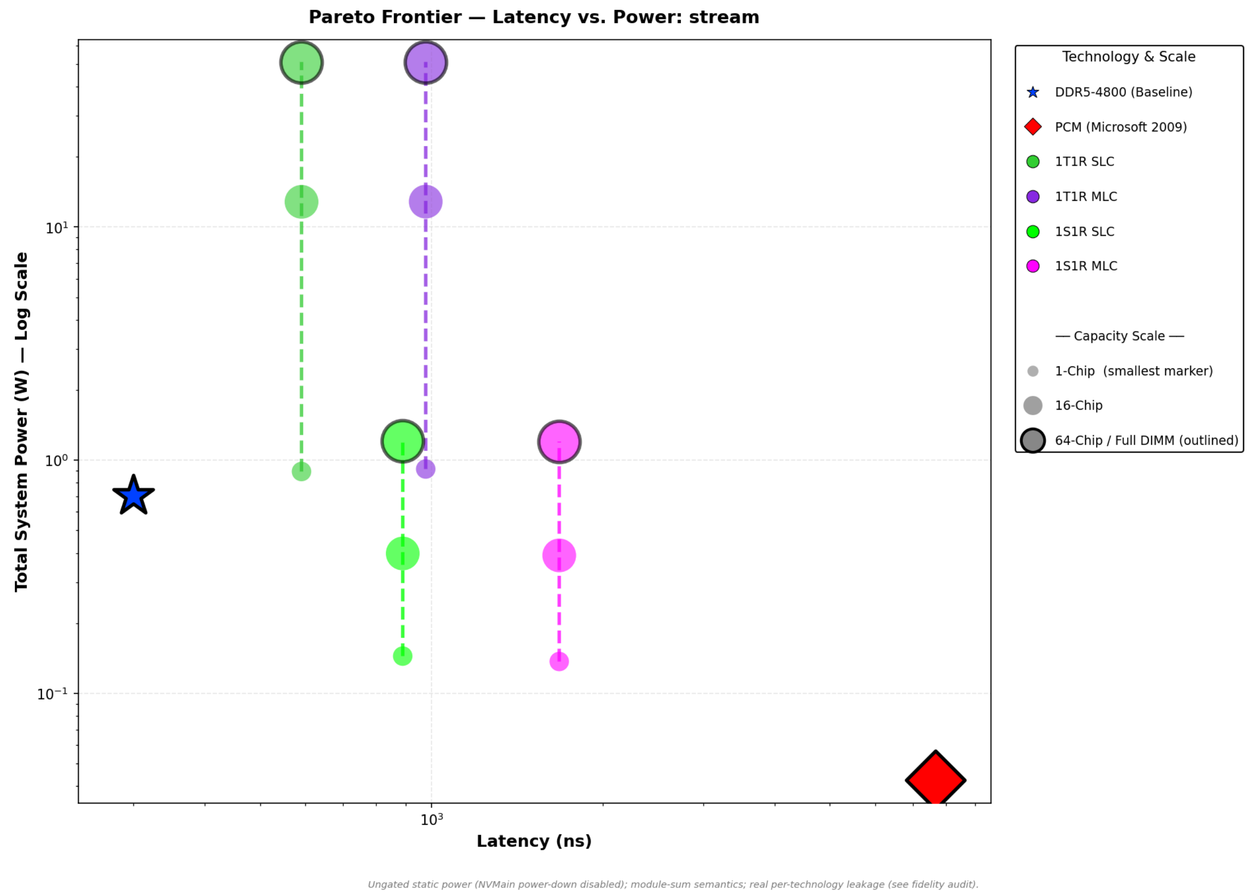


Figure 24 & 25: Pareto scaling trajectories under continuous memory-streaming workloads (LBM SPEC2017 and STREAM).

Finally, evaluating heavy streaming workloads (Figures 24 and 25) shows each family riding its own leakage tier as capacity scales. The selector-gated configurations remain within commodity power envelopes at every scale point (1.31 W full-DIMM under LBM), while the transistor-gated family scales linearly into infeasibility (51.0 W). In both cases the power cost of scaling is linear peripheral leakage rather than compounding dynamic or thermal penalties - the memory controller's activity level perturbs module power by under 0.3% for 1T1R and by at most ~11% for 1S1R (LBM) - never enough to change tier - and, per the mechanism established above for gcc, whatever latency benefit added ranks provide depends on whether the workload's address footprint actually spans multiple ranks, not on an abstract notion of workload parallelism.

3.3. Global Viability (Hero Graphs)

With the scaling dynamics established, the final evaluation must address the core question: does 22nm ReRAM present a globally viable alternative to DRAM?

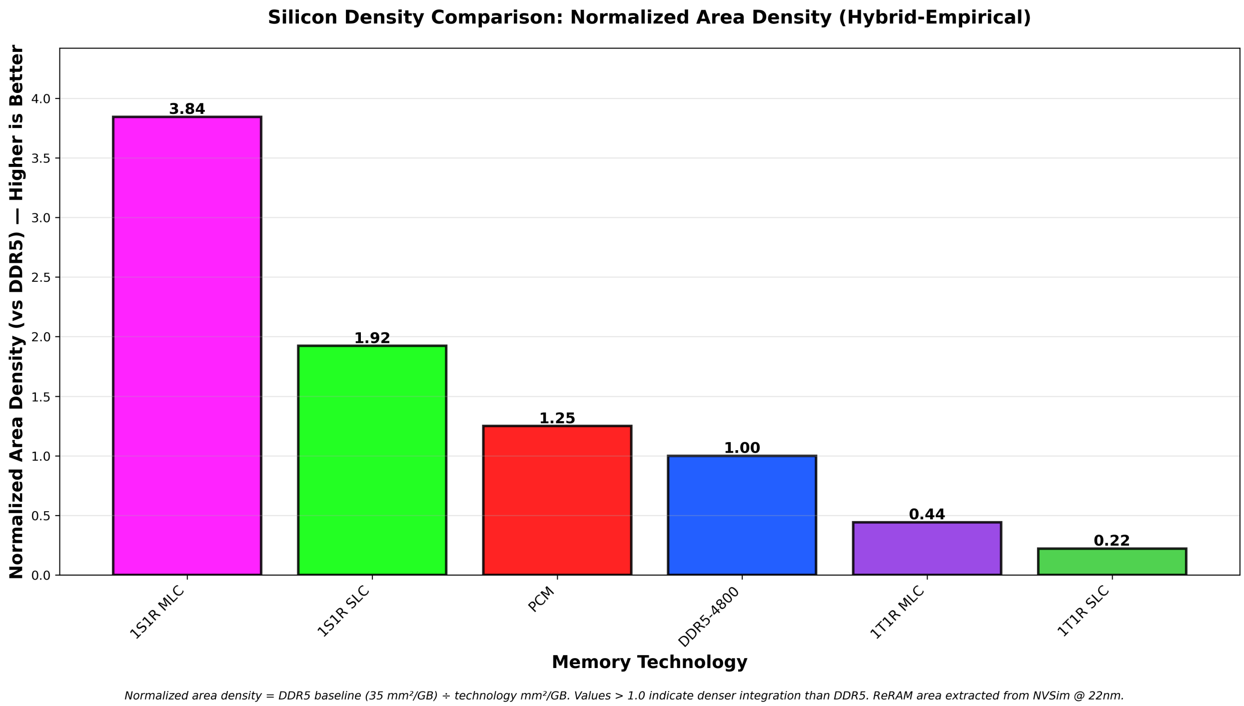


Figure 26: Die-Level Density Comparison, normalized to DDR5 = 1.00 (higher is better): NVSim-characterized die areas per GB versus a commodity DDR5 baseline of 35 mm² per GB [18].

To alleviate the severe memory supply-demand gap, MBMM must provide superior physical scaling, and density is best read at three levels. At the die level - what a module vendor can actually buy - Figure 26 normalizes NVSim's characterized die areas against a commodity DDR5 baseline of 35 mm² per GB (1.00) - a figure sitting inside the 33.1-37.6 mm²/GB band measured across Micron, Samsung, and SK hynix production 16 Gb DDR5 dies [18]: the selector cross-point delivers 1.92x DDR5's density as SLC and 3.84x as MLC, while the transistor-gated 1T1R lands at 0.22x (0.44x as MLC) - a $20F^2$ access-transistor cell fabricated at 22nm cannot compete with 10 nm-class DRAM on area, which reinforces 1T1R's latency-niche verdict. At the cell level, the comparison becomes node-independent and exact: DRAM stores one bit per $6F^2$, 1T1R one per $20F^2$ (Appendix A), and the selector cross-point one per $4F^2$ (two per $4F^2$ as MLC) - so at any matched process node, 1S1R MLC is 3.0x denser than DRAM and 1S1R SLC 1.5x denser, while 1T1R is 3.3x less dense under the planar-access-transistor assumption used throughout this book (a DRAM-process recessed-channel alternative closes this gap in principle but is not modeled here - Appendix A, Section 4.2). That the measured die-level advantage (1.92x) exceeds the 1.5x cell-level bound reflects real-die overheads on the DRAM side - peripheral, spare, and interface area that pure cell arithmetic ignores. The third level is headroom: the die-level ratios are achieved across an asymmetric process comparison - DDR5 fabricated on 10 nm-class nodes (1γ [17]) with no remaining capacitor runway [5], the ReRAM figures from a mature 22nm logic-compatible process. A $4F^2$ cross-point cell scales quadratically with feature size: a port to a 12 nm-class node would multiply the density ratios by roughly $(22/12)^2 \approx 3.4$x before any stacking, and cross-point arrays additionally admit back-end-of-line 3D deck stacking [9], a lever planar DRAM does not possess. Node scaling further compounds through the endurance capacity law of Section 3.1.4: a denser module is, by that law, directly a longer-lived module.

Table 6 makes the third level concrete: the measured die-level ratios of Figure 26, scaled quadratically to 16nm- and 12nm-class nodes, with the back-end-of-line deck-stacking lever that only the cross-point possesses.

Table 6: Projected die-level density versus DDR5 (= 1.00) under quadratic feature-size scaling and 3D deck stacking.

Config.	22nm (measured)	16nm (projected)	12nm (projected)	12nm, 2 decks	12nm, 4 decks
1S1R SLC	1.92	3.63	6.45	12.9	25.8
1S1R MLC	3.84	7.26	12.9	25.8	51.6
1T1R SLC	0.22	0.42	0.74	N/A	N/A
1T1R MLC	0.44	0.83	1.48	N/A	N/A

Projection: measured 22nm die-level anchors (Figure 26) x $(22/F)^2$, with the DDR5 baseline held fixed at its 10 nm-class density per the end-of-roadmap capacitor argument [5][17]. Deck rows multiply the 12nm figure by 2 and 4; 1T1R cannot deck-stack - its access transistor is a front-end-of-line device.

These projections are bounded claims, not measurements: the 16nm and 12nm columns are geometry rather than NVSim characterizations, they optimistically assume peripheral circuits shrink with the cell array, and selector device physics below the 2x-nm regime is unvalidated. Every ratio in the table is also linear in the assumed cell area, and that sensitivity is worth stating explicitly because the cell areas are assumptions, not measurements (Appendix A): demonstrated 1T1R endpoints span a 14nm FinFET embedded macro at $112F^2$ [20] - which would drag 1T1R's 22nm ratio from 0.22x down to roughly 0.04x - and a DRAM-process recessed-channel part at $6F^2$ [33] - which would lift it to roughly 0.7x; on the selector side, $4F^2$ is the idealized floor, and a realized cell at twice that area would halve every 1S1R entry, dropping 22nm SLC to DDR5 parity (0.96x) while MLC retains 1.92x before node scaling restores the margin. The density verdict is therefore robust for the selector cross-point but conditional on cell-area realization for 1T1R. The counterpoint is equally real, and moving on three fronts: DRAM's escape from the planar capacitor is on the incumbents' public roadmap, with Samsung outlining vertical-channel-transistor 3D DRAM for the second half of the decade [19], and the challenger ecosystem is moving faster still - a monolithic 3D X-DRAM built on 3D NAND-style vertical processing passed proof-of-concept validation in April 2026, reporting sub-10 ns access and second-scale retention [25]. Notably, even that challenger's headline retention claim is a 15x longer refresh interval, not refresh elimination: charge-based storage refreshes by construction, so while 3D DRAM contests the density column of this comparison, the zero-refresh power asymmetry this work quantifies survives every DRAM roadmap, planar or vertical. The third front attacks the economics around the cell rather than the cell itself: Intel's July 2026 patent filing for Cross-Batch Memory (XBM) proposes moving the 1T1C cell into back-end-of-line thin-film transistors and replacing HBM's silicon interposer with direct die-to-die UCIe links, targeting HBM4's footprint at lower packaging cost - a response to the same supply-side crunch behind this project's premise (Section 1.1), though the design remains patent-stage, discloses no independent performance data, and is not targeted for commercialization before 2030 [34][35]. The window Table 6 quantifies is therefore real but not permanent - it is the interval in which a cross-point that ports to logic-compatible nodes today faces a DRAM that must first re-architect its cell to go vertical. Every density multiplier in the table also compounds through the endurance capacity law of Section 3.1.4: each projected doubling of module capacity is, by that law, directly a doubling of worst-case lifetime.

The same geometry extends beyond density to power - and this is the extrapolation with architectural teeth. 1S1R's ungated draw is 97% peripheral leakage, and periphery is standard CMOS that shrinks with the process: to first order, static power per gigabyte falls by the same (22/F)^2 factor that density rises. Holding the 8 GB module and its measured 1.082 W static anchor fixed, a 16nm-class port lands at roughly 0.61 W ungated - at parity with DDR5's 0.651 W calibration floor with no gating at all (the exact crossover sits at a 16.6nm feature size) - and a 12nm-class port at roughly 0.36 W, 1.8x below the floor. Under this arithmetic the gating requirement of Section 3.1.2 is not a permanent architectural tax but a 22nm artifact: one full shrink buys ungated parity, the second buys outright dominance, while the same scaling leaves the transistor-gated family unsalvageable (50.9 W falls only to ~15 W at 12nm - still 23x the DDR5 floor). Latency, by contrast, neither improves nor collapses under scaling: filament switching is voltage- and material-limited rather than lithography-limited - HfOx devices demonstrated at 10 x 10 nm retain nanosecond-class switching [14] - so the latency profile of Section 3.1.1 carries to first order, with thinner-wire RC the main unmodeled risk. Endurance splits both ways: device variability - the field's named barrier to large arrays and multilevel operation, rooted in the stochastic filament process [14] - is unvalidated at scaled dimensions, but the capacity law converts every density multiplier in Table 6 into an equal lifetime multiplier at fixed footprint. The caveats of the preceding paragraph apply with full force - this is bounding geometry, not simulation, and per-transistor leakage densities worsen at advanced nodes, making area-proportional scaling an optimistic first order - but the direction is uniform: every process generation moves the comparison in ReRAM's favor, against a DDR5 baseline whose capacitor cannot follow [5][17].

A fair challenge to these projections is why no sub-22nm ReRAM exists to measure, given that logic CMOS ships below 5nm. The answer is primarily market structure, not device physics. ReRAM's commercial role today is embedded flash replacement on mature 40-22nm foundry processes [24] - the node range where embedded flash itself stops scaling - so no product pull exists toward leading nodes, even though devices have been demonstrated at 10 x 10 nm - with sub-10 nm filaments observed - in the laboratory [14] and a 14nm-FinFET megabit macro has been demonstrated [20]. The one scaled, selector-gated cross-point that did reach volume production - Intel and Micron's 20nm-class 3D XPoint - was discontinued in 2022 on cost-per-bit economics, squeezed between DRAM and NAND, not on device failure [23]: an instructive precedent that the barrier at scaled nodes has been the business model, not the physics. What scaling genuinely risks at the device level is variability - the cycle-to-cycle and device-to-device spread rooted in the stochastic motion of the oxygen vacancies that form the filament, which the field itself names the major remaining barrier to large arrays and multilevel operation [14] - which is exactly why Table 6's 16nm and 12nm columns are labeled projections rather than measurements, and why the memory supply-demand crisis that motivates this work (Section 1.1) may be the demand-side event that finally changes the economics.

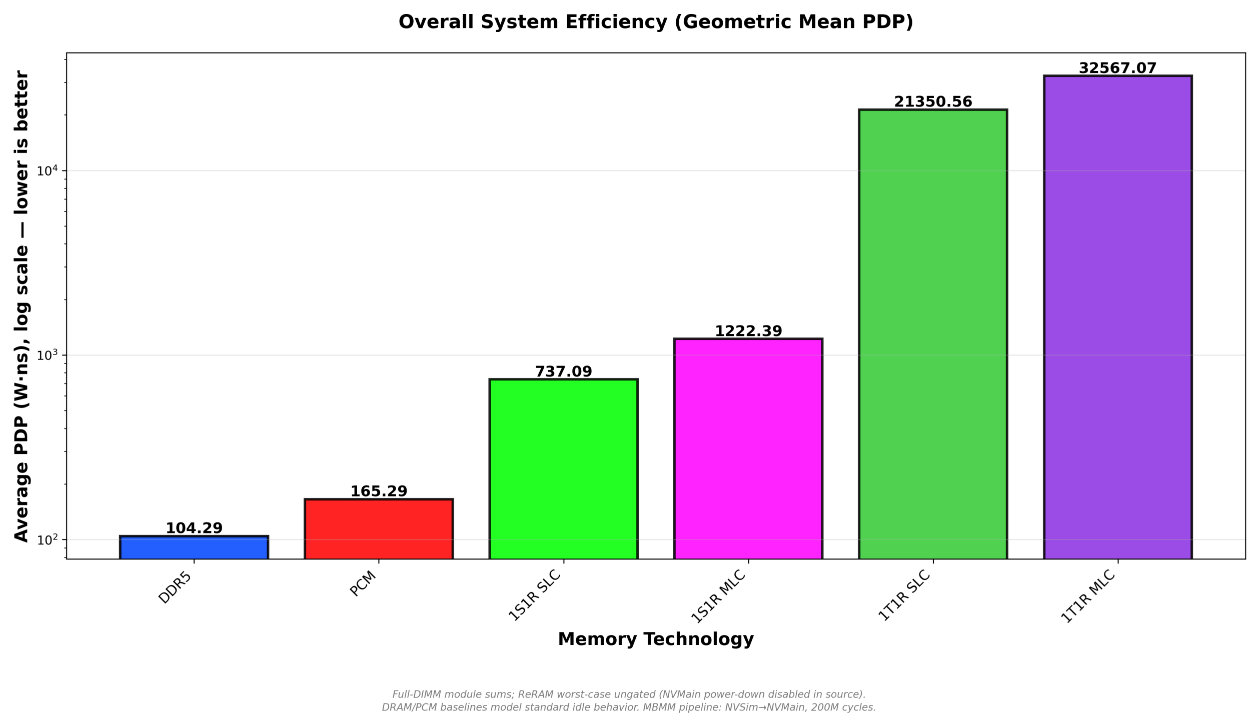


Figure 27: Overall System Efficiency (Geometric Mean PDP).

However, density is a liability if the system is too slow or burns too much power. The Geometric Mean PDP across all six workloads (Figure 27) ranks the architectures under the repaired, whole-module power model - with the mandatory caveat that DDR5's idle-gating mechanism is now restored and real (Section 3.1.6, item 5) while ReRAM's power-down energy remains an explicit, no-benefit-claimed placeholder (Appendix A). The headline is an opportunity: 1S1R SLC - drawing 1.12 W with zero refresh - lands 1.8x from DDR5's 0.623 W restored-floor, and within 11% of outright parity at the spec-limit ceiling; that gap is precisely the component an idle-gating policy attacks, because 1S1R's draw is 97% static while DDR5 spends 33-45% of its module power on refresh it can never shed. Power parity for 1S1R SLC is therefore a projection contingent on idle-gating the toolchain cannot yet simulate - bounded by the arithmetic below, not demonstrated - and a concrete research and product opening for gating-capable controllers and selector-first DIMM architectures. The full ungated ranking, matching Figure 27 bar-for-bar:

DDR5 104.3 W·ns (158.1 at the Micron calibration ceiling - the Micron-side configuration was reconstructed from its original documented current-magnitude derivation and re-run under the item (12) DDR5 timing correction, confirming the same ~4.8% relative increase the hynix-floor figure carried; see Section 3.1.6, item 8), PCM 165.3, 1S1R SLC 737.1, 1S1R MLC 1,222.4, 1T1R SLC 21,350.6, and 1T1R MLC 32,567.1. The selector's 47x standby discipline cleanly partitions the ReRAM family: ungated 1T1R is not a viable DDR5 successor at any cell density (50.9 W per module, 65-78x DDR5) - the contrast that elevates leakage discipline to the deciding architectural requirement. PCM, for its part, survives on its 0.04 W floor at latency costs relative to 1T1R SLC that range from ~4-6x under parallel AI to 13-16x under streaming and 49x under compute-bound GCC (now that it runs at its cited 400 MHz basis - Section 3.1.6, item 9): it is not displaced, but it marks the floor-power bound that gated ReRAM must beat. ReRAM's path past both DDR5 and PCM therefore runs exclusively through the idle-gating capability that the current toolchain cannot yet simulate (Section 4.1).

How much of that draw a gating policy could actually reclaim is bounded by how idle real memory is - a fraction that is strongly workload-dependent, so one measured example is quoted to make the point, not as a universal constant. Profiling Microsoft's Bing web-search and Cosmos data-analytics servers, Malladi et al. report 67-97% processor utilization against only 2-6% memory bandwidth utilization, with web search drawing under 6% of peak channel bandwidth [21]: in that deployment class - a large and commercially central one - the memory channel sits idle the overwhelming majority of the time. Memory-bound workloads occupy the other extreme, and it is a growing one: transformer token generation is dominated by matrix-vector products of such low arithmetic intensity that memory bandwidth, not compute, is its binding constraint - and with peak hardware FLOPS scaling at 3.0x per two years against DRAM bandwidth's 1.6x, the imbalance worsens by design [22]. This suite brackets that extreme deliberately: LBM and STREAM saturate the channel, and GPT-2 is precisely the decode-class workload Gholami et al. analyze. The arithmetic is honest about what this means - a channel-saturating deployment gates little (f approaches 0), leaving the ungated comparison to stand as measured, 1S1R at 1.72x the DDR5 floor and 1.11x the ceiling - while Section 3.1.1's latency verdict already assigns sustained high-parallelism serving to DDR5. The two verdicts are mutually consistent: the workload class where gating cannot rescue ReRAM power is the same class ReRAM already loses on latency, and even there the ungated penalty is bounded at 1.7x, not the 65-78x of the transistor-gated alternative. The bounding arithmetic for the 1S1R SLC module follows directly from its measured composition (1.082 W static + ~0.036 W dynamic under GCC): gating the static component for an idle fraction f leaves (1-f) x 1.082 + 0.036 W of module power. DDR5's own idle-gating mechanism is now restored and simulated (Section 3.1.6, item 5), moving its GCC calibration floor from 0.651 W to a real, measured 0.623 W - break-even against that lower floor requires f = 46% (up slightly from the pre-restoration 43%, since DDR5 now has a little less room to close) - a threshold still far below the web-serving example above, and one that much busier deployments still clear - and just f = 10% against the 1.008 W vendor ceiling (the Micron-calibration variant, not re-run this session and therefore still ungated); a policy capturing 90% idleness, plausible for the measured web-serving class, lands at roughly 0.14 W, about 4.3x below DDR5's restored floor. Two honesty bounds and one physical asymmetry frame this: bandwidth utilization overstates achievable gating, since power-down entry and exit residency consume part of every idle window; DRAM also possesses power-down states that this comparison does not exercise (the asymmetry caveat of Section 3.1.6, item 5, cuts both ways); but the floor asymmetry is physical - an idle DRAM module must keep refreshing to retain data, while a gated non-volatile module retains it at zero power. These are bounding calculations, not simulations; simulating the actual gating policy is the top future-work item (Section 4.1).

Table 7 places every axis of the comparison side by side - the tabular summary of the entire evaluation.

Table 7: Cross-technology summary, full-DIMM configurations.

Technology	GCC latency (ns)	GCC power (W)	Geo-mean PDP (W·ns)	Die density (× DDR5)	Worst-case lifetime @128 GB	Role
DDR5-4800	90.2	0.623 (46.7% refresh)	103.3	1.00	n/a (volatile)	commodity baseline
PCM	6,403.8	0.040	165.3	1.25	not evaluated	floor-power NVM; 4-49x latency cost
1T1R SLC	136.2	50.870	21,508.4	0.22	17.3 yr	latency-optimized niche; infeasible ungated
1S1R SLC	192.3	1.118	738.1	1.92	24.8 yr	flagship; power parity contingent on future gating work
1T1R MLC	185.1	50.877	32,632.8	0.44	3.1 yr	infeasible ungated
1S1R MLC	286.5	1.130	1,221.1	3.84	5.2 yr	read-only capacity tier (frozen weights)

Power/PDP: idle-gating is now restored and live for every technology (Section 3.1.6, item 5). DDR5 realizes a real, measured power reduction from it (0.651 to 0.623 W under GCC); ReRAM's power-down mechanism fires mechanically (86-89% of full-DIMM GCC cycle-slots gated) but carries an explicit, no-real-number-available placeholder energy equal to its own standby power, so its reported power is unchanged by design, while its latency/PDP absorb a small, real power-down transition-latency cost; PCM shows no power-down activity in either run, plausibly structural to its FRFCFS-WQF controller (Appendix A). Die-level density = NVSim-characterized die area per GB vs a commodity DDR5 baseline (Figure 26); node-

independent cell-level bounds: DRAM $6F^2$/bit, 1T1R $20F^2$/bit, 1S1R $4F^2$/bit (2 bits/cell as MLC) - 1T1R's $20F^2$ here is a planar/FinFET logic-transistor assumption, conditional rather than a hard ceiling: a real $6F^2$ DRAM-process recessed-channel 1T1R part has been demonstrated (Appendix A, ref [33]), which would close most of 1T1R's density disadvantage against 1S1R if adopted. Lifetime: LBM worst case, uniform wear leveling, SLC 10^7 / MLC 10^6 endurance; under the matched-host window the write streams are service-limited and differ per configuration - 1T1R SLC absorbs 3,269,479 LBM writes per 83.33 ms (39.2 M/s) versus 2,284,570 for 1S1R SLC (27.4 M/s), so the slower writer wears correspondingly slower (1S1R SLC worst case 24.8 yr at 128 GB versus 1T1R's 17.3).

4. Conclusion and Future Work

This research delivers two things: a rigorous cross-layer characterization of 22nm ReRAM as a DDR5 alternative - whose headline is that selector-gated ReRAM carries a credible, bounded projection to DDR5 power parity, contingent now not on unsimulated idle-gating (that mechanism is restored and mechanically live for every technology) but on a real ReRAM power-gating energy characterization this project could not locate in the literature - and a quantified fidelity audit of the standard NVSim-to-NVMain toolchain in which thirteen of fourteen discovered failure modes were repaired - spanning the ReRAM configurations, the metrics pipeline, and both non-ReRAM baselines - each repair validated by exact anchor arithmetic and an independent blind re-verification, with the affected simulations re-run. Using a cross-layer simulation pipeline spanning device physics (NVSim), cycle-accurate memory simulation (NVMain 2.0), and real workload traces from gem5 and SCALE-Sim, I evaluated 20 memory configurations across 6 workloads. The key quantitative findings are: 1T1R SLC ReRAM operates within 1.50x of DDR5-4800's wall-clock latency under compute-bound execution - where it runs 49x faster than the legacy PCM baseline (13-16x under sustained streaming) - while trailing DDR5 by 4.6x under parallel AI inference (Section 3.1.1); DDR5, now benefiting from its own restored idle-gating mechanism, still spends 33-47% of its module power on refresh across every workload while ReRAM's refresh cost is identically zero (Figure 7); and under the repaired power model the technologies separate into leakage classes in which the selector-gated 1S1R module is the opportunity - 1.12 W, 1.8x DDR5's restored 0.623 W conservative calibration floor and within 11% of parity at the ceiling, with ReRAM's own idle-gating mechanically active but placeholder-valued (no modeled benefit yet, Appendix A) - against 50.9 W for the transistor-gated 1T1R module (65-78x DDR5, infeasible at DIMM scale) and 0.040 W for PCM, so the 47x selector leakage discipline that NVSim characterizes at the device level (794.7 vs. 16.9 mW per chip) becomes the deciding architectural fact of the evaluation (ReRAM figures are worst-case ungated; the DRAM and PCM baselines model standard idle behavior). A critical and previously unquantified result is the endurance characterization: under real workload write rates with uniform wear leveling, lifetime scales linearly with module capacity - the modeled 8 GB SLC module yields 1.1 years under worst-case sustained streaming (below the 5-10 year server target), while server-class 64-128 GB modules reach 9-17 years, and compute-bound and read-dominant workloads exceed the target by an order of magnitude at any capacity

(Table 5). The deployment recommendation follows directly from the repaired data, and it inverts the intuitive choice: 1S1R SLC is the flagship configuration - scale-viable power, 1.9x DDR5's die-level density (3.8x as MLC), and a modest ~1.5x average-latency cost against 1T1R (1.3-1.7x across the suite) - while 1T1R SLC, despite the best raw latency in the ReRAM family, is infeasible ungated at DIMM scale and is relegated to a latency-optimized niche pending aggressive idle-gating; MLC density remains restricted to read-dominant applications (static weight storage in AI inference pipelines) where the 3.263x write-latency penalty is rarely triggered. Together these findings define a concrete agenda for research and product work: gating-capable memory controllers (Section 4.1), selector-first DIMM architectures, and MLC read-only capacity tiers for frozen model weights. Finally, the mbmm_master.py orchestration framework, which bridges NVSim device characterization to NVMain system simulation in a single reproducible pipeline - now with per-technology leakage faithfully propagated - is contributed as an open-source Physical-to-System memory simulator for the non-volatile memory research community.

Each results axis resolves to a one-line verdict. Latency (3.1.1): the MLC write penalty is workload-dependent, not uniform - absorbed by controller queueing under compute-bound traces, compounding to 1.7x/2.8x under saturating streams - which confines MLC to read-dominant roles. Power (3.1.2): the earlier technology-blind “Standby Convergence” gave way to a leakage-class hierarchy - 50.9 W 1T1R, 1.12 W 1S1R (both mechanically power-gated now, at placeholder no-benefit energy), 0.623 W DDR5 (restored idle-gating floor), 0.040 W PCM - so ReRAM power *is* leakage, and the selector's 47x standby discipline is the largest technology-differentiating fact this project measured. Efficiency (3.1.3): that same leakage term inverts the intra-ReRAM hierarchy - 1S1R SLC dominates 1T1R SLC by 29x on geometric-mean PDP - leakage class, not cell speed, decides efficiency. Endurance (3.1.4): lifetime scales linearly with capacity, so the constraint binds only for small modules under sustained write streaming. Robustness (3.1.5): the operating point is stable - read latency invariant and system-level PDP within 4% across a +/-20% ReadVoltage sweep. Scaling (3.2): added ranks pay off only where memory-level parallelism exists to use them - the Flatline Paradox - making single-chip or 8-chip configurations Pareto-optimal for write-heavy deployments.

The scope of these claims is bounded by four disclosed limitations, gathered here deliberately in one place: validation is internal to the toolchain - parameters are anchored to real silicon and vendor datasheets, but no end-to-end result is checked against measured hardware (Section 2.2); the power-parity path rests on bounding arithmetic over idle-gating the simulator cannot yet exercise (Sections 3.1.6 item 5 and 3.3); endurance projections assume ideal uniform wear leveling, with the hot-spot exposure bounded but not simulated (Section 3.1.4); and the traces are cache-less, first-10M-instruction captures with one AI trace of unverifiable provenance (Sections 2.1, 3.1.6 item 6). None of these caveats inverts a headline - the leakage-class hierarchy is measured device physics propagated to module scale, the density ratios survive their cell-area sensitivity bounds (Section 3.3, cell-area sensitivity passage), and the endurance threshold moves but does not vanish under pessimistic leveling (Section 3.1.4 hot-spot bound) - but together they draw the boundary between what this book demonstrates and what remains to be demonstrated. That boundary is, by construction, the future-work agenda of Sections 4.1-4.2.

4.1. Future Work: Simulator and Infrastructure Enhancements

Power-Down Restoration (done this cycle; one gap remains): NVMain's previously-disabled power-down state machine (`MemoryController::HandleLowPower()`) has been restored - a shared mechanism, so it now fires for every technology, not ReRAM alone. DDR5 already carried real, JEDEC-datasheet-backed power-down currents and realizes a genuine, measured power reduction from them (0.651 to 0.623 W under GCC); the 1.8x gap separating 1S1R SLC from DDR5's restored floor is therefore now compared against a real, gated DDR5 number, not an ungated one. The remaining gap is on ReRAM's side: no NVSim datapoint decomposes its leakage into a gatable-periphery-vs- ungatable-crossbar split, and a dedicated literature search (Appendix A) found no citable ReRAM power-gating energy or fraction to characterize what a real gated ReRAM module would actually draw - so ReRAM's power-down energy is set to an explicit, disclosed placeholder equal to its own standby energy (no modeled savings), keeping the mechanism mechanically live (86-89% of cycle-slots gated, confirmed directly) while claiming no unearned benefit. **Sourcing a real ReRAM power-gating characterization - what fraction of its leakage is peripheral and gatable, and at what entry/exit energy cost - is now the single highest-leverage item of future work**, not restoring the mechanism itself. PCM shows no power-down activity at all in either run, plausibly because its `FRFCFS-WQF` controller's write-queue-flush design keeps its request queue non-empty far more of the time than the plain `FRFCFS` controller ReRAM/DDR5 use, starving the power-down entry condition - a second, smaller open item, not yet root-caused to full confidence.

- **Parameter Optimization**: Conduct a sensitivity analysis on NVSim parameters, such as ReadVoltage and WritePulseWidth, to identify the efficiency-optimal operating region for LOP (Low Operating Power) devices that maximizes sensing margin while minimizing energy.
- **Native MLC Logic**: Resolve the NVSim C++ "Floating Point Exception" (FPE) specifically within the Mat.cpp sensing logic to replace my current "Analytical Penalty Method" with native circuit-level characterization of iterative sensing.
- **Standalone Device-to-System Simulator Wrapper**: Generalizing "The Bridge" (Section 2.1) - the cross-layer ETL pipeline that translates NVSim's device-level physical metrics into NVMain's cycle-accurate architectural parameters - into a reusable, project-independent simulator wrapper. The toolchain audit (Section 3.1.6) already demonstrated that this translation layer is precisely where correctness risk concentrates; packaging it as a standalone, hardened tool would let other device-level-to-system-level simulation pairings benefit from the same repairs and validation discipline, beyond this project's specific NVSim/NVMain pairing.
- **FPGA-Based Hardware-in-the-Loop Validation of Power-Down Policy**: Every result in this book is validated only against NVMain's own internal consistency (Section 2.2) - no fabricated ReRAM hardware exists to check system-level, as opposed to device-level, behavior against. A tractable next step, complementary to Power-Down Restoration above, is an FPGA emulation

of just the idle-gating/power-down state machine - using the same literature-calibrated device parameters already characterized in this book, but exercised under real hardware clock-edge timing rather than software simulation - to check whether the controller's power-down and wake behavior matches its simulated prediction. This would strengthen the system-level validation claim without requiring access to fabricated ReRAM silicon, which remains out of reach; it would not, by itself, validate NVSim's own device-level circuit numbers against real silicon, since the emulated cell still runs on the same literature-derived parameters.

4.2. Future Work: Architectural Extensions and System Scaling

- **The L3 Cache Mitigation:** Evaluating the integration of a large last-level cache (LLC) or near-memory write buffer to absorb and coalesce ReRAM write penalties, a mitigation strategy historically proven essential for legacy phase-change memory **[11]**, specifically for write-intensive AI workloads like AlexNet OFMAP.

- **Memory Controller Queue Analysis (The Flatline Paradox):** Deep-diving into the NVMain memory controller interface, queue depths, and bus saturation limits to definitively resolve the multi-rank scaling flatline.

- **Endurance-Aware Scheduling:** Implementing wear-leveling algorithms at the controller level to protect the finite lifecycle of the 22nm cells.

- **Macro-Scale 1T1R Array Limits:** While this evaluation restricted 1T1R subarrays to 1024x1024 cells to maintain baseline architectural parity with the 1S1R models, 1T1R arrays are not fundamentally bound by the same sneak-path leakage limits. Future research should evaluate the latency, power, and routing tradeoffs of enlarging 1T1R arrays to their absolute commercial foundry manufacturing limits (e.g., TSMC maximum macro bounds) to push DIMM capacity to the extreme.

- **Recessed-Channel 1T1R Density:** this evaluation models 1T1R with a planar/FinFET, width-driven CMOS access transistor ($20F^2$, Appendix A), the only access-device topology NVSim supports for this project. A commercial part has demonstrated a DRAM-process buried recessed-channel access transistor instead, reaching a $6F^2$ cell - parity with DRAM [33] - though with a different switching-layer material (full discussion in Appendix A). Closing this gap for the oxide-RRAM material system modeled in this book would require either patching NVSim with a recessed-channel access-device model or deriving an equivalent RC/ drive-current model outside NVSim and feeding it back in; either is real device-physics work, not a configuration change, and could overturn this book's “1T1R is not density-competitive” verdict if the recessed-channel geometry proves compatible with oxide-RRAM integration.

- **Node Scaling and 3D Stacking**: this evaluation deliberately fixed ReRAM at 22nm - a manufacturability-anchored choice [8] - while DRAM competes from 10 nm-class nodes with no remaining capacitor runway [5], [17]. Future work should project the $4F^2$ cross-point's quadratic density dividend at 1x-nm feature sizes, quantify the multiplicative effect of back-end-of-line deck stacking [9], and re-derive the NVSim characterization at scaled nodes - where access-transistor leakage is expected to worsen while the selector's leakage discipline widens - propagating every density gain through the endurance capacity law, which converts capacity directly into lifetime.

- **Interface Parity: Dual-Channel and Faster-PHY ReRAM**: Every ReRAM configuration in this book runs behind a deliberately conservative single-channel 800 MHz interface (Section 2.3) - a modeling choice made to isolate the device-technology comparison from a confounding interface-engineering difference, not a limitation of the memristor media itself. Part of the 4.6x AI-inference latency gap (Section 3.1.1) is attributable to this asymmetry against DDR5's dual-channel architecture. Future work should evaluate a dual-channel ReRAM interface and/or a faster PHY paired with the same characterized media - both explicitly flagged elsewhere in this book as available mitigations, not hidden costs. The FPGA-based hardware-in-the-loop validation platform proposed above (Section 4.1) is a natural fit for this exploration: the same emulated controller could be extended from single-channel to dual-channel scheduling to measure the real benefit under real hardware timing, rather than purely in software simulation.

4.3 Data Availability and Reproducibility

To ensure the scientific reproducibility of the 22nm ReRAM models, the complete simulation stack, including the mbmm_master.py orchestration script, modified NVSim/NVMain C++ source files, and JSON configuration data, has been made publicly available.

> **Project Repository:** https://github.com/uvKogan/MBMM

References

[1] M. Ahmad, "NAND Flash's Reversal of Fortune Amid the AI Boom," *EE Times*, Jan. 26, 2026. [Online]. Available: https://www.eetimes.com/nand-flashs-reversal-of-fortune-amid-the-ai-boom/. [Accessed: Apr. 25, 2026].

[2] A. J. Fleischer, "How AI Broke the Memory Market," *Octopart Pulse*, Mar. 13, 2026. [Online]. Available: https://octopart.com/pulse/p/how-ai-broke-memory-market. [Accessed: Apr. 25, 2026].

[3] X. Dong et al., "NVSim: A Circuit-Level Performance, Energy, and Area Model for Emerging Nonvolatile Memory," *IEEE Trans. Comput.-Aided Design Integr. Circuits Syst.*, vol. 31, no. 7, pp. 994-1007, July 2012.

[4] M. Poremba, T. Zhang, and Y. Xie, "NVMain 2.0: A user-friendly memory simulator to model (non-)volatile memory systems," IEEE Comput. Archit. Lett., vol. 14, no. 2, pp. 140-143, July-Dec. 2015.

[5] International Roadmap for Devices and Systems (IRDS), "More Moore," 2024. [Online]. Available: https://irds.ieee.org/images/files/pdf/2024/2024IRDS_MM.pdf.

[6] L. R. Upton et al., "EMBER: A 100 MHz, 0.86 mm2, multiple-bits-per-cell RRAM macro in 40 nm CMOS with compact peripherals and 1.0 pJ/bit read circuitry," in Proc. IEEE 49th Eur. Solid State Circuits Conf. (ESSCIRC), Sep. 2023, pp. 469-472.

[7] C. Matsui, A. Yamada, N. Misawa, and K. Takeuchi, "ReRAM resistance design of LRS and HRS for ultrahigh-capacity digital memory and analog Computation-in-Memory," IEICE Trans. Fundamentals, vol. E109.A, no. 3, pp. 596-603, Mar. 2026. DOI: 10.1587/transfun.2025VLP0011

[8] C. Xue et al., "16.1 A 22nm 4Mb 8b-Precision ReRAM Computing-in-Memory Macro with 11.91 to 195.7TOPS/W for Tiny AI Edge Devices," *2021 IEEE International Solid-State Circuits Conference (ISSCC)*, San Francisco, CA, USA, 2021, pp. 245-247.

[9] S. H. Jo et al., "3D-stackable crossbar resistive memory based on Field Assisted Superlinear Threshold (FAST) selector," *2014 IEEE International Electron Devices Meeting (IEDM)*, San Francisco, CA, USA, 2014, pp. 6.7.1-6.7.4.

[10] JEDEC Solid State Technology Association, "DDR5 SDRAM Standard," JESD79-5D, Nov. 2025. [Online]. Available: https://www.jedec.org/standards-documents/docs/jesd79-5d.

[11] B. C. Lee, E. Ipek, O. Mutlu, and D. Burger, "Architecting phase change memory as a scalable dram alternative," in *Proc. 36th Annu. Int. Symp. Comput. Archit. (ISCA)*, 2009, pp. 2-13.

[12] A. Samajdar, J. M. Joseph, Y. Zhu, P. Whatmough, M. Mattina, and T. Krishna, "A Systematic Methodology for Characterizing Scalability of DNN Accelerators Using SCALE-Sim," in *Proc. IEEE Int. Symp. Perform. Anal. Syst. Softw. (ISPASS)*, 2020, pp. 58-68.

[13] B. Q. Le et al., “Resistive RAM With Multiple Bits Per Cell: Array-Level Demonstration of 3 Bits Per Cell,” *IEEE Trans. Electron Devices*, vol. 66, no. 1, pp. 641–646, Jan. 2019.

[14] H.-S. P. Wong et al., “Metal–Oxide RRAM,” *Proc. IEEE*, vol. 100, no. 6, pp. 1951–1970, June 2012. DOI: 10.1109/JPROC.2012.2190369

[15] M. Lanza et al., “Standards for the Characterization of Endurance in Resistive Switching Devices,” *ACS Nano*, vol. 15, no. 11, pp. 17214–17231, Nov. 2021. DOI: 10.1021/acsnano.1c06980

[16] Data Center Knowledge, “Data Center Hardware Refresh Cutback by Microsoft – What’s Next?,” Aug. 25, 2022. [Online]. Available: https://www.datacenterknowledge.com/hyperscalers/data-center-hardware-refresh-cutback-by-microsoft-what-s-next-. [Accessed: Jul. 10, 2026].

[17] TrendForce, "Micron Races Ahead in 10nm-Class DRAM with 1γ DDR5 Samples Delivered to Intel & AMD," Feb. 26, 2025. [Online]. Available: https://www.trendforce.com/news/2025/02/26/news-micron-races-ahead-in-10nm-class-dram-with-1γ-ddr5-samples-delivered-to-intel-amd/. [Accessed: Jul. 11, 2026]. **[18]** J. Choe, "Comparing DDR5 Memory From Micron, Samsung, SK Hynix," EE Times, Feb. 15, 2022. [Online]. Available: https://www.eetimes.com/comparing-ddr5-memory-from-micron-samsung-sk-hynix/. [Accessed: Jul. 12, 2026]. **[19]** A. Shilov, "Samsung Puts 3D DRAM on the Roadmap, Stacked DRAM to Follow," Tom's Hardware, Apr. 3, 2024. [Online]. Available: https://www.tomshardware.com/pc-components/dram/samsung-outlines-plans-for-3d-dram-which-will-come-in-the-second-half-of-the-decade. [Accessed: Jul. 12, 2026]. **[20]** J. Yang et al., "A 14nm-FinFET 1Mb Embedded 1T1R RRAM with a 0.022μm² Cell Size Using Self-Adaptive Delayed Termination and Multi-Cell Reference," in Proc. IEEE Int. Solid-State Circuits Conf. (ISSCC), 2021. DOI: 10.1109/ISSCC42613.2021.9365945 **[21]** K. T. Malladi, F. A. Nothaft, K. Periyathambi, B. C. Lee, C. Kozyrakis, and M. Horowitz, "Towards Energy-Proportional Datacenter Memory with Mobile DRAM," in Proc. 39th Annu. Int. Symp. Computer Architecture (ISCA), 2012, pp. 37-48. DOI: 10.1109/ISCA.2012.6237004 **[22]** A. Gholami, Z. Yao, S. Kim, C. Hooper, M. W. Mahoney, and K. Keutzer, "AI and Memory Wall," IEEE Micro, vol. 44, no. 3, pp. 33-39, May/June 2024. DOI: 10.1109/MM.2024.3373763 **[23]** Tom's Hardware, "Intel Kills Optane Memory Business Entirely, Pays $559 Million to Exit," Jul. 28, 2022. [Online]. Available: https://www.tomshardware.com/news/intel-kills-optane-memory-business-for-good. [Accessed: Jul. 13, 2026]. **[24]** TechInsights, "Advanced TSMC 22ULL Embedded RRAM Chip Unveiled." [Online]. Available: https://www.techinsights.com/blog/advanced-tsmc-22ull-embedded-rram-chip-unveiled. [Accessed: Jul. 13, 2026]. **[25]** Tom's Hardware, "Neo Semiconductor's Revolutionary 3D X-DRAM for AI Processors Has Passed Proof-of-Concept Validation," Apr. 24, 2026. [Online]. Available: https://www.tomshardware.com/tech-industry/artificial-intelligence/neo-semiconductors-revolutionary-3d-x-dram-for-ai-processors-has-passed-proof-of-concept-validation-company-secures-funding-to-develop-next-gen-memory-hbm-alternative. [Accessed: Jul. 13, 2026]. **[26]** N. Binkert et al., "The gem5 Simulator," ACM SIGARCH Computer Architecture News, vol. 39, no. 2, pp. 1-7, May 2011. DOI: 10.1145/2024716.2024718 **[27]** J. D. McCalpin, "Memory Bandwidth and Machine Balance in Current High Performance

Computers," IEEE Computer Society Technical Committee on Computer Architecture (TCCA) Newsletter, pp. 19-25, Dec. 1995. **[28]** Standard Performance Evaluation Corporation, "SPEC CPU 2017 Benchmark Suite." [Online]. Available: https://www.spec.org/cpu2017/. [Accessed: Jul. 13, 2026]. **[29]** Micron Technology, Inc., "16Gb DDR5 SDRAM Addendum: MT60B4G4, MT60B2G8, MT60B1G16, Die Revision A," Doc. No. CCM005-0005-1684161373-30, Rev. D, Feb. 2023. [Online]. Available: https://www.micron.com/products/memory/dram-components/ddr5-sdram/part-catalog **[30]** SK hynix Inc., "16Gb DDR5 SDRAM," datasheet. [Online]. Available (registration required): https://product.skhynix.com/support/downloads.go

[31] A. Levy, L. R. Upton, M. D. Scott, D. Rich, W.-S. Khwa, Y.-D. Chih, M.-F. Chang, S. Mitra, B. Murmann, and P. Raina, "EMBER: Efficient Multiple-Bits-Per-Cell Embedded RRAM Macro for High-Density Digital Storage," *IEEE J. Solid-State Circuits*, vol. 59, no. 7, pp. 2081-2092, July 2024. DOI: 10.1109/JSSC.2024.3387566

[32] J. Izraelevitz, J. Yang, L. Zhang, J. Kim, X. Liu, A. Memaripour, Y. J. Soh, Z. Wang, Y. Xu, S. R. Dulloor, J. Zhao, and S. Swanson, "Basic Performance Measurements of the Intel Optane DC Persistent Memory Module," arXiv:1903.05714, Aug. 2019.

[33] R. Fackenthal, M. Kitagawa, W. Otsuka, K. Prall, D. Mills, K. Tsutsui, J. Javanifard, K. Tedrow, T. Tsushima, Y. Shibahara, and G. Hush, "19.7 A 16Gb ReRAM with 200MB/s Write and 1GB/s Read in 27nm Technology," in Proc. IEEE Int. Solid-State Circuits Conf. (ISSCC), Feb. 2014, pp. 338-339. DOI: 10.1109/ISSCC.2014.6757460

[34] Intel Corporation, "Ultra High Bandwidth Memory with Backend Transistors," U.S. Patent Application Publication No. US 2026/0191095 A1, Appl. No. 19/001,921, filed Dec. 26, 2024, published Jul. 2, 2026. [Online]. Available: https://www.freepatentsonline.com/y2026/0191095.html. [Accessed: Aug. 22, 2026].

[35] TrendForce, "Intel Patent Reveals XBM Matching HBM4 Footprint Without Interposers; Commercialization Seen After 2030," Jul. 8, 2026. [Online]. Available: https://www.trendforce.com/news/2026/07/08/news-intel-patent-reveals-xbm-matching-hbm4-footprint-without-interposers-commercialization-seen-after-2030/. [Accessed: Aug. 22, 2026].

[36] B. C. Lee, E. Ipek, O. Mutlu, and D. Burger, "Architecting Phase Change Memory as a Scalable DRAM Alternative," in *Proc. 36th Annu. Int. Symp. Computer Architecture (ISCA)*, 2009, pp. 2-13. DOI: 10.1145/1555754.1555758

[37] M. K. Qureshi, V. Srinivasan, and J. A. Rivers, "Scalable High Performance Main Memory System Using Phase-Change Memory Technology," in *Proc. 36th Annu. Int. Symp. Computer Architecture (ISCA)*, 2009, pp. 24-33. DOI: 10.1145/1555754.1555760

[38] P. Zhou, B. Zhao, J. Yang, and Y. Zhang, "A Durable and Energy Efficient Main Memory Using Phase Change Memory Technology," in *Proc. 36th Annu. Int. Symp. Computer Architecture (ISCA)*, 2009, pp. 14-23. DOI: 10.1145/1555754.1555759

[39] C. Xu, D. Niu, N. Muralimanohar, R. Balasubramonian, T. Zhang, S. Yu, and Y. Xie, "Overcoming the Challenges of Crossbar Resistive Memory Architectures," in *Proc. 21st IEEE*

Int. Symp. High Performance Computer Architecture (HPCA), 2015, pp. 476-488. DOI: 10.1109/HPCA.2015.7056056

[40] E. Kültürsay, M. Kandemir, A. Sivasubramaniam, and O. Mutlu, "Evaluating STT-RAM as an Energy-Efficient Main Memory Alternative," in *Proc. IEEE Int. Symp. Performance Analysis of Systems and Software (ISPASS)*, 2013, pp. 256-267. DOI: 10.1109/ISPASS.2013.6557176

[41] Y. Choi et al., "A 20nm 1.8V 8Gb PRAM with 40MB/s Program Bandwidth," in Proc. IEEE Int. Solid-State Circuits Conf. (ISSCC), San Francisco, CA, Feb. 2012, pp. 46-48.

[42] D. Kau, S. Tang, I. V. Karpov, R. Dodge, B. Klehn, J. A. Kalb, J. Strand, A. Diaz, N. Leung, J. Wu, S. Lee, T. Langtry, K.-W. Chang, C. Papagianni, J. Lee, J. Hirst, S. Erra, E. Flores, N. Righos, H. Castro, and G. Spadini, "A Stackable Cross Point Phase Change Memory," in *2009 IEEE International Electron Devices Meeting (IEDM)*, Baltimore, MD, Dec. 2009, pp. 617-620. DOI: 10.1109/IEDM.2009.5424263

[43] International Technology Roadmap for Semiconductors, 2011 Edition, Process Integration, Devices, and Structures (PIDS), Semiconductor Industry Association, 2011, pp. 10-11 (subthreshold source/drain leakage current design targets: HP logic 100 nA/µm, LOP logic 5 nA/µm, LSTP logic 10 pA/µm).

[44] C. Auth et al., "A 22nm High Performance and Low-Power CMOS Technology Featuring Fully-Depleted Tri-Gate Transistors, Self-Aligned Contacts and High Density MIM Capacitors," in *2012 Symposium on VLSI Technology (VLSIT) Digest of Technical Papers*, June 2012, pp. 131-132.

Appendix A: Simulation Parameters and Literature Grounding

- **Process Node:** 22nm FinFET LOP (Low Operating Power). *Justification:* LOP was selected over High Performance (HP) to guarantee electrical convergence in the NVSim solver and to accurately reflect the strict thermal envelopes required for high-capacity Main Memory modules.
- **Resistance Targets:** $10^5\Omega$ LRS and $10^9\Omega$ HRS. *Reference:* The $10^5\Omega$ LRS floor is **Matsui et al.'s [7]** direct recommendation for high-capacity *digital* ReRAM memory, to mitigate bitline IR-drop on dense arrays; the paired $10^9\Omega$ HRS is adopted from the same paper's analog Computation-in-Memory (CiM) design point as a representative high-resistance target, since [7] does not separately specify an HRS floor for the digital-memory case. *Sensitivity:* A dedicated NVSim sweep across HRS $\in \{25, 100, 1000, 10000\} \times$ LRS (i.e. $2.5 \times 10^6\Omega$ to $10^9\Omega$, LRS held fixed at $10^5\Omega$) shows this choice has negligible practical consequence: modeled leakage power is bit-for-bit identical across the full two-orders-of-magnitude range for both 1T1R (794.656 mW) and the selector-gated cell (16.907 mW) - the 47x leakage-class separation reported in Section 3.1.6 is driven entirely by the CMOS-transistor-vs-selector access-device model, not by HRS - and read latency varies by at most 1.7% (9.472 ns at $25\times$ to 9.631 ns at $1000\times$-$10000\times$) across the swept range. The exact HRS value is therefore not load-bearing for any headline finding in this book.
- **Access-Device Leakage Model (the 47x figure):** The transistor-vs-selector leakage-class separation reported throughout Section 3.1.2/3.1.3 rests on two independently-sourced leakage currents, not one. The selector side is the $10^9\Omega$ HRS target above, from Matsui et al. [7]. The transistor side - previously cited only as "NVSim's internal 22nm process model," with no external reference - is independently corroborated by two convergent sources: the ITRS 2011 Edition (PIDS chapter) subthreshold-leakage design target for 22nm *Low Operating Power* logic, $5\ \mathrm{nA}/\mu\mathrm{m}$ **[43]**, and C. Auth et al.'s (Intel) measured off-state leakage on actual fabricated 22nm FinFET silicon, $5\text{-}20\ \mathrm{nA}/\mu\mathrm{m}$ for their mid-power variant **[44]** - the ITRS LOP target sits inside Intel's independently measured band, a roadmap projection corroborated by real silicon. A rough plausibility check (not a re-derivation of NVSim's internal constant): selector leakage at $10^9\Omega$ HRS is on the order of 0.5-1 nA per device, and the cited transistor I_{off} figures (1-100 nA/µm) place a per-device transistor leakage in the same broad 1-100 nA range - consistent with a roughly 47x, not many-orders-of-magnitude, separation. Both sides of the 47x ratio are now independently cited; NVSim's own paper **[3]** was not itself accessible to confirm its exact internal constant, so this remains corroborating evidence, not a re-derivation.
- **MLC Penalties:** 1.5x Read Latency, 3.263x Write Latency, 1.1x Read Energy, 3.0x Write Energy (2 bits/cell vs. 1 bit/cell). *Reference:* Directly measured on the same EMBER macro, across its two publications by the same author group. Read energy (1.0/1.1 pJ/bit at 1/2 b per cell) is from **Upton et al. [6]** (ESSCIRC 2023), Table I. Read latency, write-verify bandwidth, and write-verify energy are all from the journal follow-up, **Levy et al. [31]** (IEEE JSSC 2024): read bandwidth 2.4/1.6 Gbps at 1/2 b per cell (Section V.A / Abstract) gives the 1.5x read-

latency multiplier via the same bandwidth-ratio method as write-latency; write-verify bandwidth 12.4/3.8 Mbps and write-verify energy 0.40/1.2 nJ/bit at 1/2 b per cell (Section V.B) give the 3.263x and 3.0x multipliers. The ESSCIRC conference paper reports no write-verify data split by bits-per-cell (only an aggregate, non-split SET/RESET pulse-energy estimate) and no 2-bit/cell read-latency figure at all - its Table I “Read Time (1 b)” row lists only 1-bit/cell numbers for every compared macro, including EMBER’s own. Earlier drafts of this book used unsourced 3x/4x placeholder multipliers attributed to “EMBER Macro analytical heuristics” that do not exist in either EMBER publication; a subsequent correction replaced these with 1.917x/3.263x/1.1x/3.0x, but the 1.917x read-latency figure was itself a data error - it paired EMBER’s own 12 ns read time with a “23 ns” figure that Table I attributes to a different, competing macro’s 1-bit/cell measurement, not EMBER’s 2-bit/cell number. The 1.5x figure above corrects this using EMBER’s own bandwidth data (Section 3.1.6, items 13 and 14 document both correction rounds and the resulting re-simulations).

- **Bit-Cell Area:** 20 F^2 for 1T1R, 4 F^2 for 1S1R. *Justification:* 20 F^2 reflects the physical reality of CMOS-compatible access transistors, whereas 4 F^2 represents the idealized limit of cross-point selector technology. The 1T1R figure is transistor-limited, not memristor-limited: the memristor occupies the back-end-of-line above the cell, while the access transistor must be sized wide enough to source the SET/RESET programming current [14]. A production-grade 14nm FinFET embedded 1T1R macro reports a 0.022 μm^2 cell - about 112F^2 at that node [20] - so 20F^2 is generous relative to planar/FinFET logic-transistor integration, which makes the density verdict against it conservative *for that architecture*. It is not generous in absolute terms: DRAM-process integration using a buried recessed-channel access transistor rather than a planar logic transistor has been demonstrated at 6F^2 - parity with DRAM's own 6F^2 cell - in a commercial 16Gb, 27nm part [33]. That device pairs the recessed transistor with a Cu-filament CBRAM switching layer rather than the HfOx/TaOx oxide-RRAM family modeled here, and NVSim's access-device model in this project (width-driven CMOS or diode only, no recessed-channel option) cannot represent it, so no simulated results are claimed for that configuration anywhere in this book (see Section 4.2 for the future-work case). [33] stands as an existence proof that 1T1R density is architecture-dependent, not a validated data point: 20F^2 should be read as a mid-range, logic-compatible assumption, not a technology ceiling. Note that the 4F^2 figure is the array-level cell footprint; peripheral circuitry (sense amplifiers, row decoders, column multiplexers) adds overhead, and the periphery-inclusive die-level density ratios are reported in Section 3.3 and Figure 26 (1S1R: 1.92x SLC / 3.84x MLC versus DDR5).
- **Memory Controller Queue Depth:** The FRFCFS memory controller (`MemControl/FRFCFS/FRFCFS.cpp`) reads a single combined read+write `QueueSize` key, falling back to a hardcoded 32 entries when unset. A separate `ReadQueueSize`/`WriteQueueSize` pair exists only in the different FRFCFS-WQF controller, which this project's ReRAM and DDR5 configurations never use - only the PCM baseline runs under FRFCFS-WQF. Every ReRAM and DDR5 configuration generated by this project's pipeline left `QueueSize` unset prior to this study, silently relying on that hardcoded default rather than a documented, deliberate choice;

`3_gen_nvmain_config.py` now writes it explicitly (still defaulting to 32, so no headline result in this book changes). *Sensitivity:* I ran a dedicated sweep (`sweep_queue_size.py`), using the same matched-host LBM window as Section 3.1.1 (Section 3.1.6, item 11), raising `QueueSize` from the 32-entry default until each configuration stopped completing within a bounded wall-clock budget. At the 1T1R SLC full-DIMM scale - the configuration behind Section 3.1.1's “1T1R SLC completes 40.0%” figure - completion rose from 40.0% (32, the reported baseline) to 40.0% (48) to 40.7% (64, the practical ceiling; 80 exceeded the simulation-time budget) of DDR5's 16,447,102-request reference, at a real latency cost: average total latency rose from 374.5 ns to 536.4 ns (+43.2%) to 686.9 ns (+83.4%) over the same range. The effect is far larger at single-chip scale, where less inherent bank/rank parallelism leaves more of LBM's demand queue-depth-limited rather than device-limited: completed requests rose monotonically from QueueSize 32 through 128 (+4.16% at 128, versus +0.7% at full-DIMM scale), at a steeper latency cost (+258% at 128 versus +83% at full-DIMM). Larger queues become impractical to simulate well before any completion ceiling is reached (90 s at single-chip, roughly 200 s at full-DIMM, both wall-clock simulation time, not modeled device time). *Conclusion:* the LBM completion gap reported in Section 3.1.1 is not a pure device-physics ceiling - it is partly a controller-queue artifact, and it is tunable, but only within a narrow, real throughput/latency tradeoff. This book's headline results retain the documented 32-entry default throughout; this sweep characterizes that choice, it does not change it.

- **Address-Footprint / Rank-Mapping Root Cause (Section 3.2's Scaling Reversal):** An earlier draft of Section 3.2 claimed gcc (compute-bound) gets “almost zero latency benefit” from chip-count scaling while the AI-inference traces get theirs “drastically slashed” via rank-level interleaving, attributing the difference to workload memory-level parallelism (MLP). This project's own generated data shows the opposite: gcc improves ~14% (152.58 to 130.91 ns, 1T1R SLC) as chip count scales, while GPT-2 IFMAP is exactly flat (458.41 ns at every chip count) and AlexNet IFMAP is nearly flat (394.73 to 397.41 ns). *Root cause, confirmed by direct experiment against real NVMain source, not inferred:* this configuration's address-decode logic (R:BK:RK:C mapping, COLS=1024, 64-byte bursts - `src/AddressTranslator.cpp`, `src/TranslationMethod.cpp`) gives each rank an addressable span of exactly $2^6 \times 2^{10}$ = 65,536 bytes (64 KB) before the decoder rolls into the next rank, independent of how many ranks the architecture provides. gcc's SPEC trace touches ~51.4 MB of address space - about 800x that span - so it genuinely exercises every additional rank as chip count scales. GPT-2 IFMAP's trace touches only ~64.0 KB, six bytes short of the exact 64 KB boundary: every request in the entire trace decodes to rank 0 regardless of architecture, so scaling chip count is architecturally inert for it. AlexNet IFMAP's ~68.3 KB footprint spills a small tail past that boundary into a second rank, producing the small, non-beneficial sensitivity observed. *Queue-depth ruled out:* a dedicated sweep (`sweep_queue_size.py`) across QueueSize values 16/32/64 and all four chip-count architectures, for both gcc and GPT-2 IFMAP (24 runs, at each trace's confirmed matched-host cycle count), shows queue depth changes absolute latency but has zero interaction with the chip-count flatline for either trace - GPT-2 IFMAP's stats are bit-for-bit identical across every architecture at every queue depth tested. *Conclusion:* the reported effect is a workload address-footprint artifact, not evidence about memory-level parallelism in general - the two AI-inference traces available to this study are almost certainly single-layer SCALE-Sim captures far smaller than a full model's real working set, not representative of “AI inference” as a workload class. Section 3.2's narrative has been corrected accordingly; a larger, more representative AI-inference trace remains future work (Section 4.1).

Appendix B: Execution Pipeline and Command Interface

To facilitate replication of the system scaling and trace execution, the following command-line interface (CLI) instructions document the exact mbmm_master.py parameters used to generate the results in Section 3.

B.1: Configuration Generation The system architecture factory dynamically maps hardware constraints (e.g., forcing a 64-bit bus width for single-chip models) sets the DIMM clock frequency to 800 MHz, and - since the matched-host correction (Section 3.1.6, item 11) - pins the host reference frequency to CPUFreq = 3000 MHz for every configuration:

Bash

```
python3 3_gen_nvmain_config.py --freq 800
```

B.2: High-Cycle Batch Simulation The master orchestration script executes the gcc_spec2017.nvt trace across the full matrix of 18 architectures (DRAM baselines, 1T1R, and 1S1R up to 64-chip Full DIMMs) with matched-host cycle budgets, cycles = ceil(250,000,000 x CLK / 3000), so that every configuration admits the identical trace population over an identical 83.33 ms window (Section 3.1.6, item 11) - 66,666,667 cycles for the 800 MHz ReRAM matrix:

Bash

```
python3 mbmm_master.py --cycles 66666667 --trace benchmarks/gcc_spec2017.nvt --models \

2D_DRAM_example 3D_DRAM_example \

reram_22nm_1t1r_slc_single reram_22nm_1t1r_slc_8chip reram_22nm_1t1r_slc_16chip
reram_22nm_1t1r_slc_full_dimm \

reram_22nm_1t1r_mlc_single reram_22nm_1t1r_mlc_8chip reram_22nm_1t1r_mlc_16chip
reram_22nm_1t1r_mlc_full_dimm \

reram_22nm_selector_slc_single reram_22nm_selector_slc_8chip
reram_22nm_selector_slc_16chip reram_22nm_selector_slc_full_dimm \

reram_22nm_selector_mlc_single reram_22nm_selector_mlc_8chip
reram_22nm_selector_mlc_16chip reram_22nm_selector_mlc_full_dimm
```

The DDR5-4800 and PCM baselines complete the 20-configuration matrix and are executed with the same interface using their model names (DDR5_4800_DRAM, pcm_microsoft_2009) and their own matched-host budgets (200,000,000 and 33,333,334 cycles; 2D/3D controls: 55,500,000 and 111,083,334); the same command is repeated for each of the six benchmark traces.

Appendix C: Reproducibility Statement

Repository: https://github.com/uvKogan/MBMM. The complete simulation stack (NVSim 22nm FinFET LOP models, NVMain 2.0 configurations, workload traces, and all post-processing scripts) is available under the MIT License.

Software versions: NVSim (patched, commit hash in repo README), NVMain 2.0 (patched, commit hash in repo README), gem5 v25.1, SCALE-Sim v2, Python 3.10+, matplotlib 3.9, pandas 2.2.

Pipeline interface: The master orchestration script mbmm_master.py accepts --cycles (simulation window), --trace (NVMain trace file), and --models (space-separated list of named configurations). A complete simulation run across all 20 configurations and 6 workloads at 200M cycles completes in approximately 8-12 hours on a standard Linux workstation. All statistical outputs (processed_bar_chart_metrics.csv, processed_hero_metrics.csv, processed_geometric_means.csv) and generated figures are written to results/ and are fully reproducible from the raw NVMain stats files.